

Patreon Catalogue

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Below are all the files included in the “**File + Commercial Tier**” on my [Patreon](#).

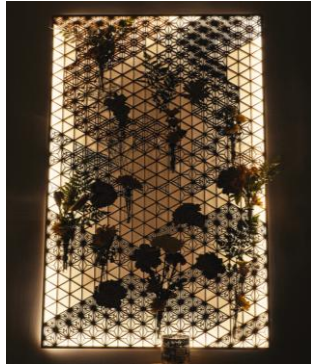
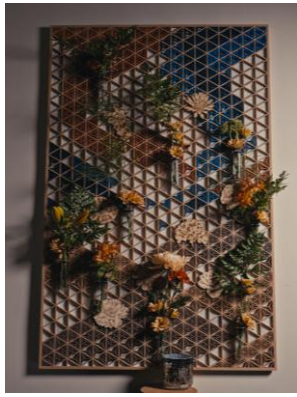
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Print Profiles Included

Size
(W x H x D)

Materials (Affiliate Links)
Filament per kg spool

Additional Files



79.4 cm x 121.8 cm x 2.4 cm

[Details](#)

Filament

[2 x latte brown](#)

[1 x caramel](#)

[1 x dark red](#)

[1 x marine blue](#)

[1 x dark blue](#)

[1 x dark brown](#)

[1 x terracotta](#)

[1 x white](#)

Reduced Filament Alternative

[2 x latte brown](#)

[1 x caramel](#)

[1 x dark red](#)

[1 x marine blue](#)

[1 x white](#)

Flower Laser Cut Templates

Test Tube Holder Inserts

Magnet-Compatible Inserts

Other

[Smooth PEI Plate](#)

[LED Strip](#)

[1" Test Tubes](#)

[Magnets \(D10 x 2mm\)](#)

[Plywood](#)

[Cork](#)



218 cm x 81.8 cm x 2.4 cm

[Details](#)

Filament

[5 x latte brown](#)

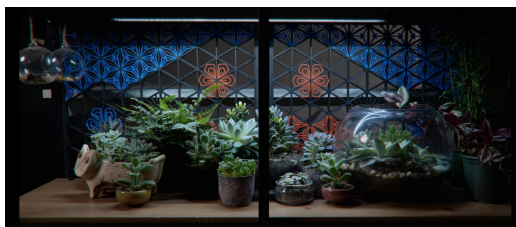
[1 x black](#)

[1 x dark red](#)

[1 x marine blue](#)

Other

[Smooth PEI Plate](#)



[Fits in a FJALLBO Shelf](#)

[Details](#)

Filament

[1 x black](#)

[1 x marine blue](#)

[1 x dark red](#)

FJALLBO compatible hangers

Other

[Smooth PEI Plate](#)

[FJALLBO Shelf](#)



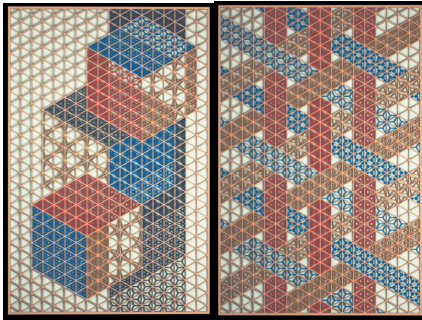
[Fits in a Gorilla Rack Shelf](#) (Non-Industrial Version)

Filament
[2 x black \(very close to 3\)](#)
[1 x marine blue](#)

Other
[Smooth PEI Plate](#)
[Gorilla Rack Shelf](#)

Keyhole compatible hangers

[Details](#)

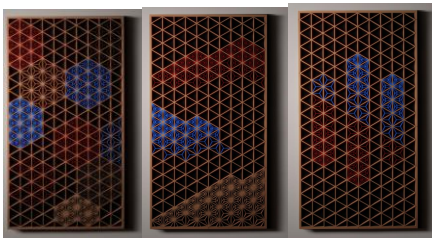


79.4 cm x 121.8 cm x 2.4 cm

Filament
[2 x latte brown](#)
[1 x caramel](#)
[1 x dark red](#)
[1 x marine blue](#)
[1 x white](#)
[1 x dark blue \(for cubes\)](#)

Other
[Smooth PEI Plate](#)

[Details](#)



44.8 cm x 81.8 cm x 2.4 cm

Filament
[1 x latte brown](#)
[1 x marine blue](#)
[1 x dark red](#)
[1 x black](#)

Other
[Smooth PEI Plate](#)

[Details](#)

Source Code Files Included

File Name	Description	Instructions
modular_kumiko_panels.scad	The modular_kumiko_panels.scad file lets you fully customize your panel. It will let you create panels that fit exact dimensions, adjust the pitch size, change the frame depth, change the border thickness, and lets you generate all the components	Click here for instructions

without having to go through the customizer on MakerWorld.

svg_insert_generator.scad

The **svg_insert_generator.scad** file will let you generate the background inserts and designs made using kumikodesigner.com in just a few steps. Faster than the manual generator alternative.

[Click here for instructions](#)

individual_insert_generator.scad

Lets you generate components individually (one pattern at a time). Additionally, it lets you generate the test tube holder components, and create inserts that can embed magnets.

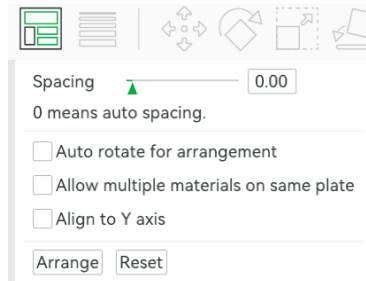
[Equivalent to MakerWorld Instructions](#)

Additional features that MakerWorld lacks:

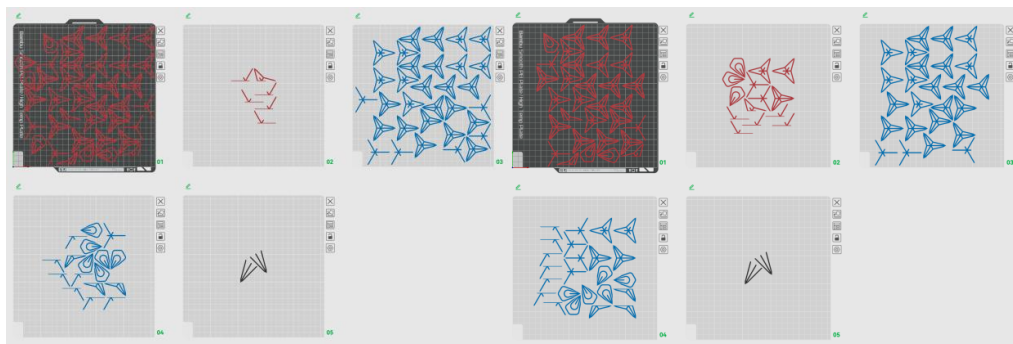
- Can generate test tube holders (Pattern = -1)
- Magnet Insert Generator

Printing Components

- **Bambu Studio users:** use the .3mf files and print the components. Print parameters are built into the objects and shouldn't need to be modified but if you accidentally change parameters, refer to the table below for parameter settings. Pattern and background inserts are left unarranged so you can reference them during the assembly process. To arrange the components, select the following option in Bambu Studio



Auto arrange inserts (**Make sure “Allow multiple materials on same plate” is deselected**)



Move any inserts close to the edges to a different plate and rearrange.

Left: before | Right: after

- Other users: Check if your printer's profile is available in Bambu Studio. If it's available, I highly recommend printing from Bambu Studio to avoid having to manually set things up. Alternatively, I provide all the frame components, inserts, and background pieces in their respective .stl files which you can use and prepare manually. Below are my recommended printing settings for each component.

Component	Walls	Infill Density	Infill Pattern	Wall Generator	Top shell Layers	Bottom Shell Layers
Subpanels	1	30%	Lightning (to save material)	Arachne *	>= 5	>= 3
Seams						
Outer_Frame		10%	Gyroid (for strength)			
Border						
Border_Background	>= 5	15%				

Hanger						
--------	--	--	--	--	--	--

Component	Layer Height	Initial Layer Height	Walls	Infill Density	Infill Pattern	Wall Generator	Top shell Layers	Bottom Shell Layers
Background Inserts	0.1	0.1	1	30%	Lightning (to save material)	Arachne *	>= 5	>= 3
Pattern Inserts	0.2	0.2						

** It is essential the wall generator is set to "Arachne". The walls made using "Classic" are too thick.*

Assembly Instructions

[Assembly instructions are available here.](#) The FJALLBO and Gorilla Shelf compatible panels use a different hanging method. See below for their specific instructions:

V2 Primary Panel Additional Instructions



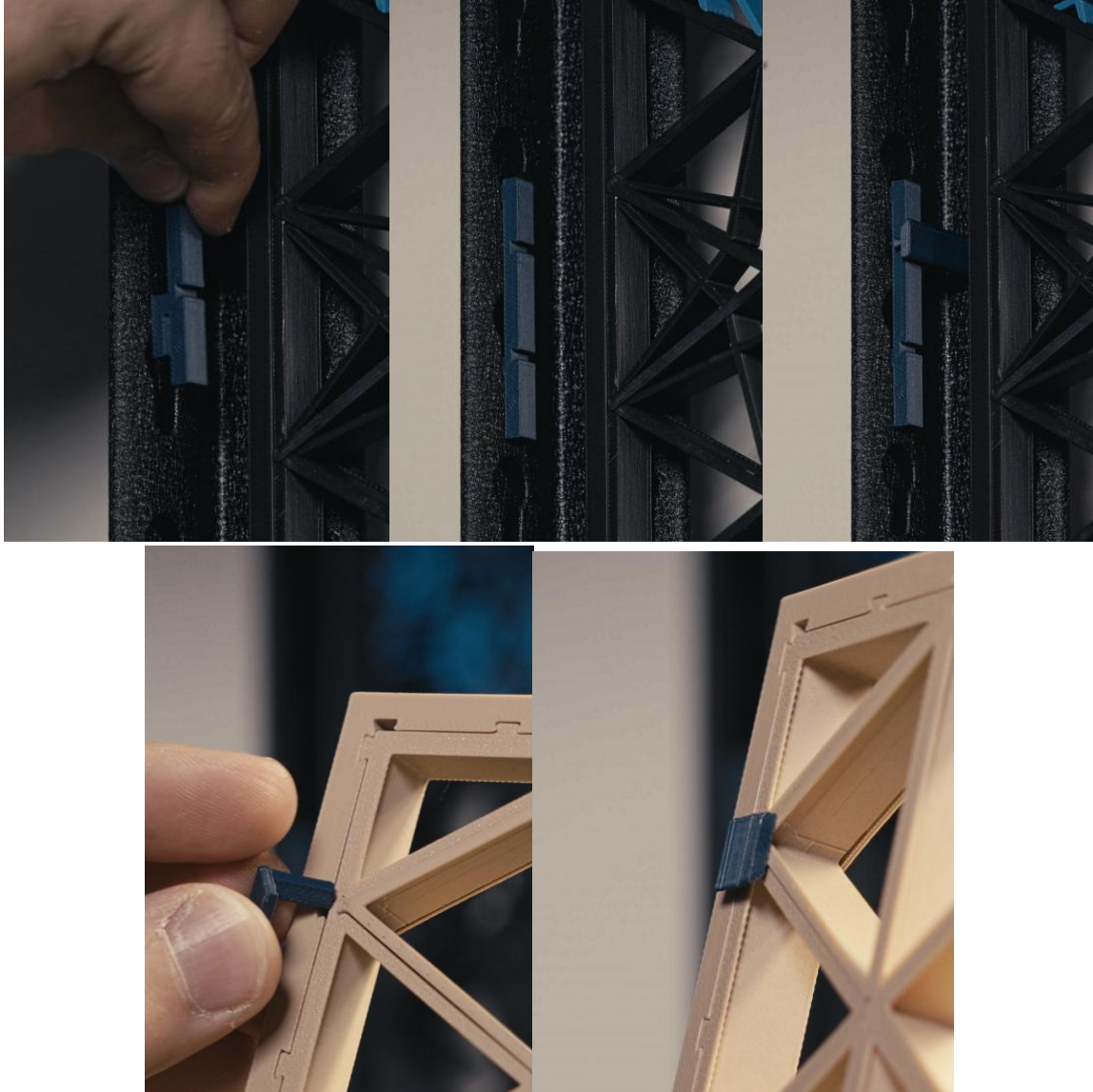
FJALLBO Compatible Panel Installation

- On vacation RN so I only have access to my phone so the instructions will be slow to type up. The gist is there are hanger components that latch onto the shelves but you should probably add some double sided tape or sticky tac to hold the components in place.



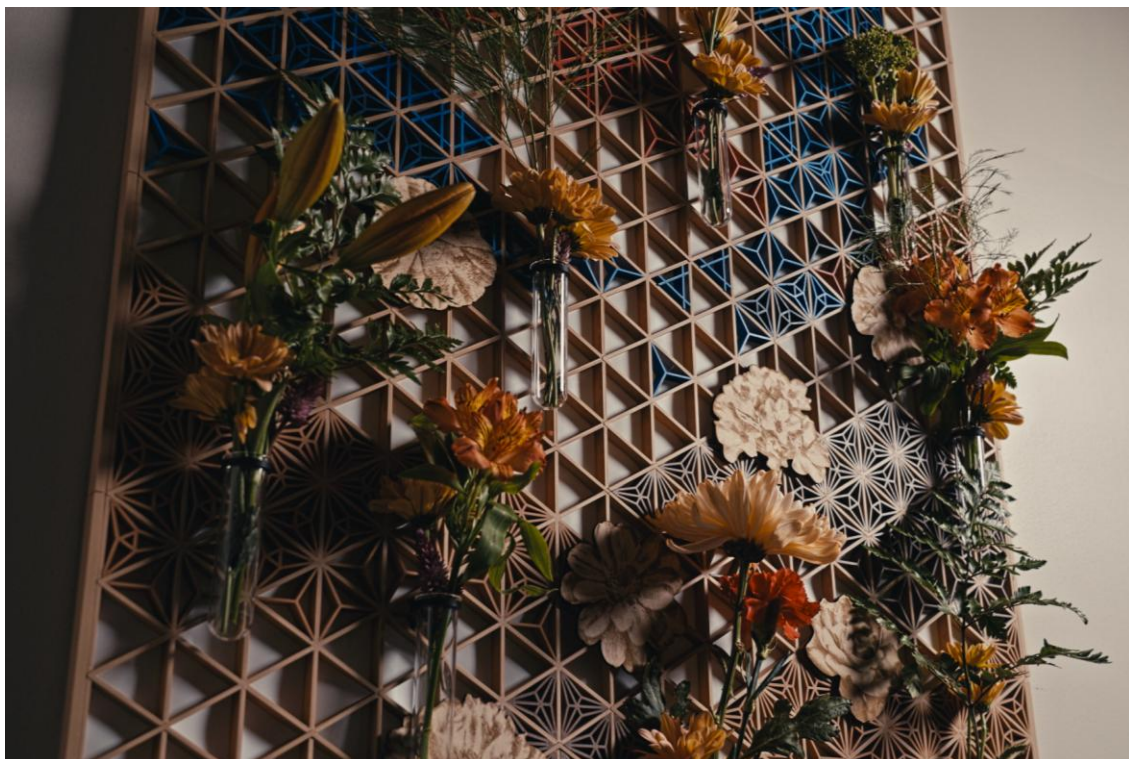
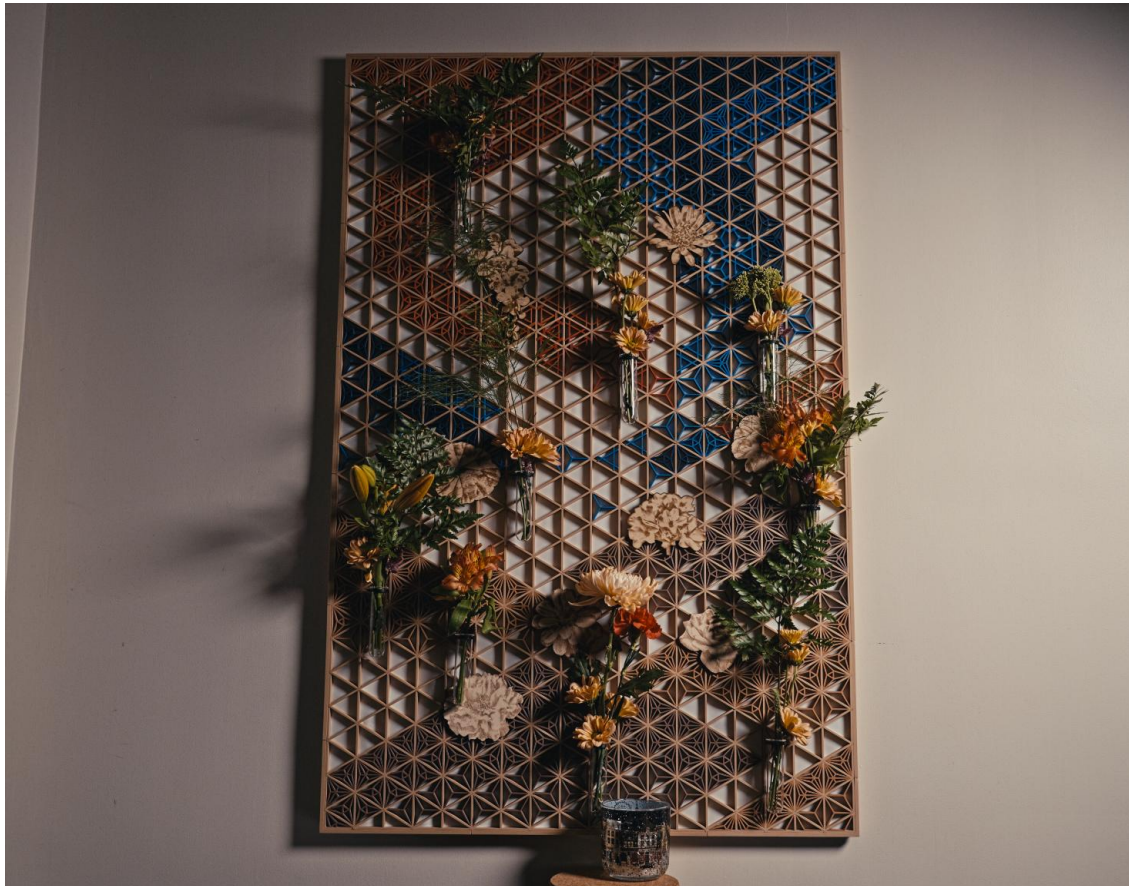
Gorilla Shelf Compatible Panel Installation

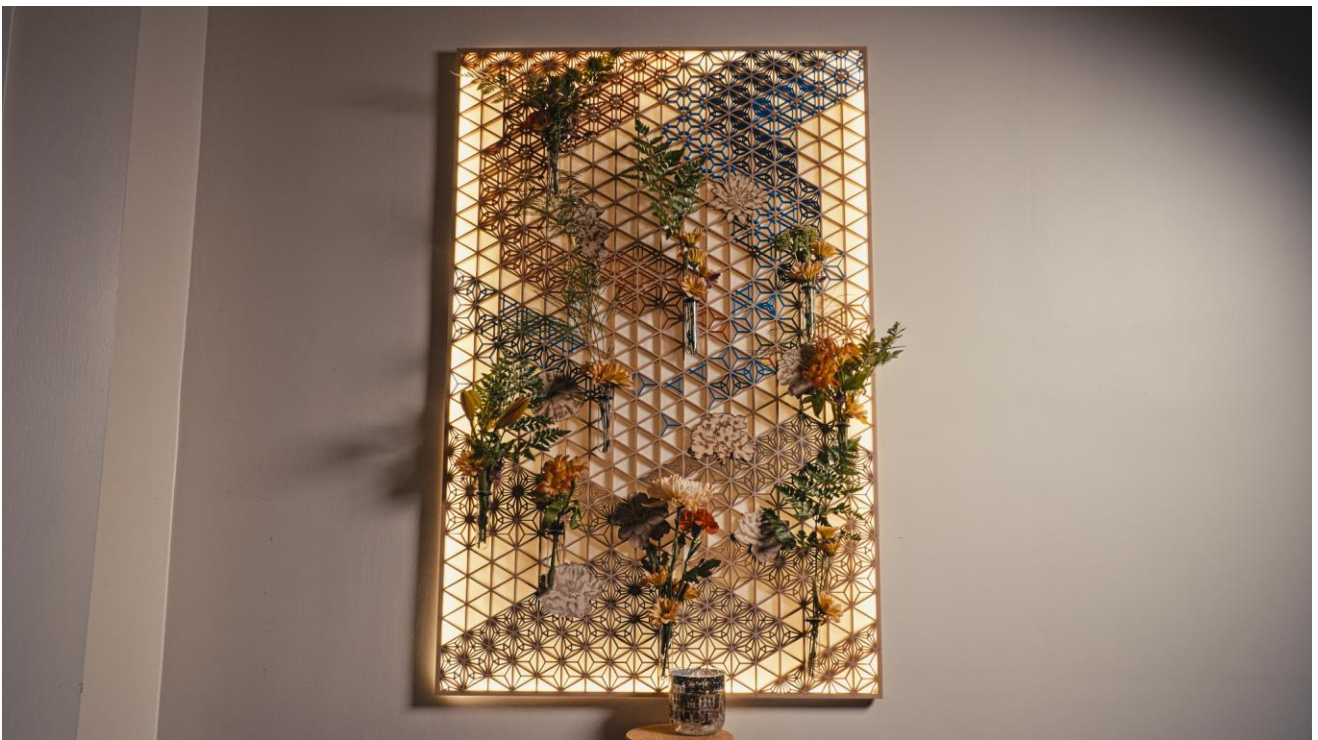
- On vacation RN so I only have access to my phone so the instructions will be slow to type up. The gist is there are hanger components that hook onto the keyholes in shelves and then a dovetailed piece is added horizontally. A matching piece is added to the panel so they can hang onto one another. Note, the dovetail piece can fit in one of two slots depending on if you have an even width (dovetail piece should be added to the same height) or an odd width panel (use different heights). Hard to highlight now since I can't edit the pics but once I'm back I'll include some more context.



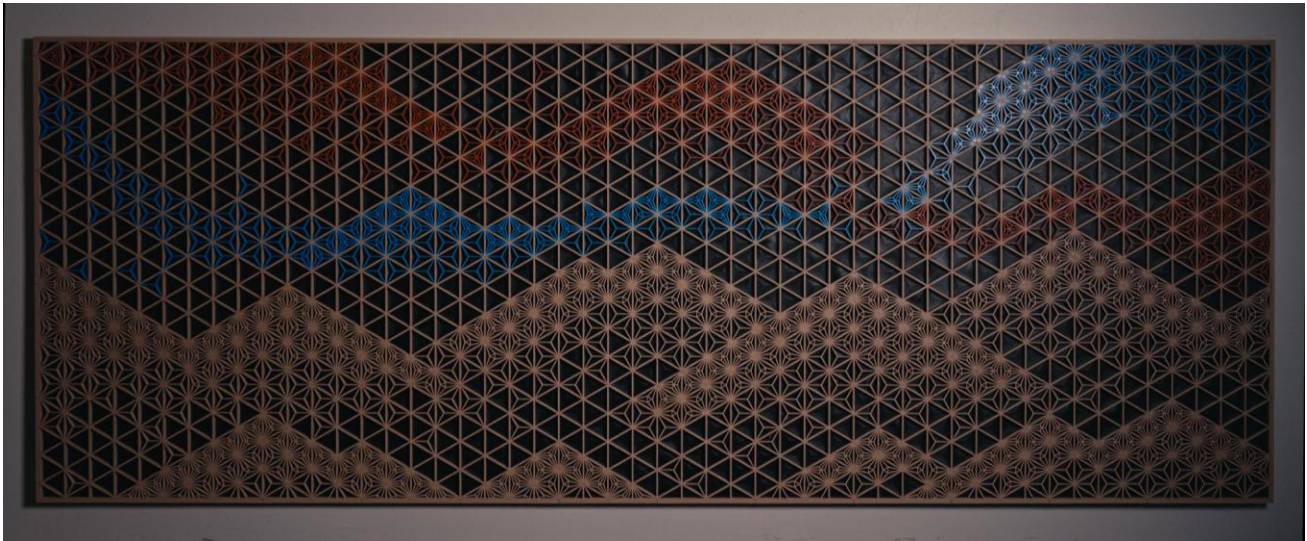
Catalogue Images

Video 2 - Primary Panel





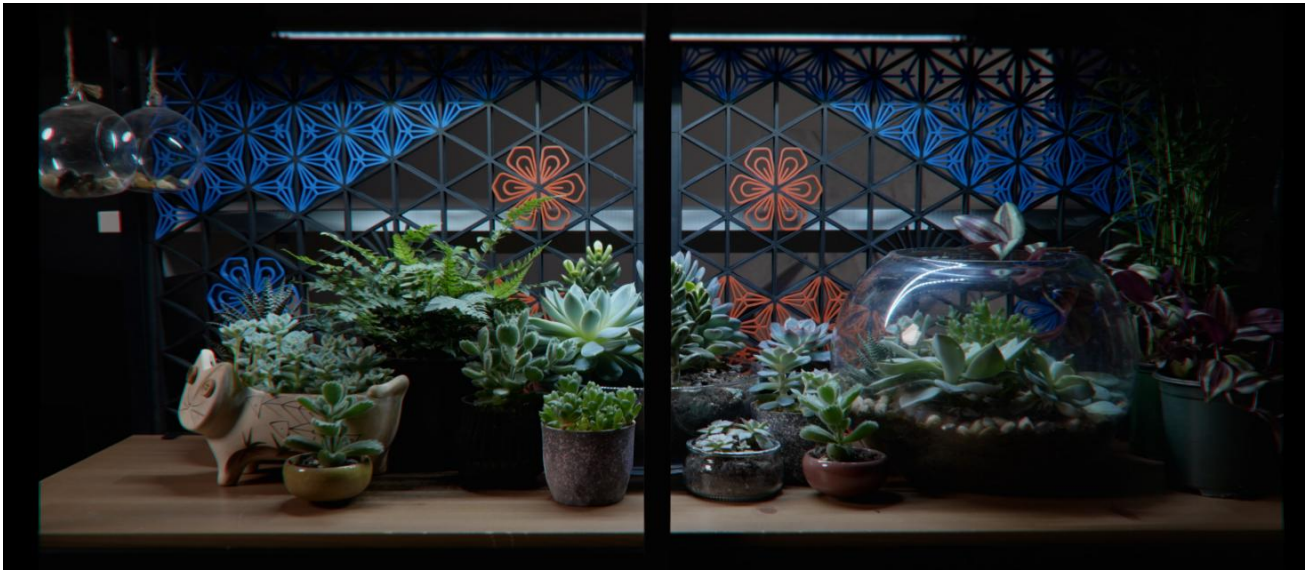
Video 1 - Primary Panel



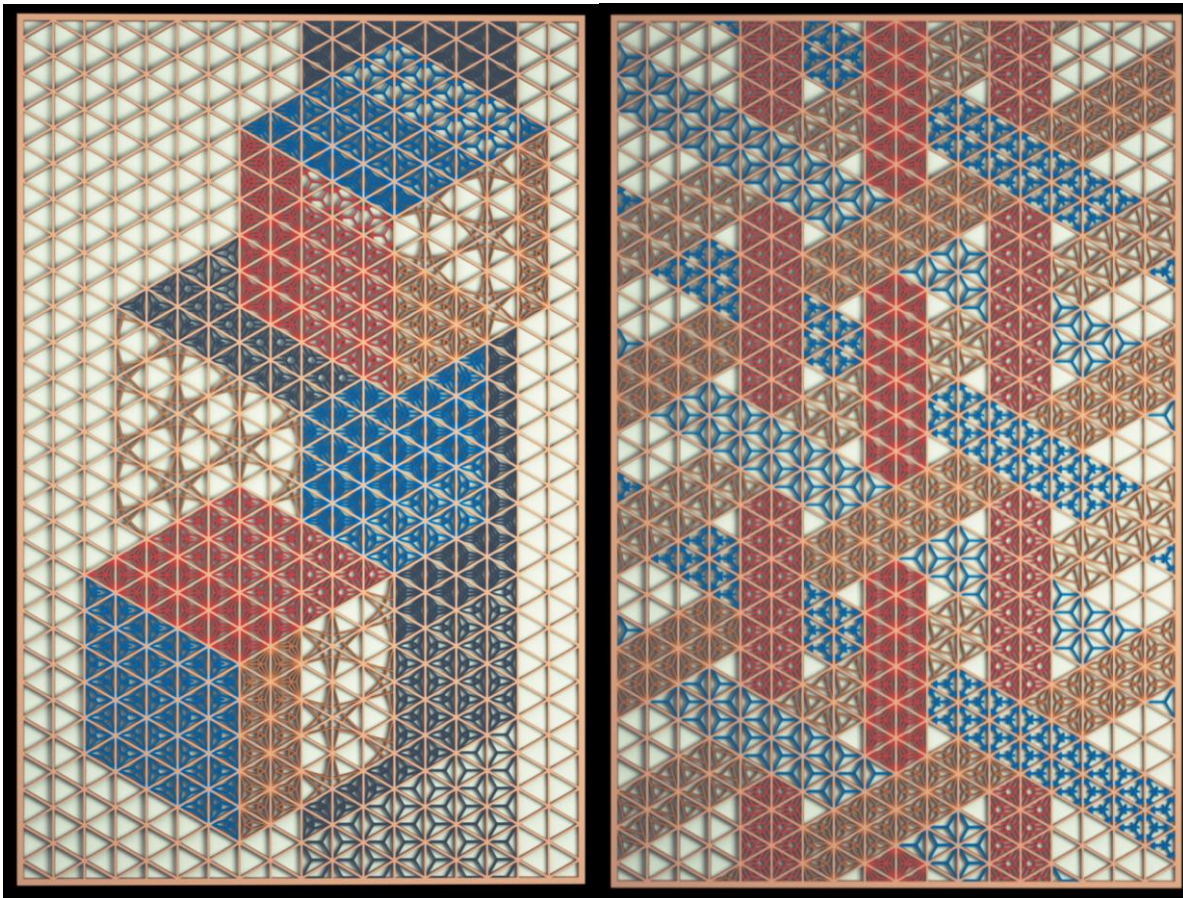
Video 2 - Gorilla Shelf Compatible Panel



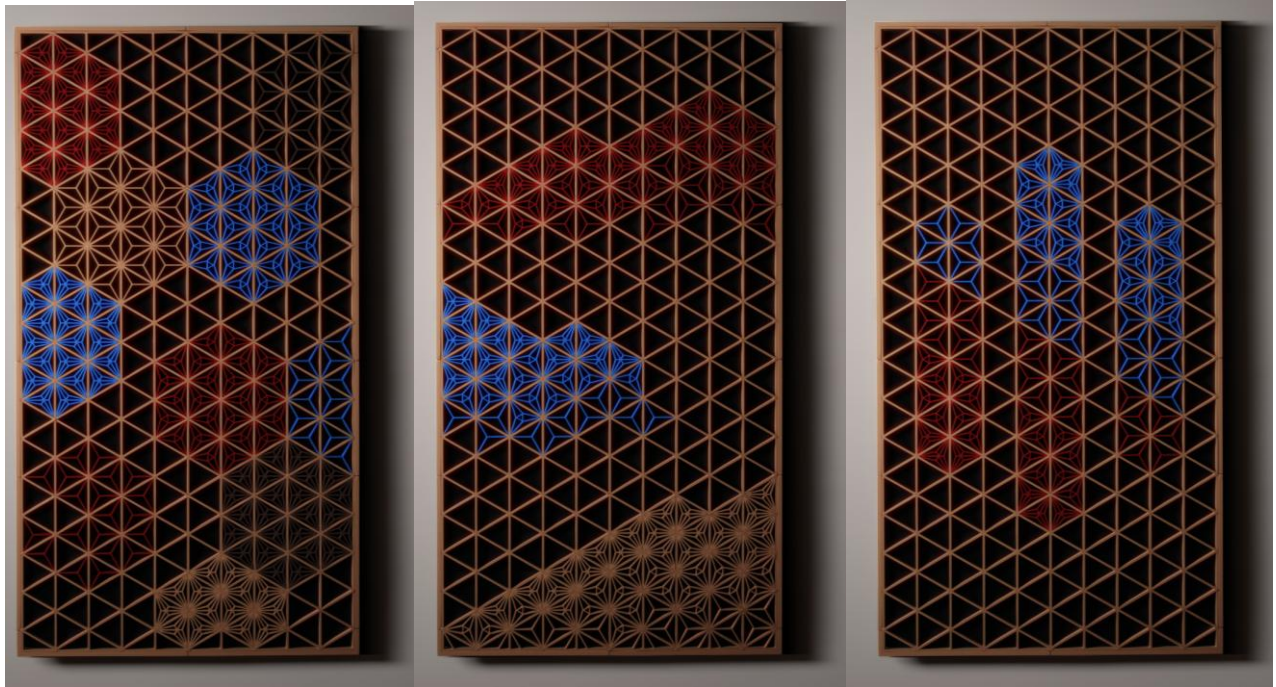
Video 2 - Ikea Fjallbo Series-Compatible Panel



Video 2 - Alternative Designs



Video 1 - Secondary Panels



Main Instructions

Kumiko Frame Generator: Instructions

Made by @paper_view - [YouTube](#) | [Patreon](#)

If you'd like to sell 3d prints of this model or any of my other models, please make sure to be an active subscriber of the commercial tier on my Patreon. [Patreon link here](#).

[Link to model](#)

Please make sure to read the instructions completely.

*Message me on MakerWorld or Patreon if you have any questions, need help, or notice any bugs. I'm quite responsive but if there's a lot of individuals contacting me it'll take me longer to respond, so your patience is appreciated. **Before reaching out, please read the instructions thoroughly, as they may answer your question.***

IMPORTANT: Do not print anything from the "Frame Reference Generator.scad" customize tool. The customizer is for design and reference only so you would know which and how many pieces are needed for printing. Only print components from the print profile.

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- [Step 5: Assembly](#)
- [Step 6: Hanging the Frame](#)
- [Glossary](#)

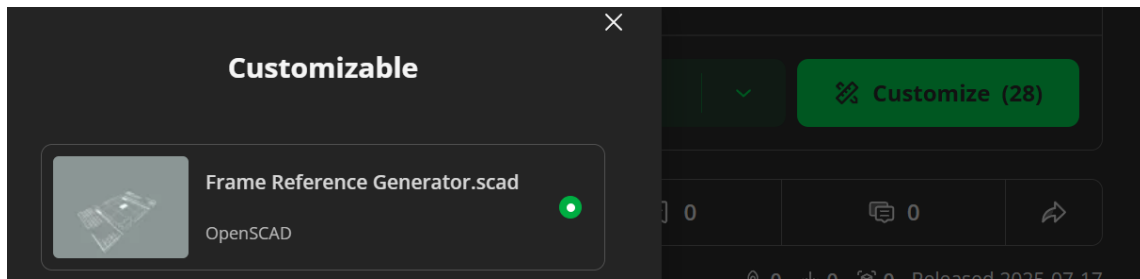
Step 0: Materials

Below are affiliate links (no extra charge to you) to the printer, filament, and other consumables that I used to make the panels in my video.

- **Printer:** [Bambulab P1P](#)
- **Build plate:** [Smooth PEI plate](#)
- **Filament:** [PLA Matte filament](#)
 - Refer to the [Patreon catalogue](#) for filament requirements for my designs

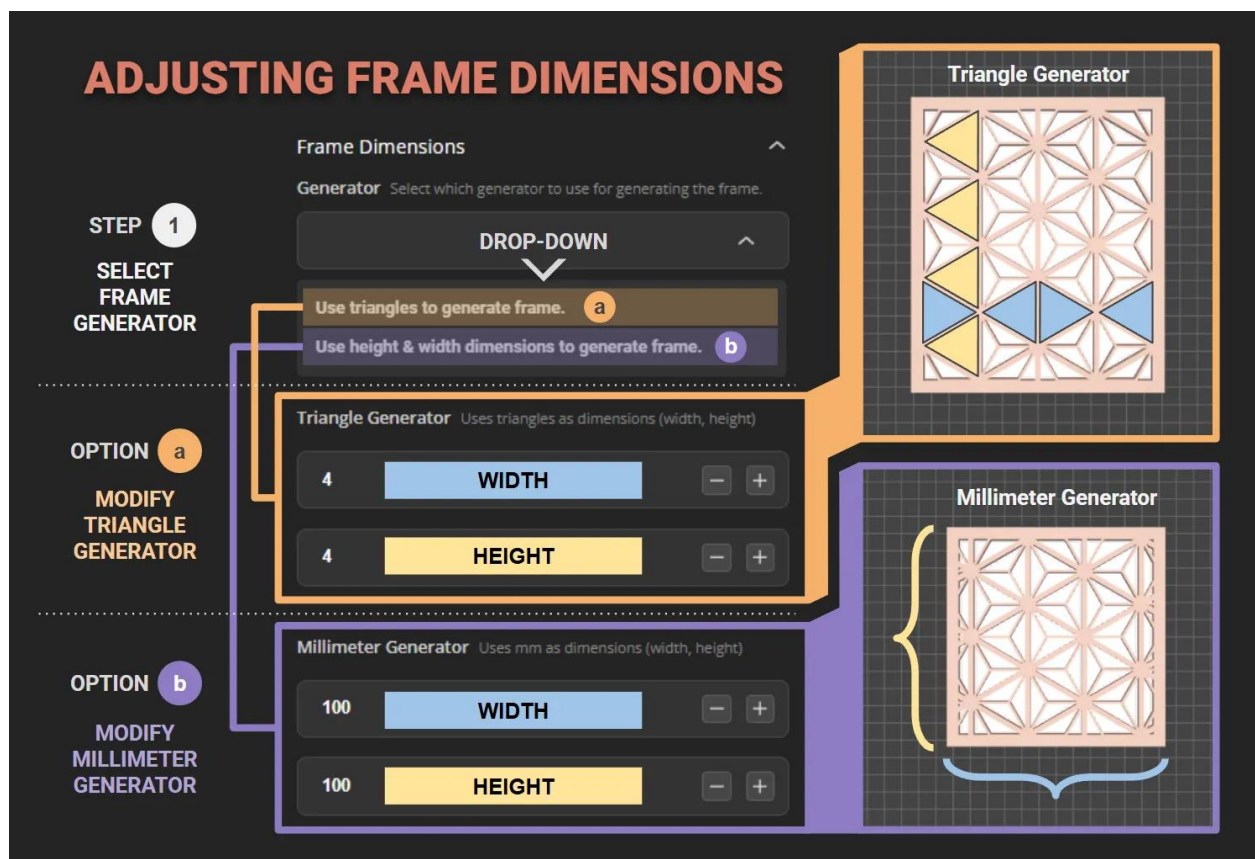
Step 1: Planning your panel

Step 1.0: Open the “Frame Reference Generator.scad” customizer.

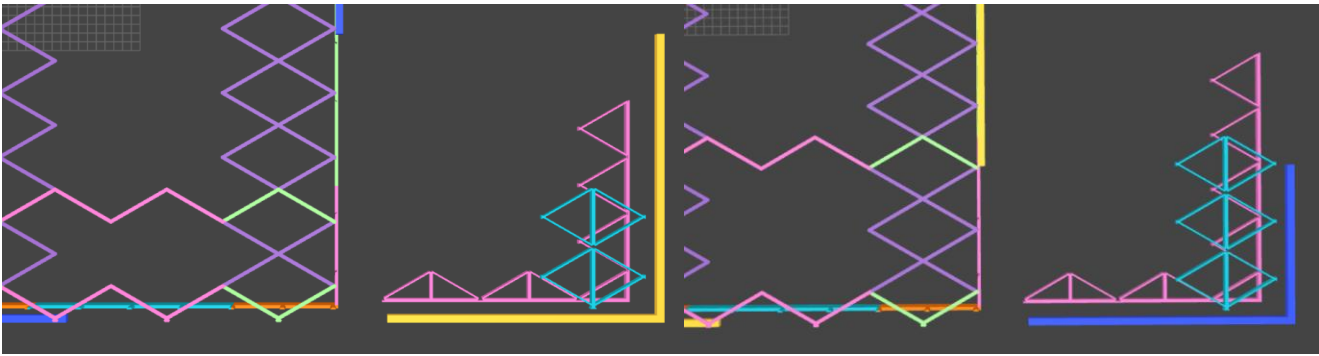


Step 1.1: Using the “Frame Reference Generator.scad” customizer, determine the size of panel you would like to produce. You can use triangles or millimeter dimensions in the customizer. Note: millimeter dimensions are rounded down to the nearest panel that would fit. Reference the text in the center of the model to understand the exact height and width of the panel.

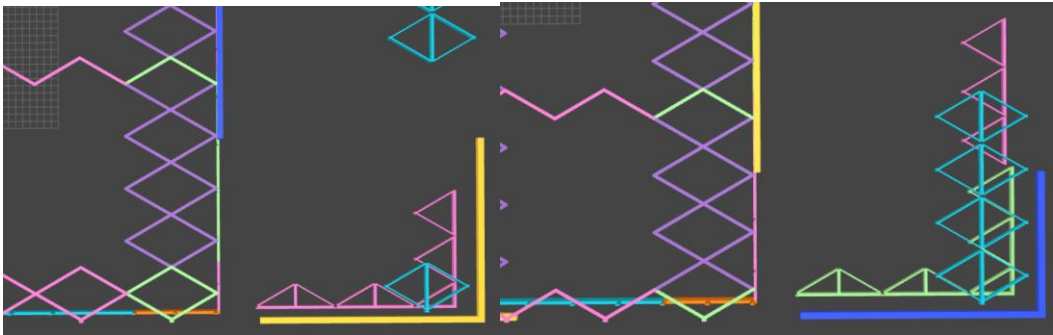
IMPORTANT: Panels that are extremely wide or tall are difficult to move and will require super glue to hold them together (anything bigger than 3 or 4 feet wide probably needs super glue). Height is less of an issue, but I would avoid making panels too tall. If you want to cover a wide area, I would recommend assembling multiple smaller panels (~3/4 feet wide).



Note: a minority of panel configurations (mainly those ending with a single  subpanel) will have an unsupported BR (bottom-right) corner which will need to be super glued into place. This isn't a big issue, but just something to consider when generating. Below are examples of a supported vs unsupported BR corner.

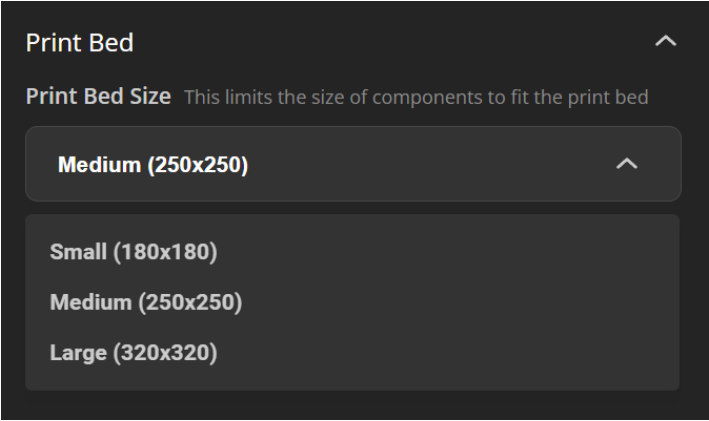


Supported (notice how the BR corner in pink extends past the horizontal line)

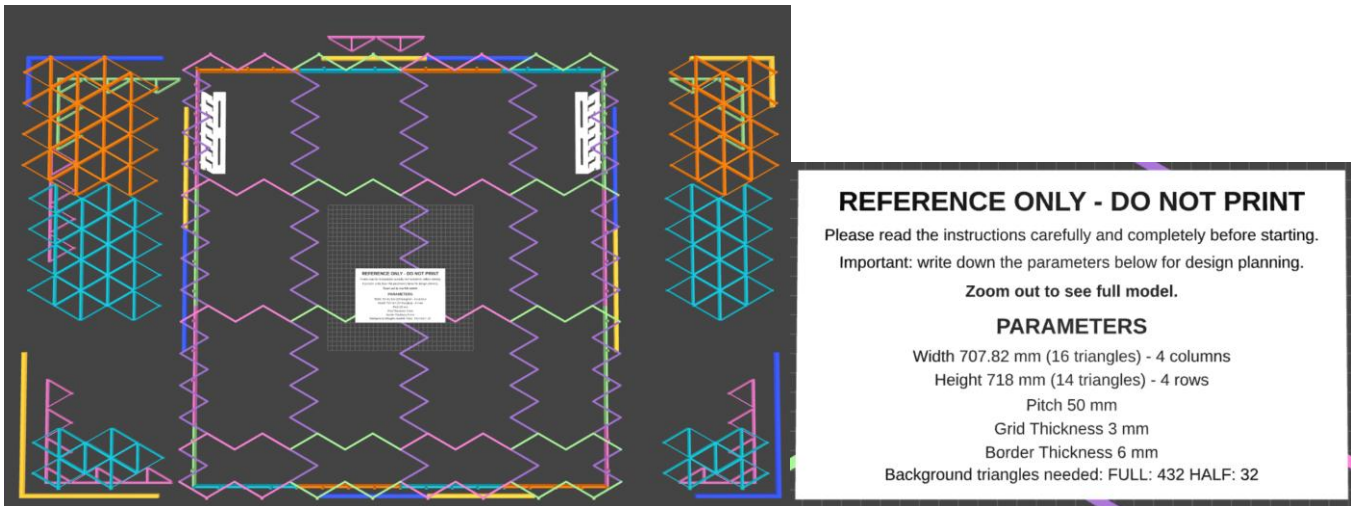


Unsupported (notice that the BR corner in pink (left) or green (right) does not extend past the horizontal line)

Step 1.2: Adjust the “Print Bed” parameter to match your printer's bed size.



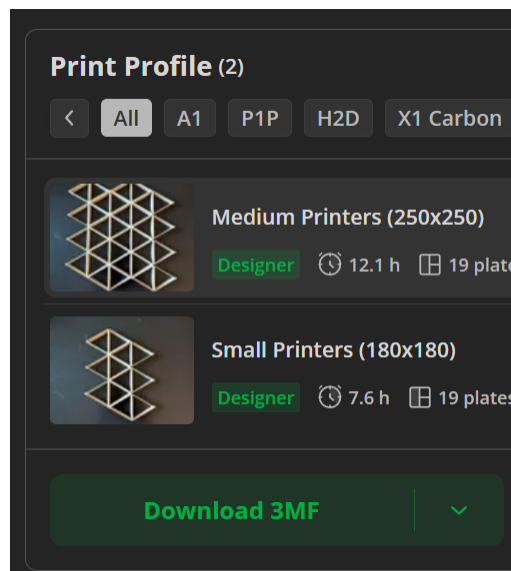
Step 1.3: Once you’ve used the customizer to generate your panel. Screenshot the model and the “PARAMETERS” section under the “REFERENCE ONLY - DO NOT PRINT” card. This is important so you can come back to the customizer to reference which components you need to print. Alternatively you can download the model but just make sure not to print it.



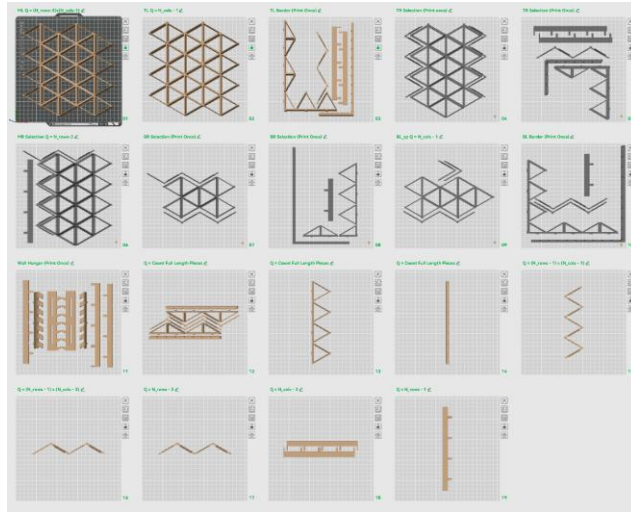
Step 2: Prepare and print frame components

Step 2.0: Due to the number of component variants needed to accommodate any frame size for a given printer's bed size, the print profile first needs to be modified to match the components needed to print the desired frame generated in **Step 1**. The below steps run you through the process of getting the print profile ready so you can bring printing.

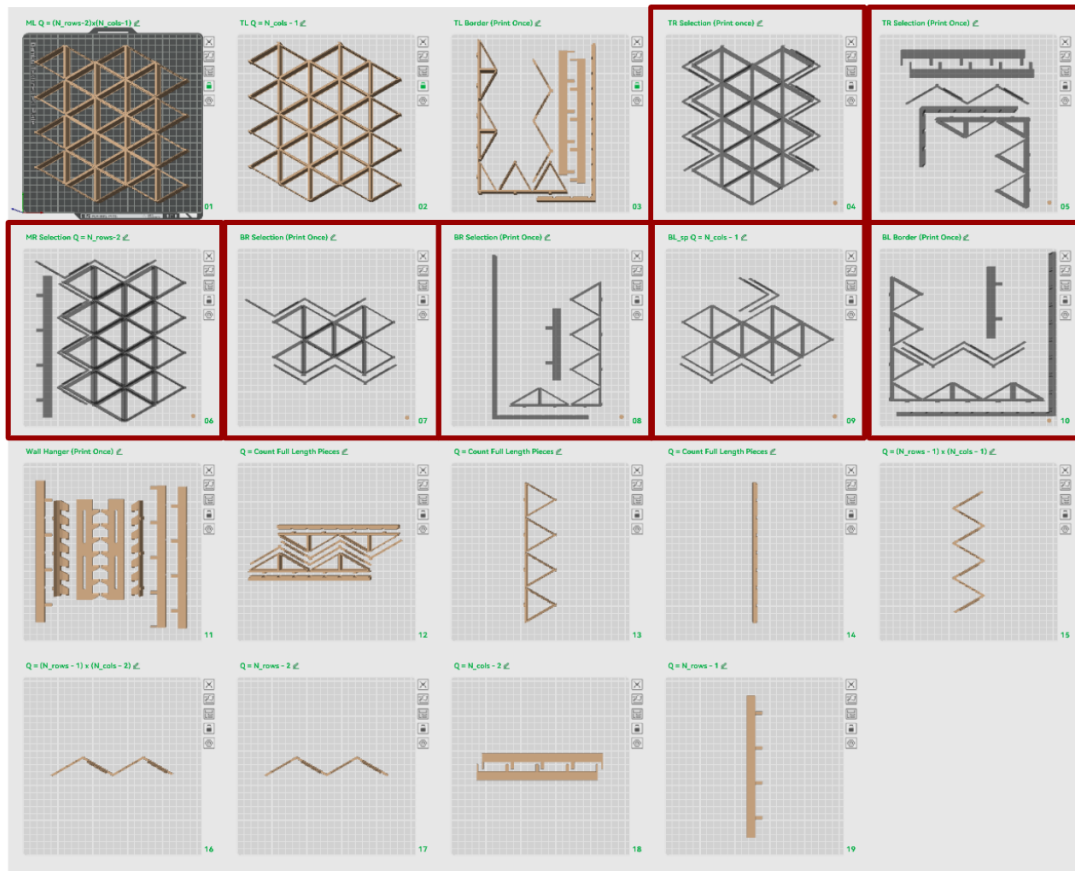
Step 2.1: Download the print profile for your 3d printer. Note: I'll be adding an H2D print profile in the next few days.



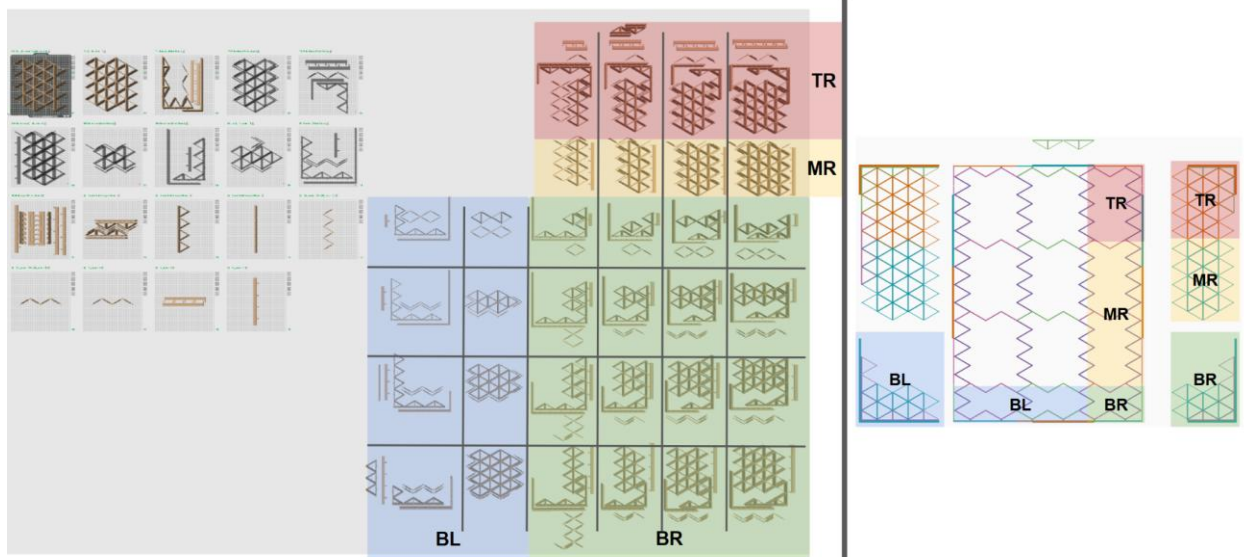
Step 2.2: The print profile has multiple print panels for different components. Each print panel needs to be printed but you would need to determine **which** and **how many** components to print for your specific frame.



Step 2.3.0: For print plates with the word “selection” in its name and grayed out components (highlighted below), you would need to select the set of components you need to print. This changes based on the dimension of the panel.



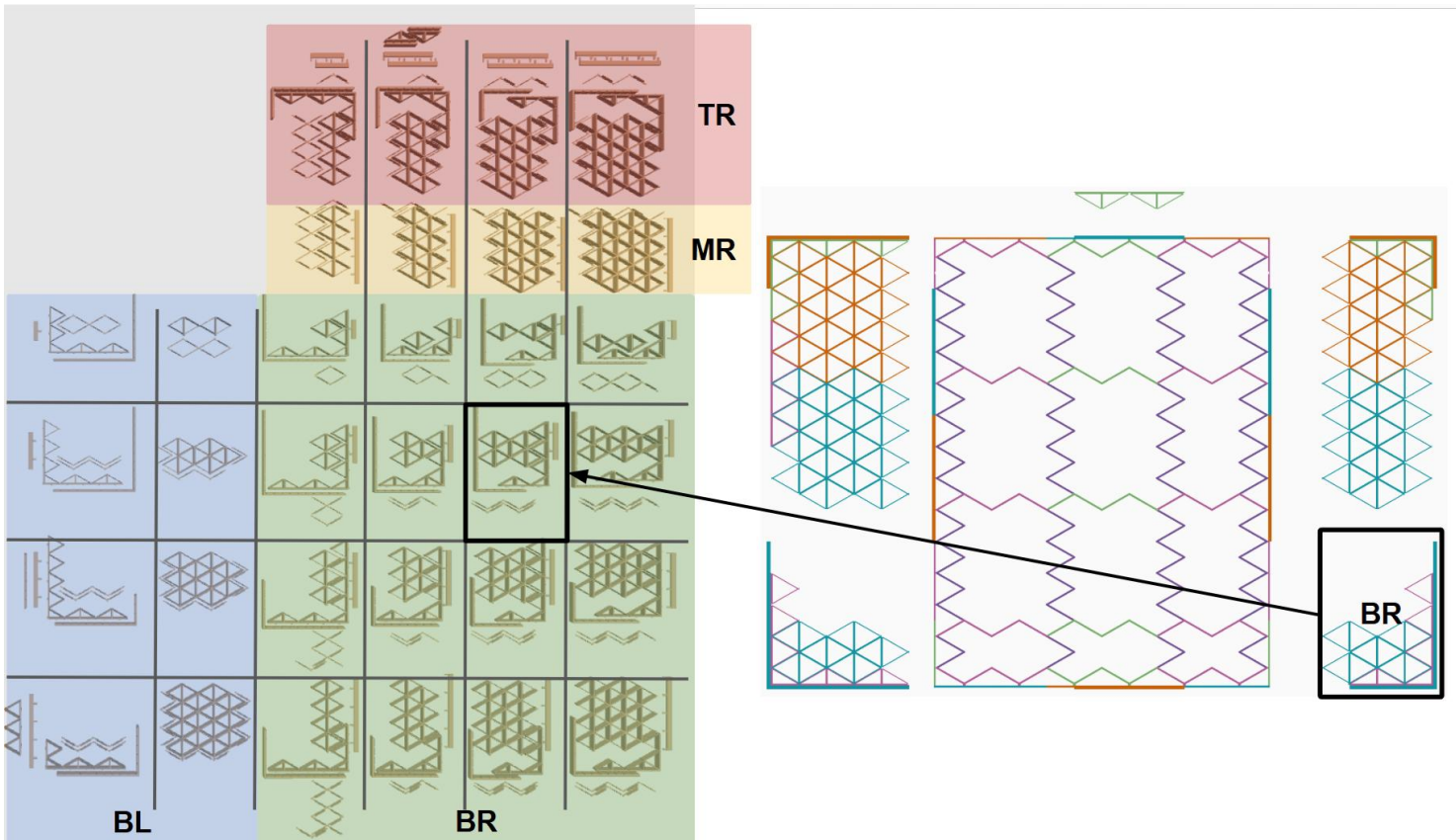
Step 2.3.1: In the print profile, on the right side of the print beds (you need to zoom out), there are arrays of grouped components which will be used to populate the panels highlighted in **Step 2.3.0**. They are labelled based on their positions in the reference frame (TR = top right, MR = middle right, BL = bottom left, BR = bottom right).



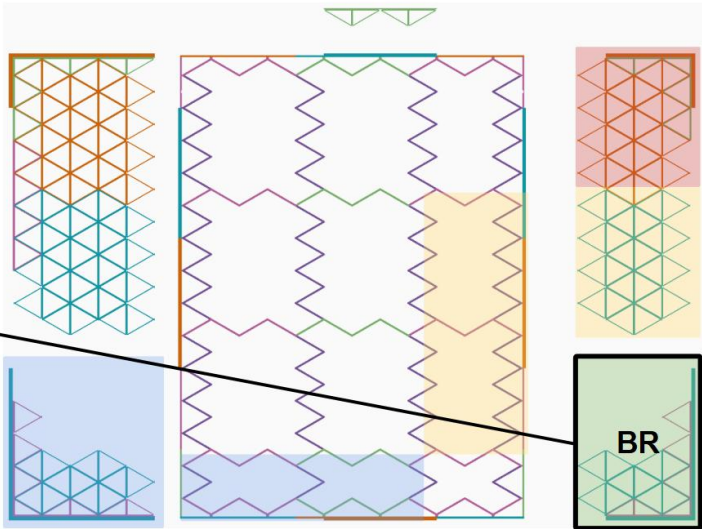
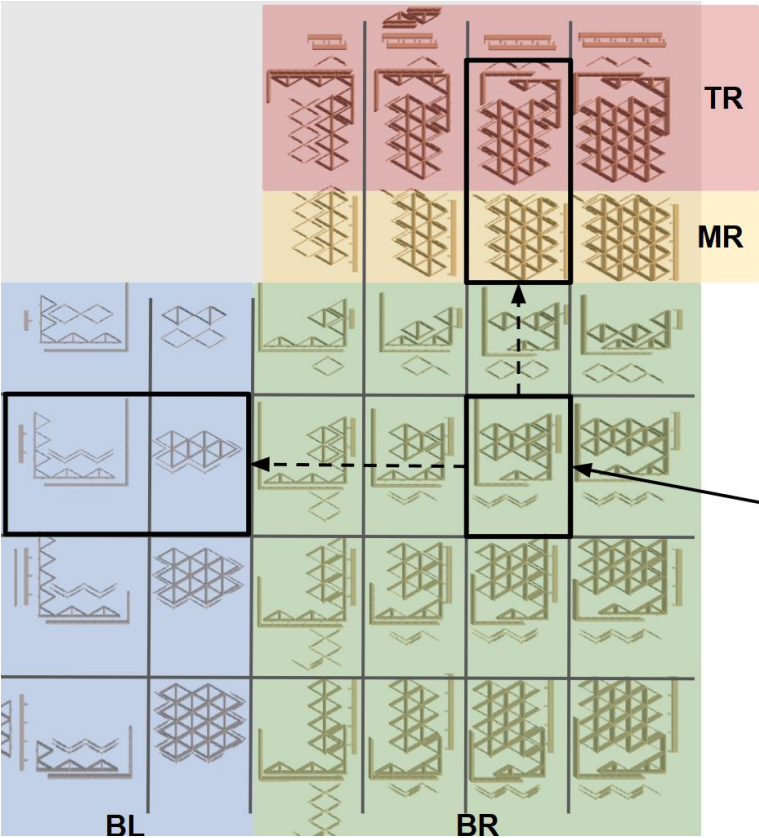
Print Profile

Customizer

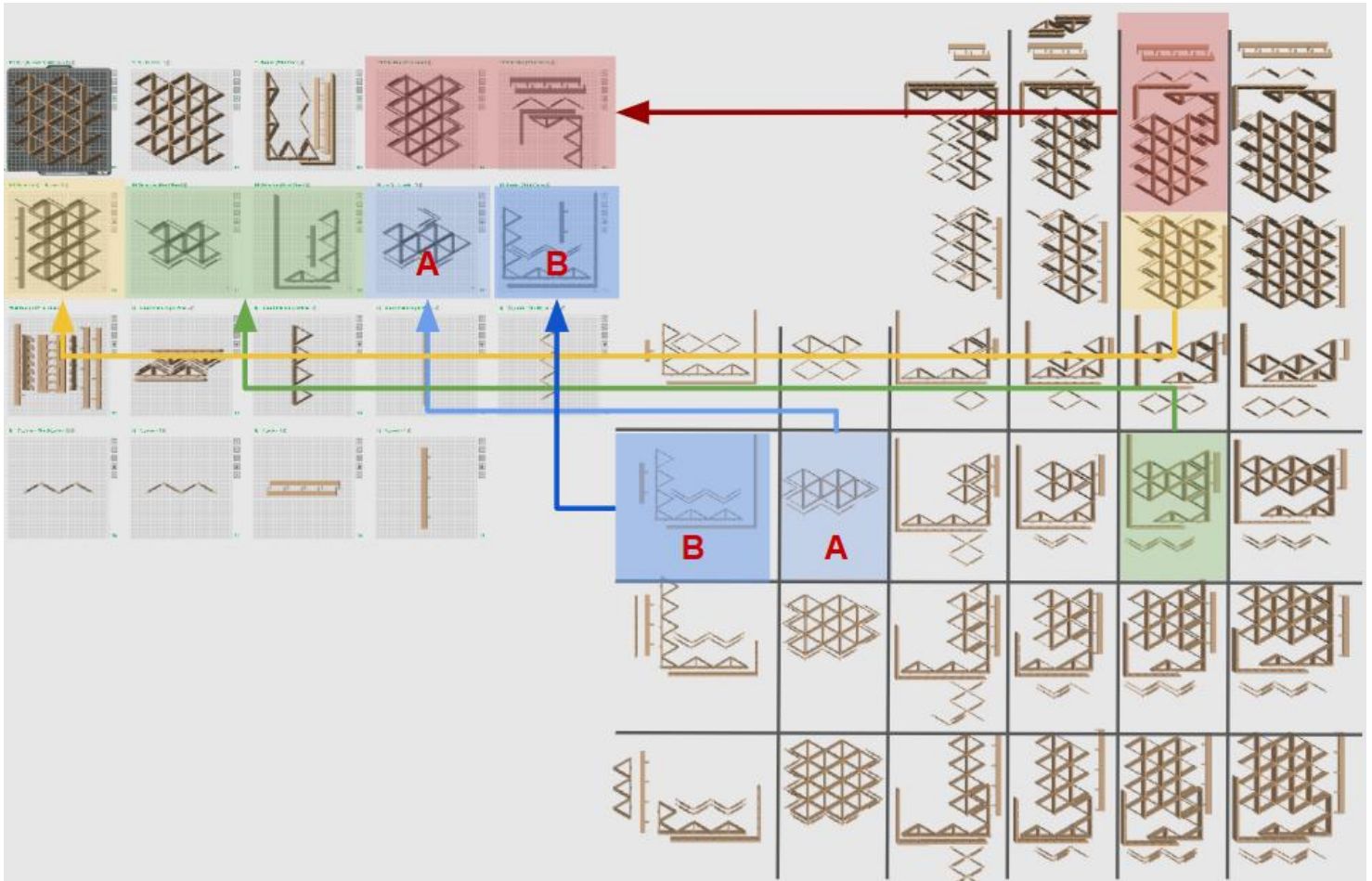
Step 2.3.2: To select the required components, pull up the reference panel from the customizer created in Step 1, look at the bottom right panel in the reference and find the identical piece from the options available in the “BR” section of the print profile. The grid of components is arranged based on the width (columns) and height (rows) of the bottom right piece (three wide and two tall in this example).



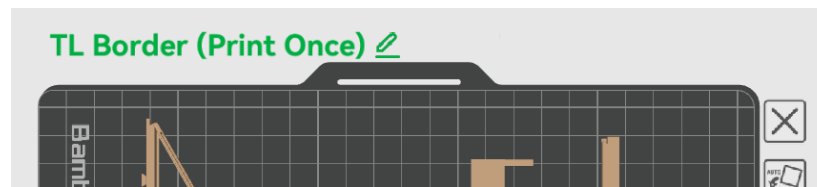
Step 2.3.3: From the component found in Step 2.3.2, move horizontally and vertically to the ends to identify the corresponding TR, MR, and BL components for your reference frame.



Step 2.3.4: Copy the components needed to the corresponding print plate. Make sure to delete what was previously there.

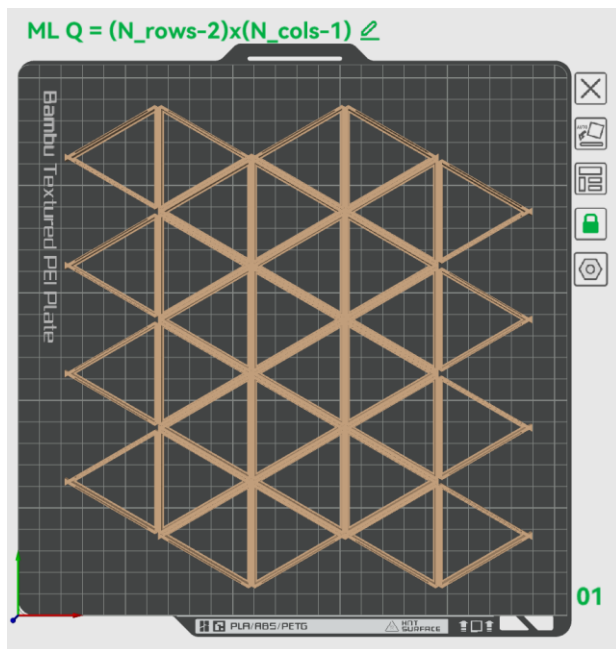


Step 2.4.0: How many times a component needs to be printed is outlined according to the title of the print plate. Some only need to be printed once, and will be indicated by “(Print once)” in the name of the print plate (TL border, TR selections, BL border, BR selections, and wall hanger).



Step 2.4.1: For some print panels you will need to calculate how many times it needs to be printed, and the formula is in the name of the print plate. N_cols and N_rows refers to the number of columns and rows of panels in the full print and are provided under “PARAMETERS” generated from the customizer in Step 1.

In the example below, you would need to print everything in the print panel “ML” 6 times -- $Q = (4-2) \times (4-1)$.



REFERENCE ONLY - DO NOT PRINT

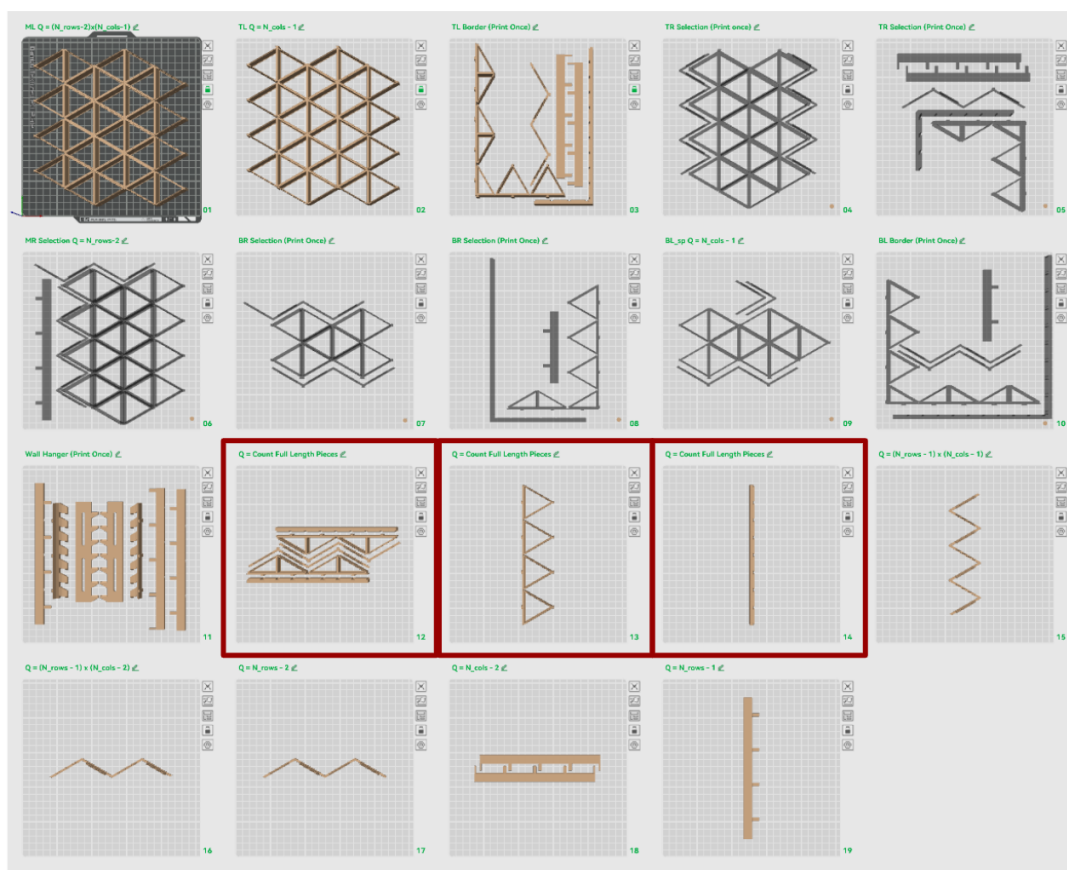
Please read the instructions carefully and completely before starting.
Important: write down the parameters below for design planning.

Zoom out to see full model.

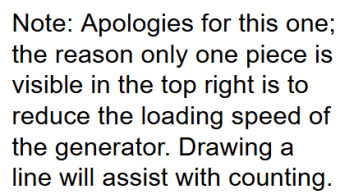
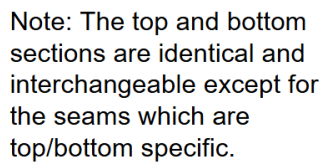
PARAMETERS

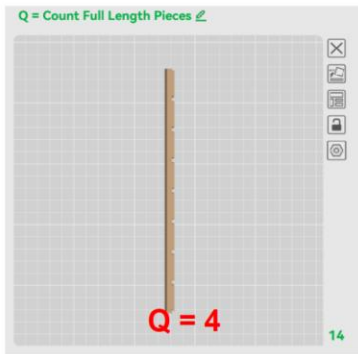
Width 707.82 mm (16 triangles) - 4 columns
Height 718 mm (14 triangles) - 4 rows
Pitch 50 mm
Grid Thickness 3 mm
Border Thickness 6 mm
Background triangles needed: FULL: 432 HALF: 32

Step 2.4.2: For some print panels, you will need to manually count (for now, as I work on a built-in calculator) how many times a component needs to be printed. Follow the diagrams below to count how many of each component are needed.

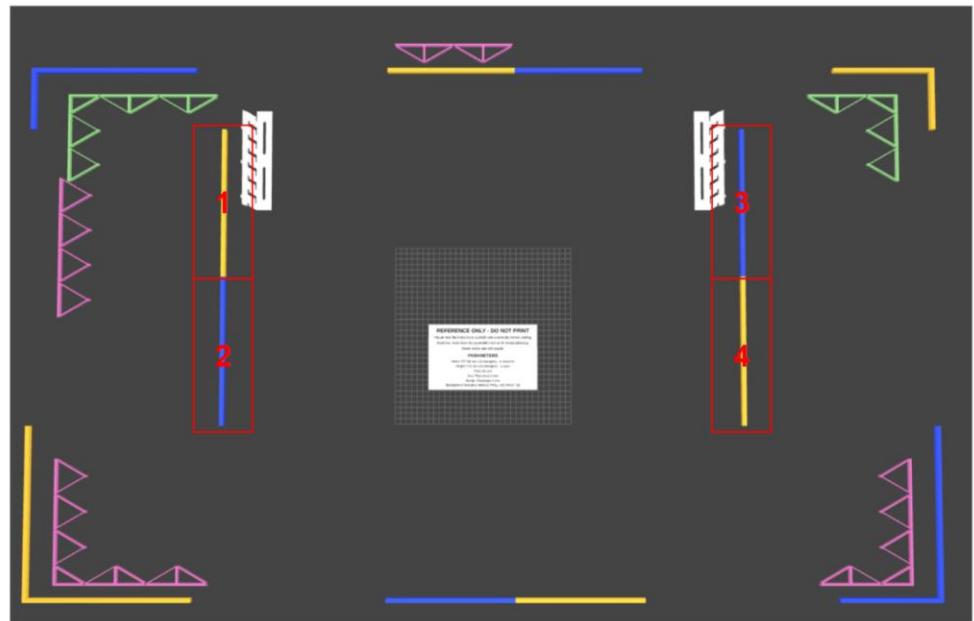


Horizontal Frame Assembly (includes horizontal border, horizontal seam, and horizontal frame)



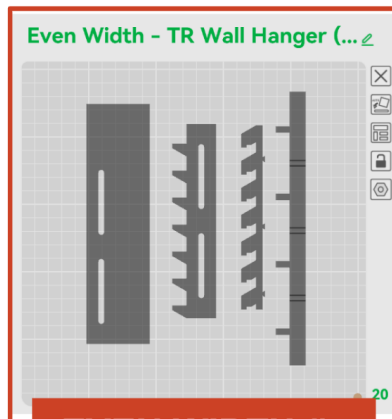


Note: Make sure to only count full-length sections.



Step 2.5: Unsuppress the corresponding TR Wall Hanger component.

- If N triangles width is even: select “Even Width”
- If N triangles width is odd: select “Odd Width”

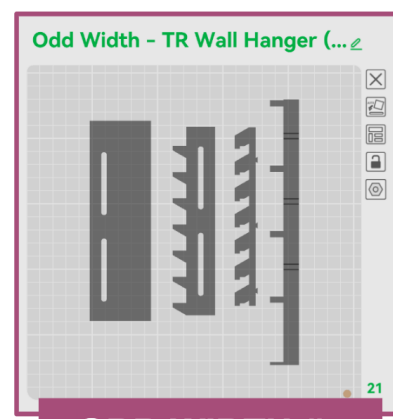


EVEN WIDTH #s

Please read the instructions carefully and completely before starting.
Important: write down the parameters below for design planning.
Zoom out to see full model.

PARAMETERS

Width 534.615 mm (12 triangles) - 3 columns
Height 718 mm (14 triangles) - 4 rows
Pitch 50 mm
Grid Thickness 3 mm
Border Thickness 6 mm (width) & 6 mm (height)
Background triangles needed: Full: 324 Half: 24



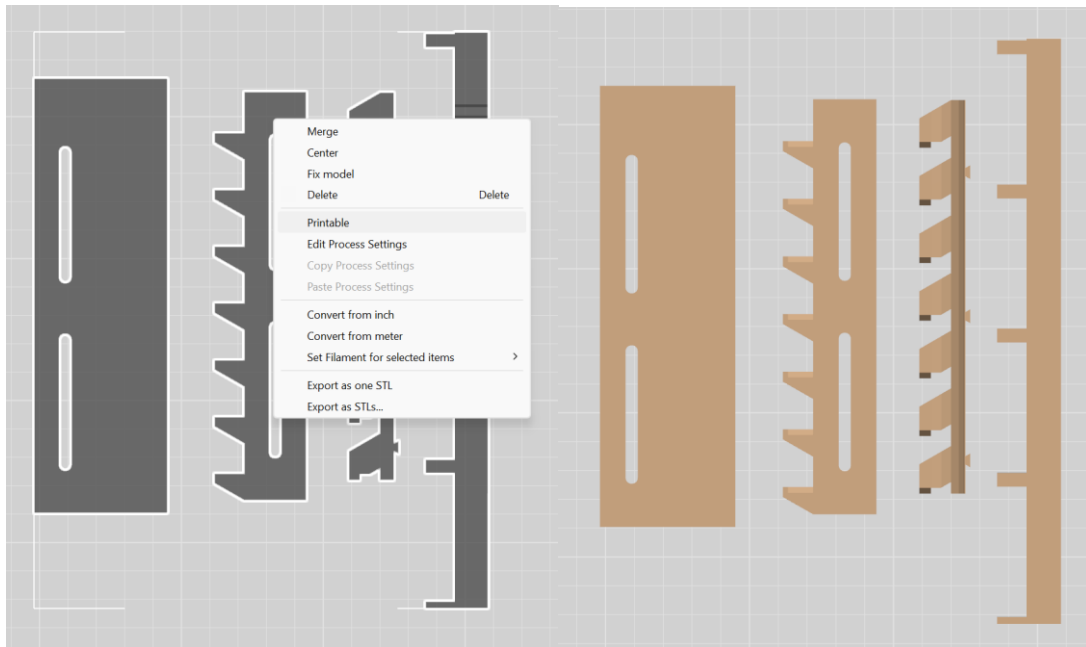
ODD WIDTH #s

Please read the instructions carefully and completely before starting.
Important: write down the parameters below for design planning.
Zoom out to see full model.

PARAMETERS

Width 664.519 mm (15 triangles) - 4 columns
Height 718 mm (14 triangles) - 4 rows
Pitch 50 mm
Grid Thickness 3 mm
Border Thickness 6 mm (width) & 6 mm (height)
Background triangles needed: Full: 405 Half: 30

How to determine whether to unsuppress the “Even” or “Odd” TR Wall Hanger



Right click and select “Printable” to unsuppress the TR Wall Hanger

Step 2.6: Print all the print plates according to the selections and the calculated quantities. I printed everything on a Smooth PEI Plate to ensure seams and borders result in a smooth matte finish. Make sure to clean your print bed every few prints. **Important: if you’re opening the print profile in a different slicer than Bambu Studio, make sure you set the correct print parameters (see below) or else your components will not fit correctly.**

Print Parameters

These are built into the .3mf print profiles I provide, but if you’re using a different slicer or generating the frame from my [source code](#), make sure to adjust to match

Component	Walls	Infill Density	Infill Pattern	Wall Generator	Top shell Layers	Bottom Shell Layers
Subpanels	1	30%	Lightning (to save material)	Arachne *	>= 5	>= 3
Seams						
Outer_Frame		10%	Gyroid (for strength)			
Border						
Border_Background	>= 5	15%				
Hanger						

** It is essential the wall generator is set to “Arachne”. The walls made using “Classic” are too thick.*

Note: After printing each panel, lay them out on a surface according to their location (ex. ML = middle left, TL = top left) and keep the components from one print panel in one location. This will make the assembly process much easier.

Step 2.7: Proceed to Step 3 while waiting for all the components to finish printing.

Step 3: Design your pattern

Note: Please consider donating to **Douwe**, the creator of the Kumiko Designer tool below. I truly couldn't have created this project without the help of their tool. Click the "Donate" button on their website.

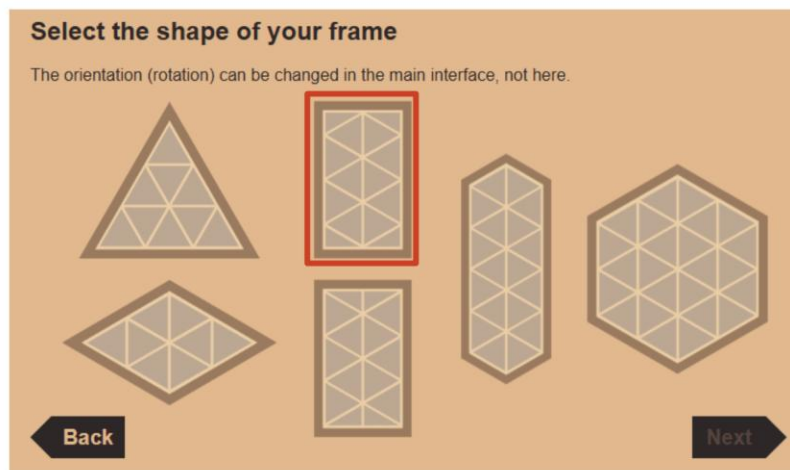
Support my work



Creating and maintaining this website requires a significant time investment for me. If you enjoy using this application, please consider making a small donation in order for me to keep the website hosted.

Donate to Douwe at <https://www.kumikodesigner.com/>.

Step 3.1: Head over to Kumiko Designer website ([link here](https://www.kumikodesigner.com/)). Click "Start designing", select "Framed Panel" and select the rectangular frame shape.



Step 3.2: Refer to the text box saved in step 1.3, and refer to Width triangles.

- If N triangles width is **even**: check "Flip orientation of half triangles"
- If N triangles width is **odd**: leave unchecked

PRINTABLE VERSION

Please read the instructions carefully and completely before starting.
Important: write down the parameters below for design planning.
Zoom out to see full model.

PARAMETERS

Width 534.615 mm (12 triangles) - 3 columns
Height 718 mm (14 triangles) - 4 rows
Pitch 50 mm
Grid Thickness 3 mm
Border Thickness 6 mm (width) & 6 mm (height)
Background triangles needed: Full: 324 Half: 24

EVEN WIDTH #s

Flip orientation of half triangles: ☒

Flip orientation of half triangles: ☐

Frame dimensions

Size A (number of triangles): 14

Size B (number of triangles): 12

Frame color:

ODD WIDTH #s

Flip orientation of half triangles: ☐

Flip orientation of half triangles: ☐

Frame dimensions

Size A (number of triangles): 14

Size B (number of triangles): 12

Frame color:

Step 3.3) Referring to the text box, input the correct "Size A" (equivalent to height) and "Size B" (equivalent to width), Pitch, and Mitsuke (equivalent to Grid Thickness) into kumiko designer. You can leave Frame width as default. **IMPORTANT: the preview box that shows the kumiko panel appears to be oriented incorrectly in kumiko designer. Please rely on the instructions here to properly orient your design.**

PRINTABLE VERSION

Please read the instructions carefully and completely before starting.
Important: write down the parameters below for design planning.
Zoom out to see full model.

PARAMETERS

Width 534.615 mm (12 triangles) - 3 columns
Height 718 mm (14 triangles) - 4 rows
Pitch 50 mm
Grid Thickness 3 mm
Border Thickness 6 mm (width) & 6 mm (height)
Background triangles needed: Full: 324 Half: 24

Frame dimensions

Height

Size A (number of triangles): 14

Size B (number of triangles): 12

Frame width (mm): 12

Frame color:

Width

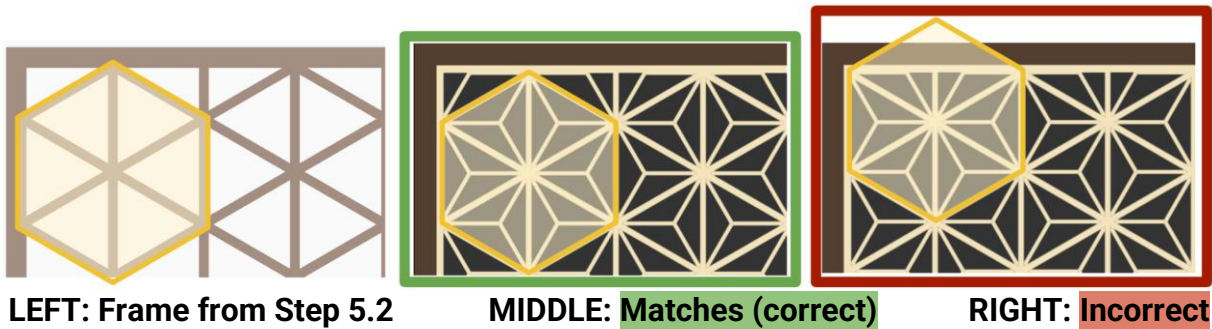
Pitch

Pitch (mm): 50

Mitsuke (mm): 3

Step 3.4) Validate the frame.

Step 3.4.1) Validate that the correct frame was selected in step 3.2. To do this, compare the top left corner of both frames and make sure they are identical.



Step 3.4.2) Validate the correct parameters were inputted in step 3.3. To do this press "p" on your keyboard for the frame parameters you inputted. Compare what's on the screen to the parameters in your text box screenshot. Ignore Frame_Depth and compare everything else.

PRINTABLE VERSION

Please read the instructions carefully and completely before starting.
Important: write down the parameters below for design planning.

Zoom out to see full model.

PARAMETERS

Width 534.615 mm (12 triangles, 3 columns)

Height 718 mm (14 triangles, 4 rows)

Pitch 50 mm

Grid Thickness 3 mm

Border Thickness 6 mm (width) & 6 mm (height)

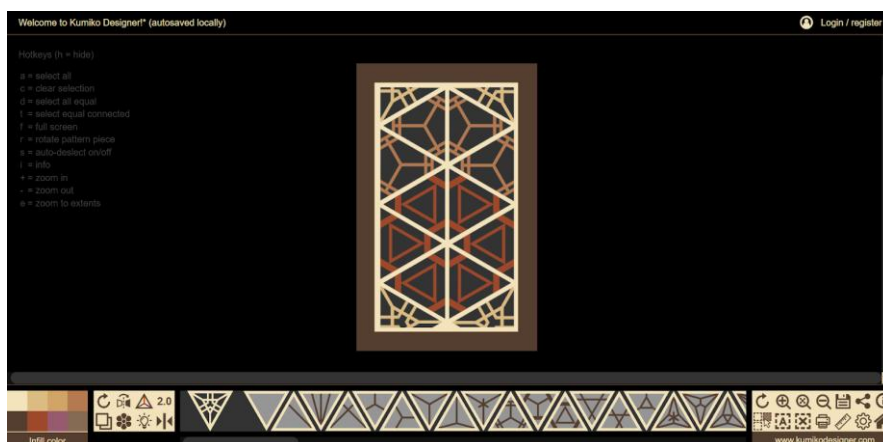
Background triangles needed: Full: 324 Half: 24

```
// Uses triangles as dimensions (width, height)
Triangle_Generator = [12,14];
// Distance between triangles in frame.
Grid_Pitch = 50;
// Depth of frame
Frame_Depth = 12;
// Thickness of frame
Grid_Thickness = 3;
// Thickness of border around frame.
Border_Thickness = 12;
```

Validate the saved parameters (LEFT) with kumikodesigner parameters (RIGHT)

Step 3.4.3) If either 3.4.1 or 3.4.2 does not match, redo step 5 and re-verify.

Step 3.5: Using the tool you can customize the inserts and colors. Consider making an account to save your designs.

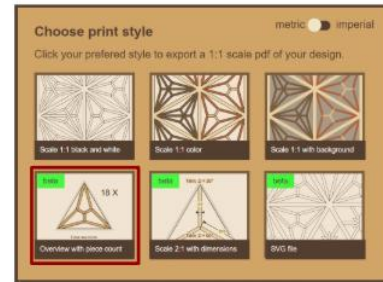


Step 3.6: Once satisfied, export the design and export the inserts required.

1



2



3

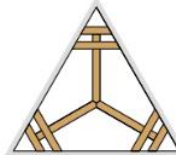
2×



2×



4×



6×

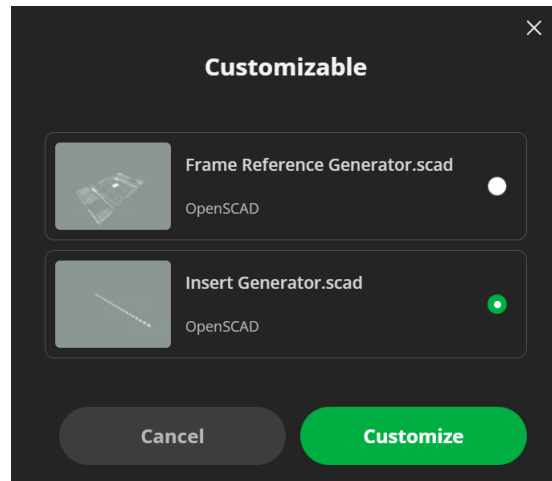


Step 4: Generate and Print Inserts

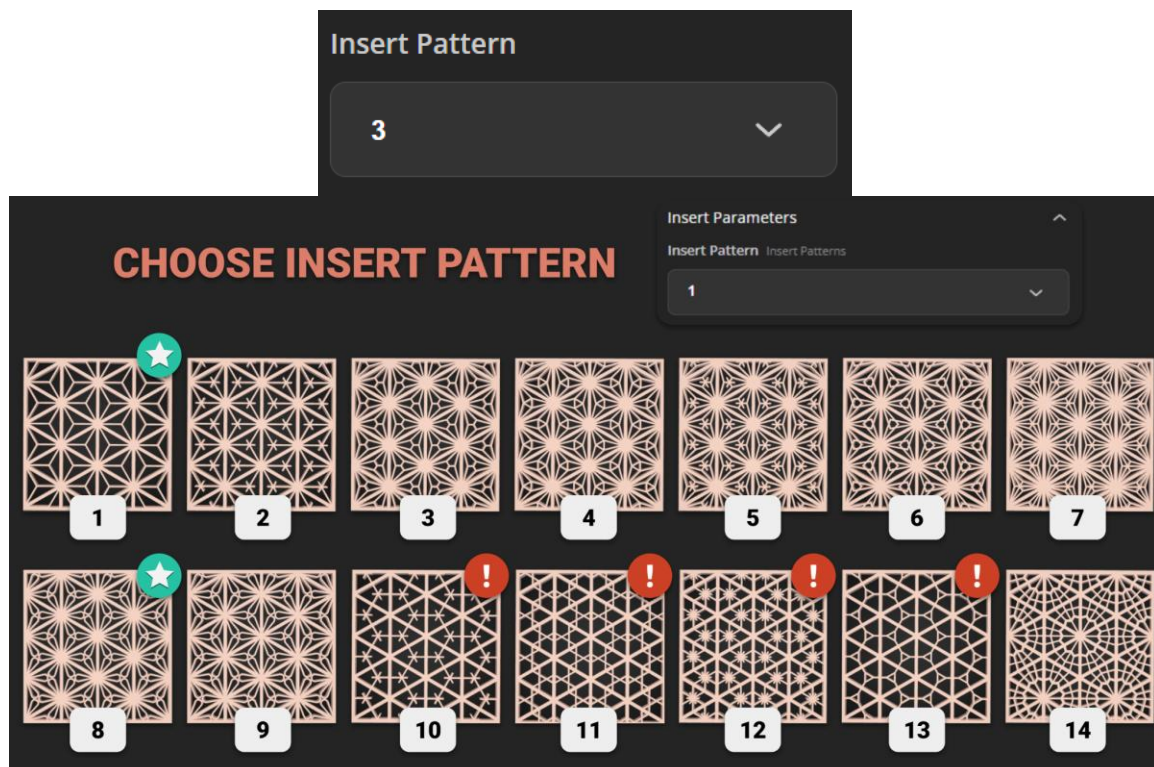
Use the “Insert Generator.scad” to download the corresponding inserts and backgrounds. I would recommend using a matte black background to create contrast between the inserts and background to make things stand out.

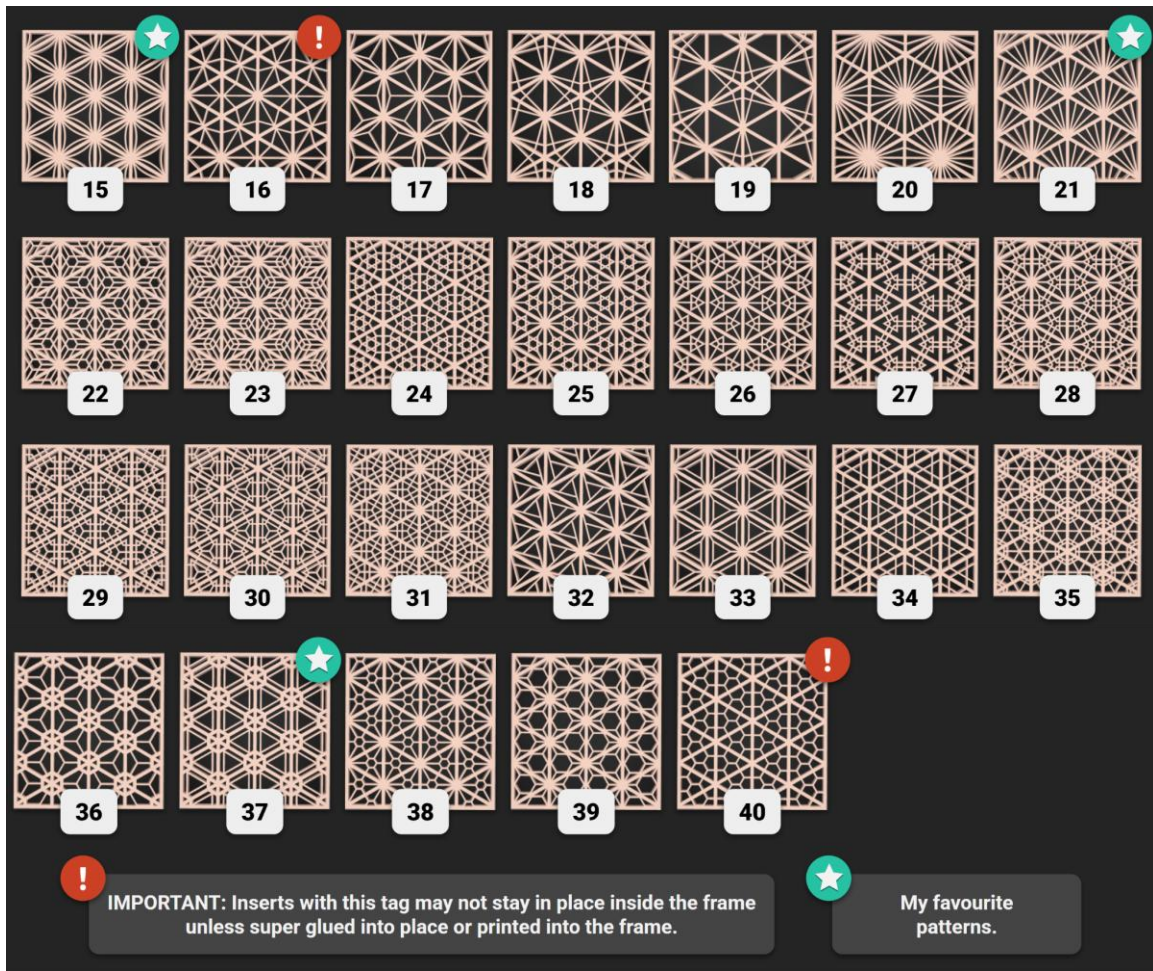
Instructions for Insert Generator:

Step 4.1: Open the insert generator:

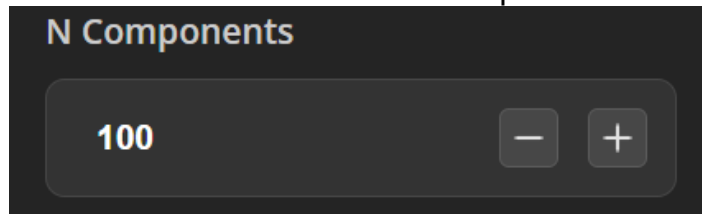


Step 4.2: Select the insert you want to print from the list of options available (see screenshot below).



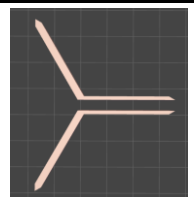
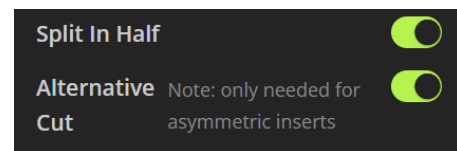
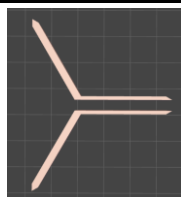
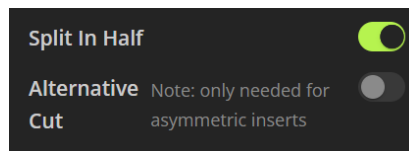
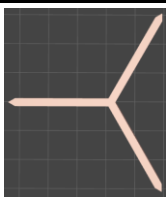


Step 4.3: Adjust “N Components” to reflect the number of duplicates needed for that pattern.



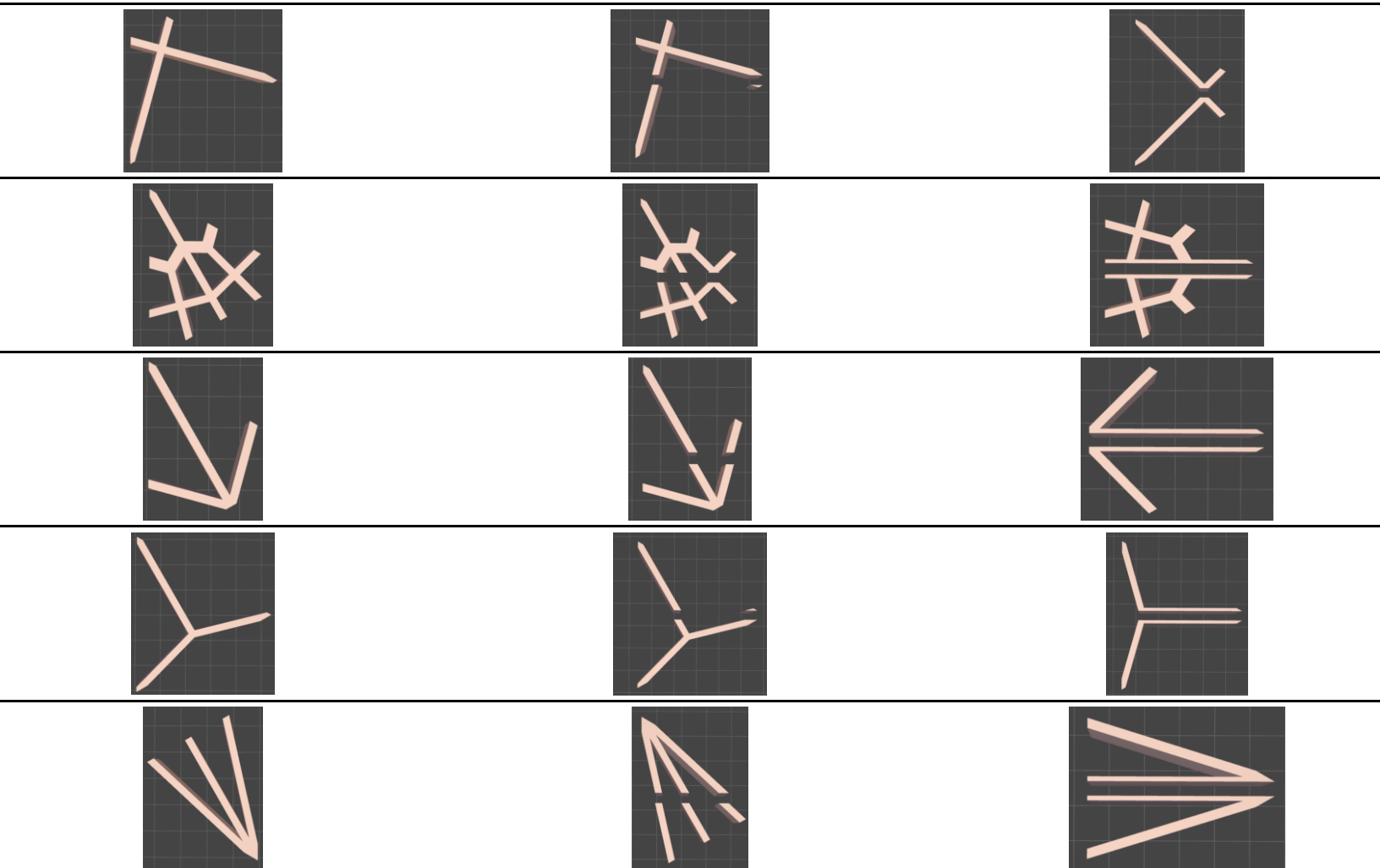
Step 4.4: If needed, split the insert and optionally select the “Alternative Cut” for the insert. Note: the alternative cut only affects asymmetric inserts (see table below for examples).

Pattern (Unsplit)

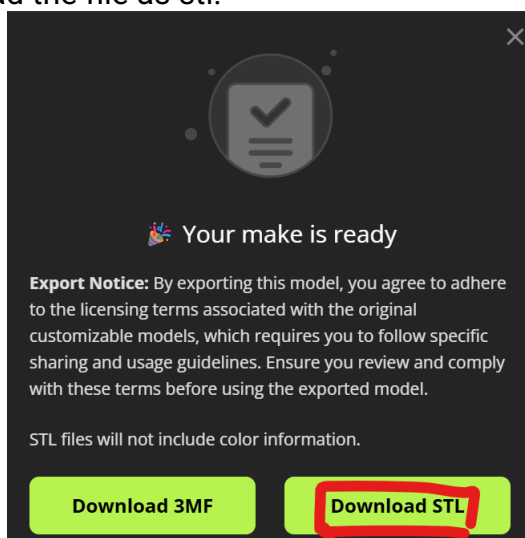


Note: since the insert above is symmetric, there is no difference whether “Alternative Cut” is selected or not.

The half-inserts below are those affected by whether “Alternative Cut” is selected or not. Refer to the model you designed in Kumiko Designer to understand the number of components of each that are required.

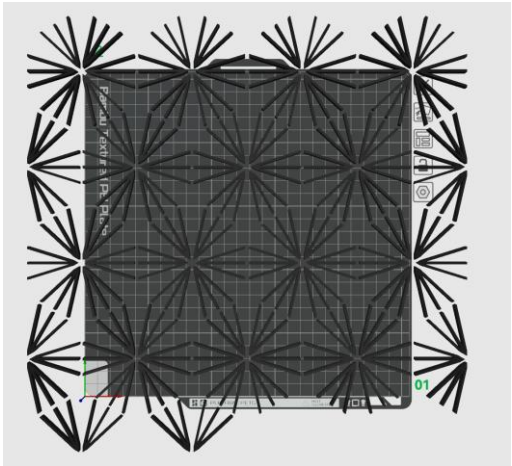


Step 4.5: Generate and download the file as stl.

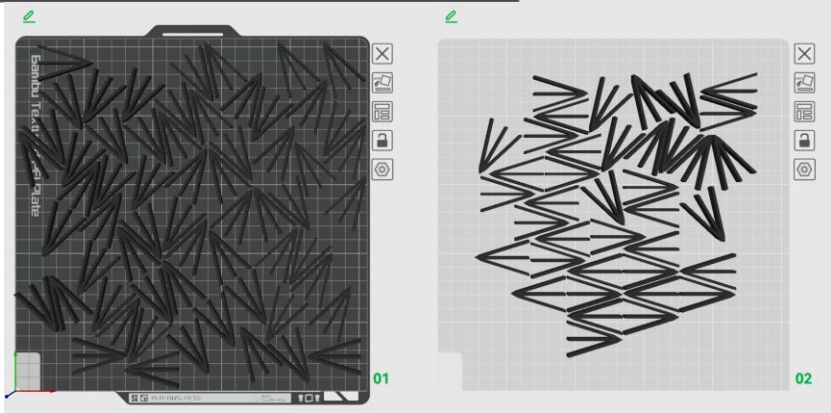


Step 4.6: Open the STL, click on the inserts, click “**Split to Objects**”, then click “**Arrange all objects**”. This will arrange all the inserts onto your print plate.

Note: I would recommend moving any inserts close to the edge of your print plate to a new print plate since I found the edges of my print plates slowly accumulate oil from handling, resulting in poor first layer adherence and print failures for inserts along the edge. Alternatively, just make sure to clean your plate thoroughly before printing, especially along the edges.



When you open the STL, the inserts may not all fit on your print plate depending on the number of components you selected.



After splitting and arranging, the inserts will now fit and are ready to print.

Step 4.7: If exporting the inserts as an stl file, make sure to adjust the printing settings to match the following:

	Layer Height	Initial Layer Height	Walls	Infill Density	Infill Pattern	Wall Generator	Top shell Layers	Bottom Shell Layers	
serts	0.1	0.1	1	30%	Lightning (to save material)	Arachne	>= 5	>= 3	All ot de "0.20

	0.2	0.2							For a
--	-----	-----	--	--	--	--	--	--	-------

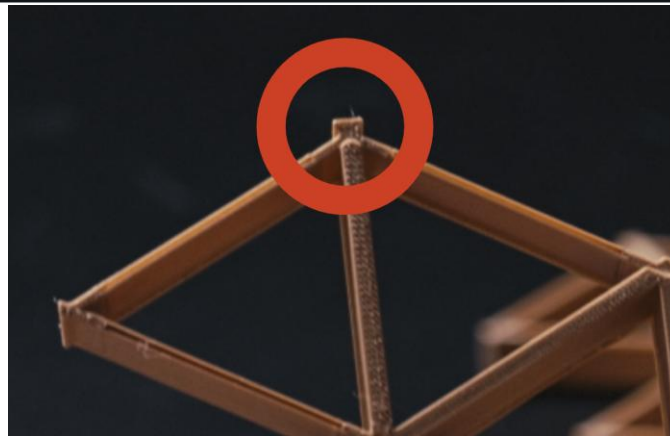
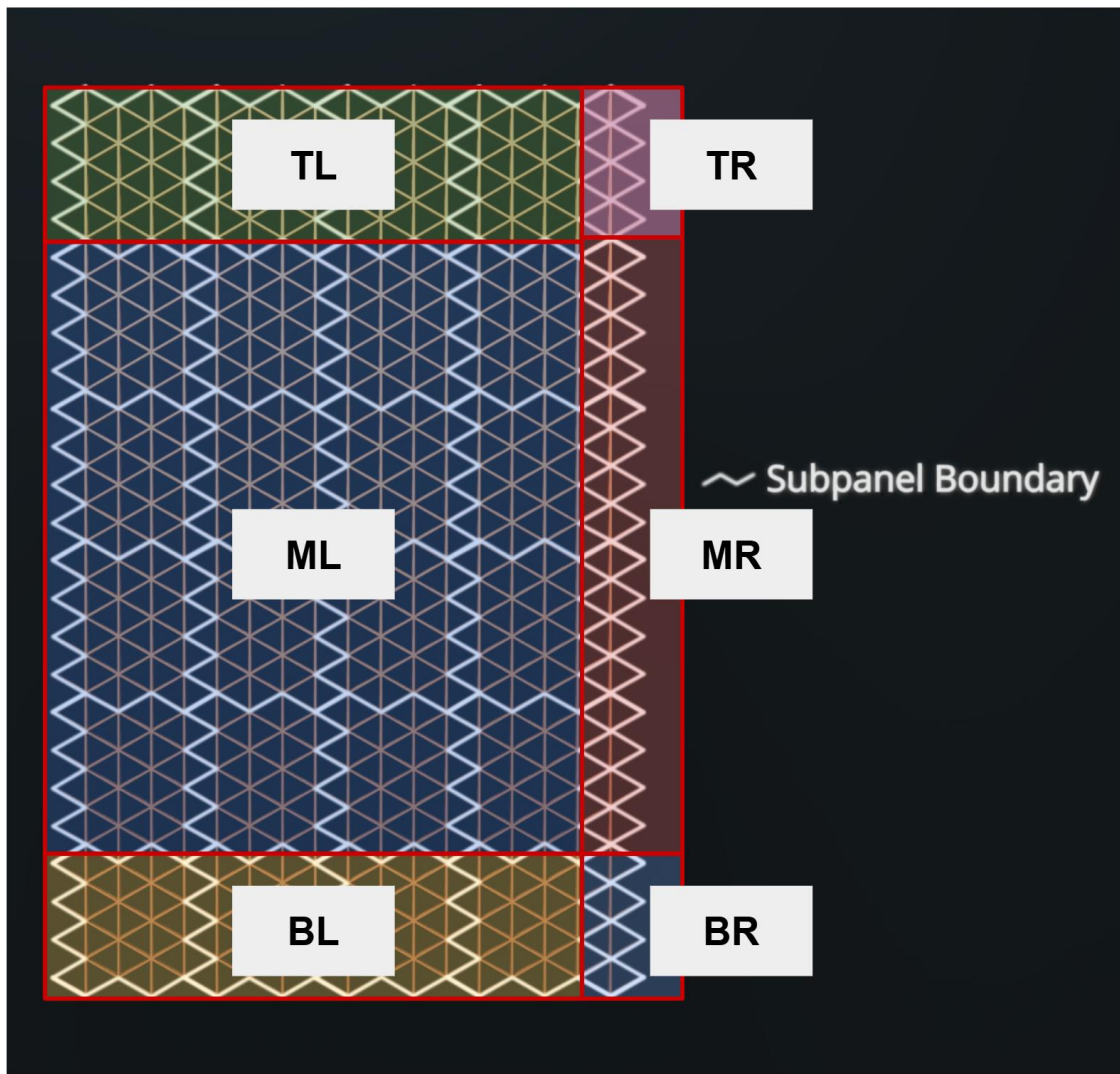
Step 4.8: Print your inserts.

Step 5: Assembly

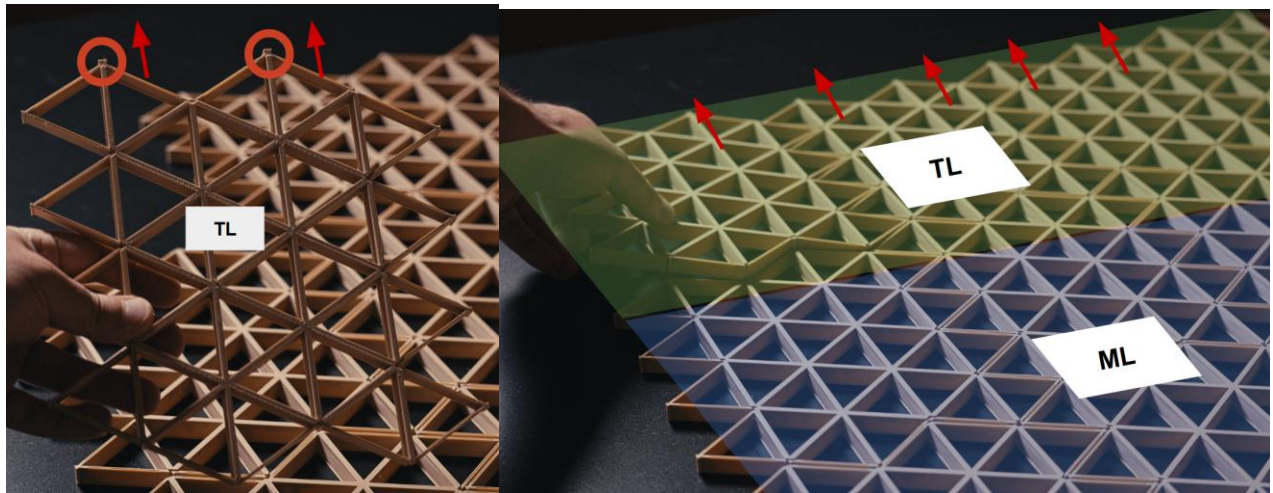
Note: You can refer to the glossary on the last page for the name of the components and their locations in the print profile and customizer.

Step 5.0: Understand and arrange your components. Throughout the assembly process please make sure to compare your current component with the reference panel you have available to understand where they should go.

Step 5.1: Begin by putting all the subpanel rows together. TL, TR, ML, MR, BL, and BR subpanels are all different and need to be arranged according to the image below. TL and TR large dovetails should all point up and BL and BR large dovetails should all point down. It’s easy to accidentally misalign them so be careful and line things up before assembling anything.

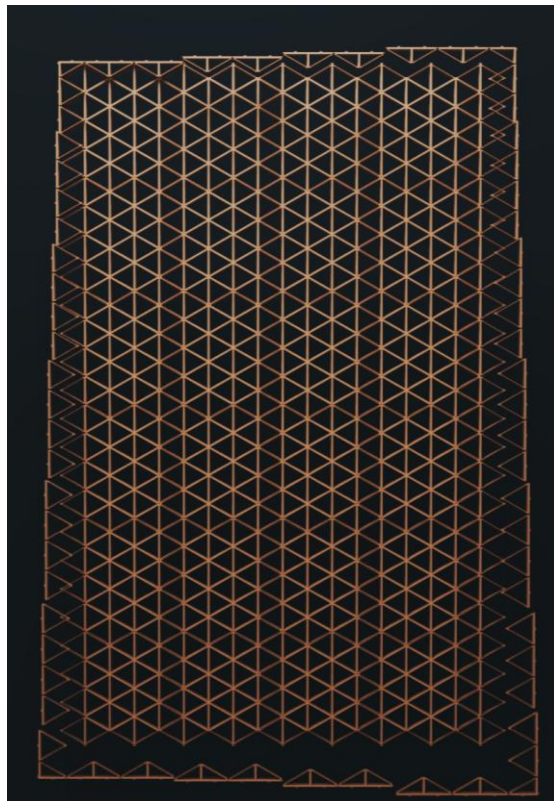


Large dovetail part of TL, TR, BL, BR. Make sure they're pointed in the correct direction (TL & TR point up | BL & BR point down)

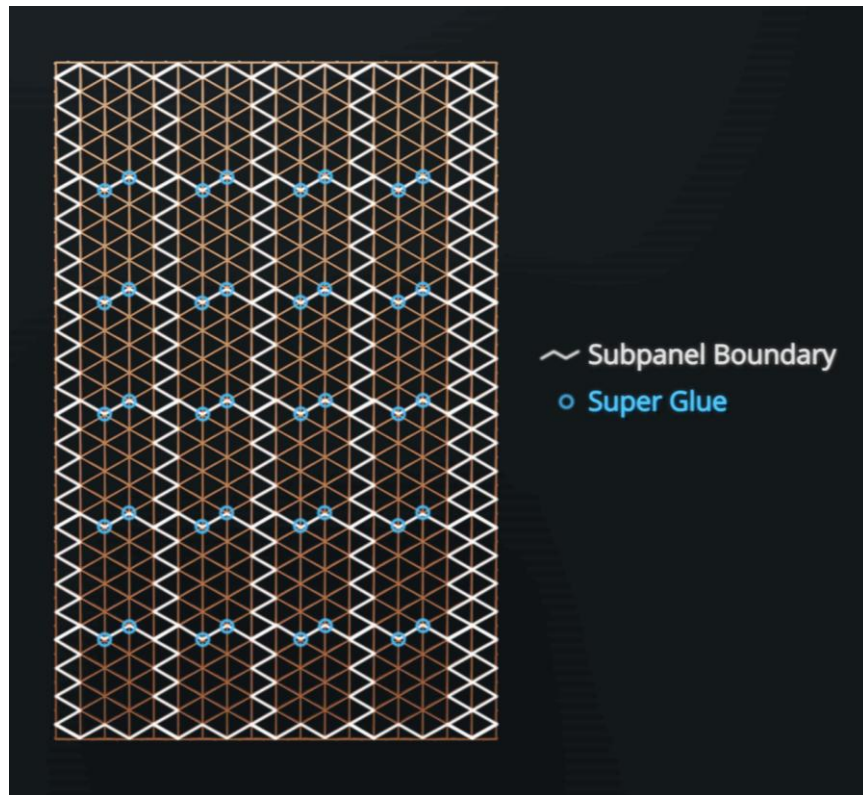


Large dovetail of several TL subpanels all pointing up.

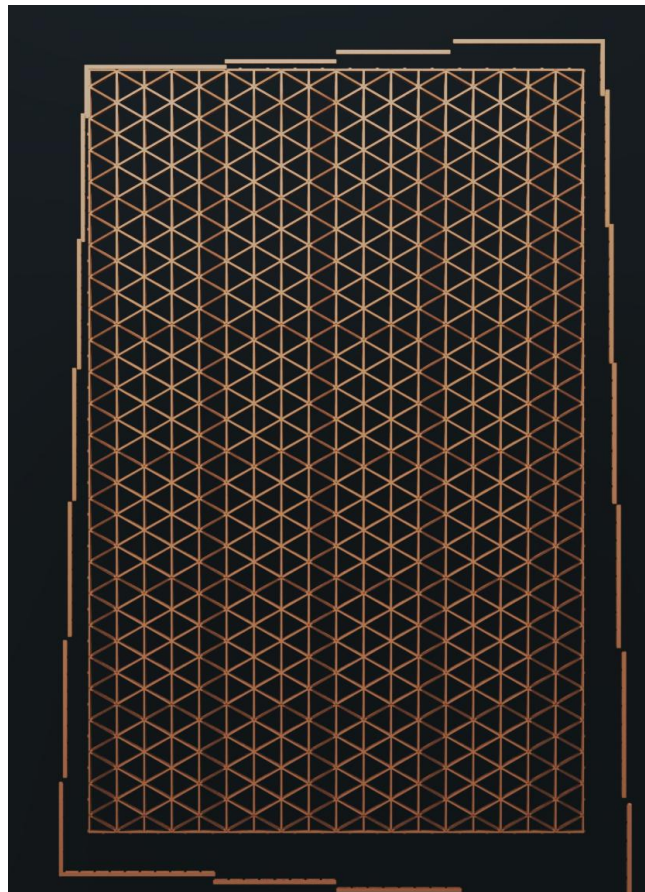
Step 5.2: Join the rows together using the outer border components. Make sure the correct outer border component is joined with the correct section.



Step 5.3: Place super glue in the positions shown below between two subpanels. This will hold subpanels together vertically and stabilize the frame. This is all I did for my 7 foot wide panel and didn't have issues. However, you have to still be careful when lifting the panel since that's when the most stress will be exerted on the center of the frame. Place cardboard or something thick to prevent the super glue from damaging the surface below.



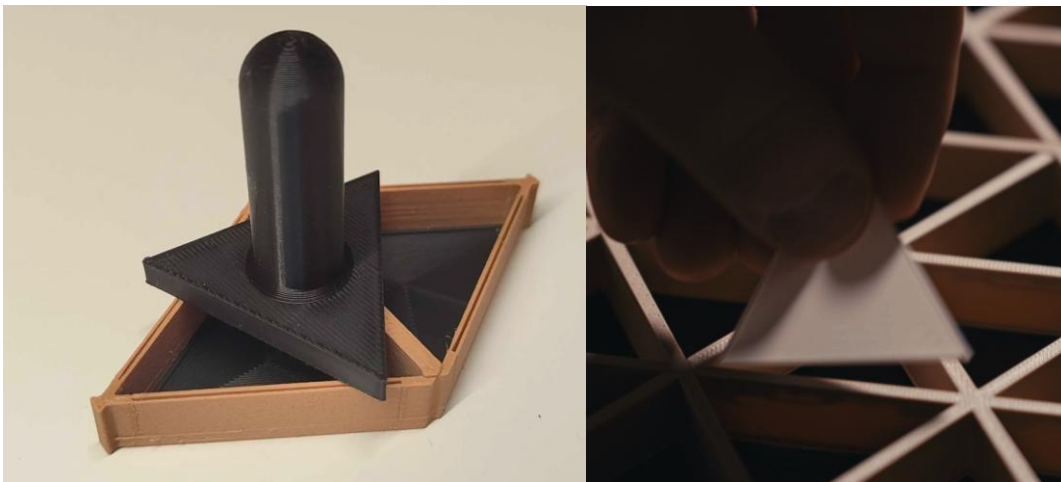
Step 5.5: Add border pieces.



Step 5.6: Add the seams. There are multiple seam variants. Below I try to highlight the main differences between each. Refer to your reference panel when adding the seams to make sure you're putting the correct variant in the correct location.

Optional: the seams can come loose when moving the panel, so if you're satisfied with the frame size, you can super glue them in place. You only need a tiny dab to hold them in place.

Step 5.7: Add background inserts. [@Minispurk](#) made a great tool that assists with pushing the background pieces to the bottom of the frame. [Model available here](#). Alternatively: using a thin tool, push the corners of the inserts to the bottom corners of the triangle. It's very easy for these inserts to be placed incorrectly which can lead to creased background pieces. If you're backlighting your panel, it's especially important that all the inserts go in flat without creases. Take your time; if you do it correctly the first time, it'll save you time since you no longer need to fix the inserts. I use my hands to orient them in place since after a certain amount of practice you can tell whether the insert was seated correctly or not. If you flip over the panel you'll notice which inserts are misplaced.

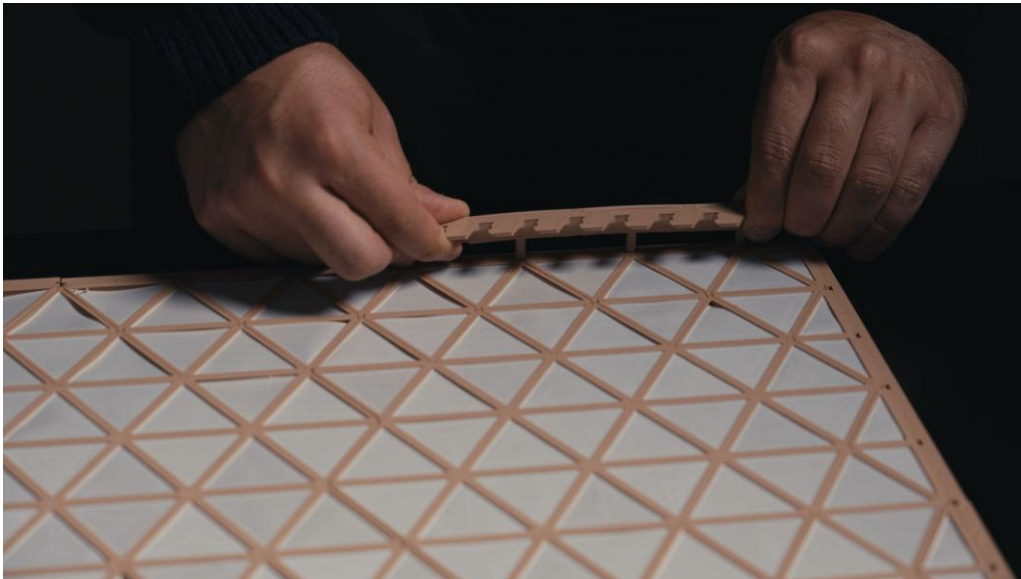


Step 5.8: Add the inserts based on your design.

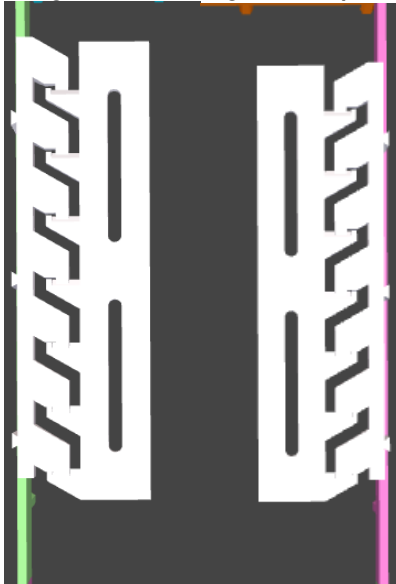


Step 5.9: Carefully flip over the panel. For large frames that sag, use a thick piece of cardboard or a piece of wood to support the middle of the frame as you turn over the panel.

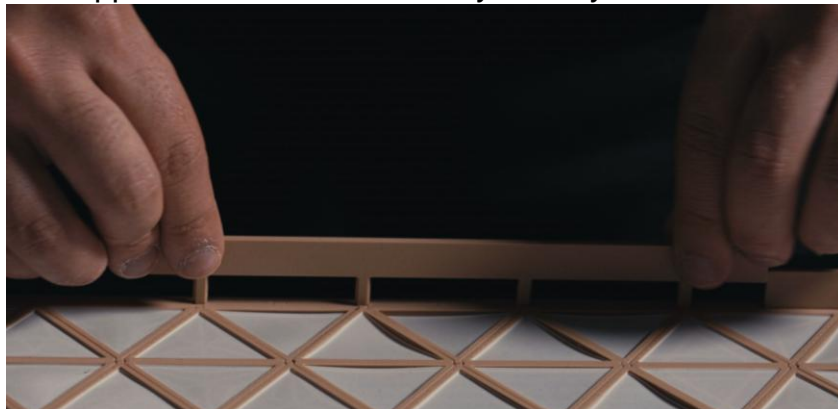
Step 5.10: Add the corresponding hanger components. Make sure to get the orientation and placement correct. Reference the customizer to understand how these pieces come together.



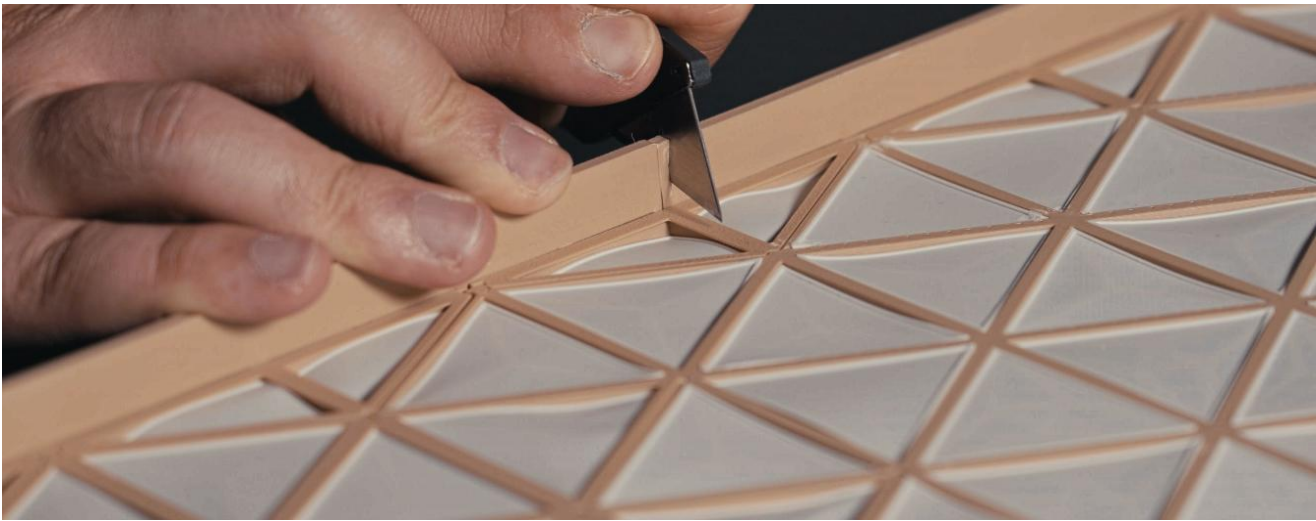
Placing the TL hanger component.



Step 5.11: Insert the remaining border background components. These also have ML, MR, BL, BR variants so if pieces don't appear to fit somewhere they're likely meant for a different location.



Optional: Add an LED strip along the border background pieces. I start at the bottom of my panel, work the LED strip around and then meet it back to the start. [5m long LED Strip available here](#) if your perimeter is > 5m you'll need to either daisy chain the LED strips or buy a longer one. I'm not too familiar with LED strips so I don't have much advice beyond that.



At the bottom of your frame, cut a small section to let LED wires pass through



Starting at the bottom, insert the LED, and work your way around the panel.

Step 6: Hanging the Frame

Step 6.1: Using a level, mark where the top and sides of the frame should sit (this should be based on the distance between Border components).

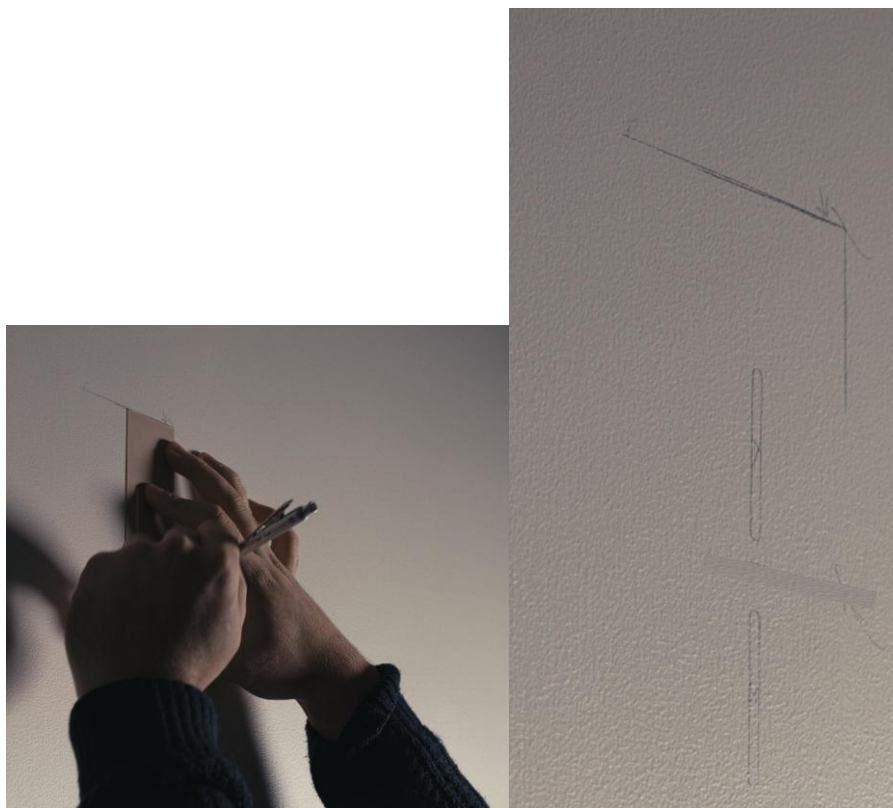


Measure from corner to corner



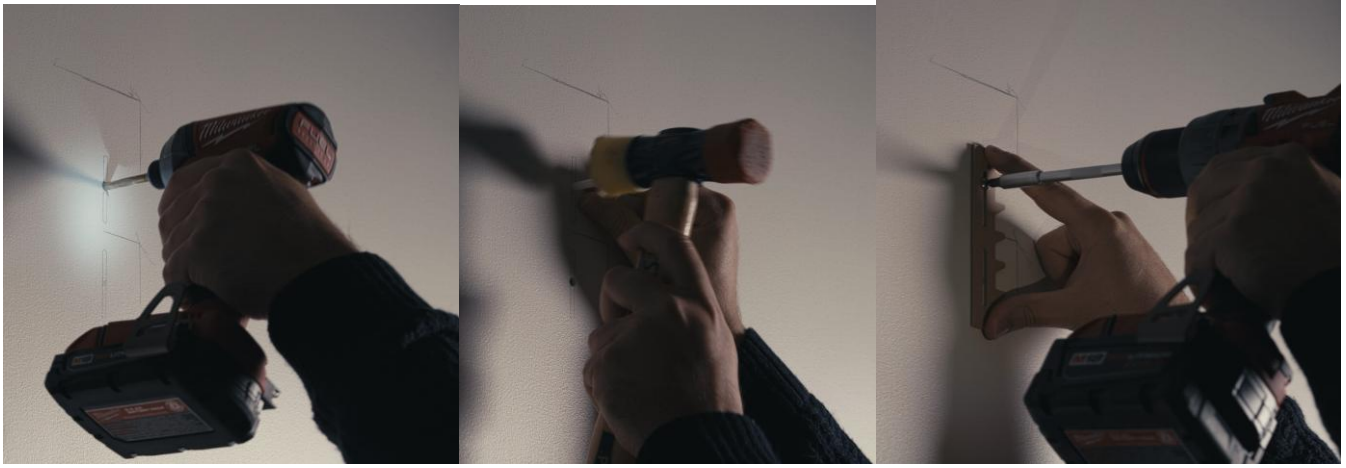
Mark corners on wall

Step 6.2: Using the corresponding template, mark the center of the slots. Note: if your panel has an odd width, you will have a different template for the right hanger.



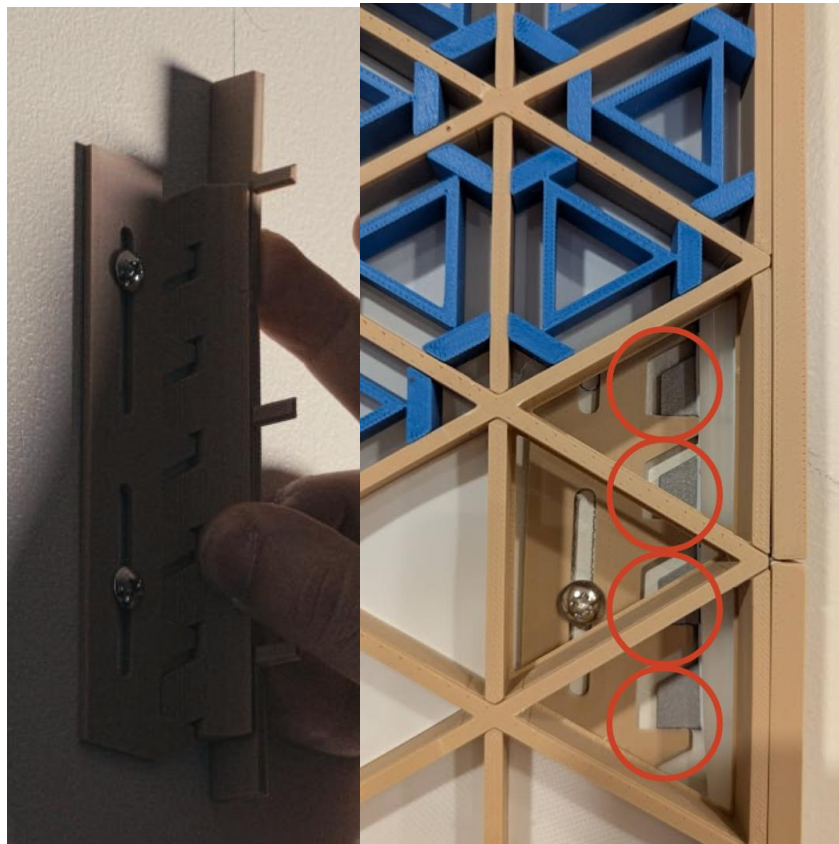
Center points made using template

Step 6.3: Drill the holes and use an anchor to install the wall cleat.



Drill, anchor, screw

Step 6.4: Lift the panel carefully--using a piece of wood or cardboard to support the centre of the frame--and hang the panel onto the cleat. Adjust the cleats as necessary to level the panel. **Make sure all cleats engage with each other.** Avoid hanging the panel only by the top rungs of the wall hanger as this might lead to your panel falling from the wall.



Visualization of how the wall component and panel component hanger should engage. It's extremely important that all rungs engage. You can remove a couple inserts to verify (just be careful when removing them).

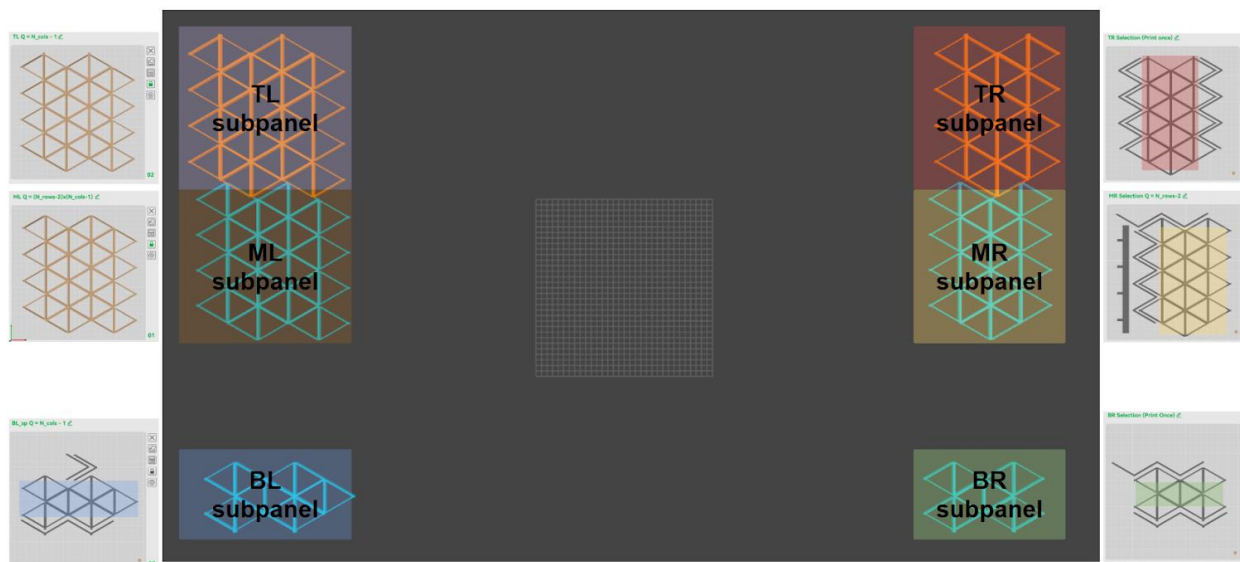
Enjoy! Leave a comment with your panel on the [MakerWorld project](#)!
I love seeing all the panels people have made.

Message me on MakerWorld or Patreon if you have any questions, need help, or notice any bugs.

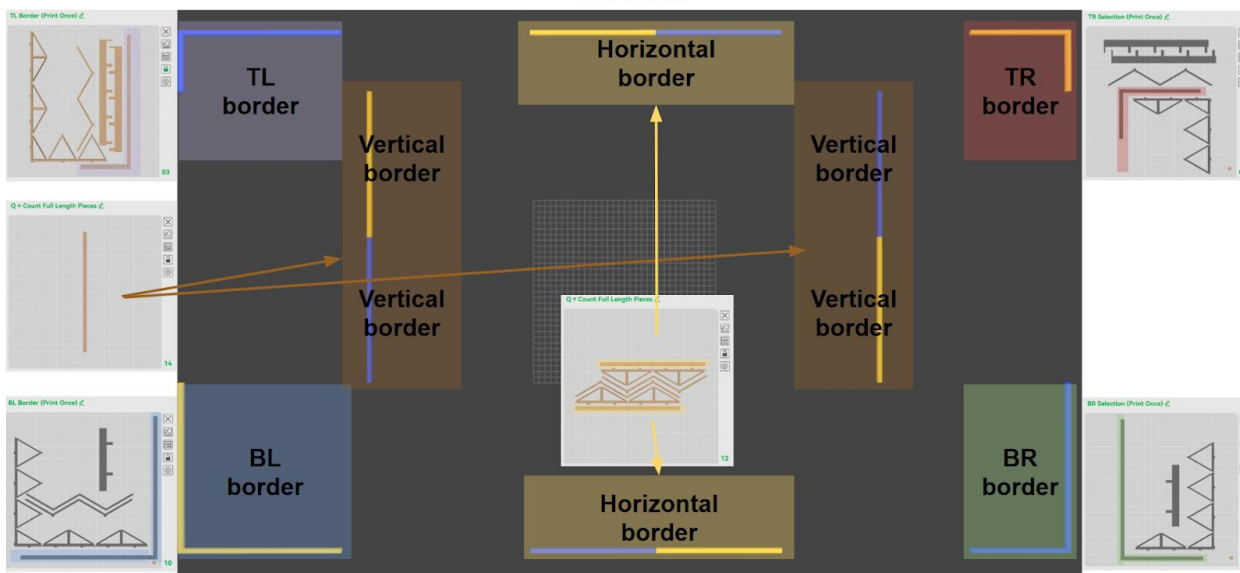
I'm quite responsive but if there's a lot of individuals contacting me it'll take me longer to respond, so your patience is appreciated. **Before reaching out, please read the instructions thoroughly, as they may answer your question.**

Glossary

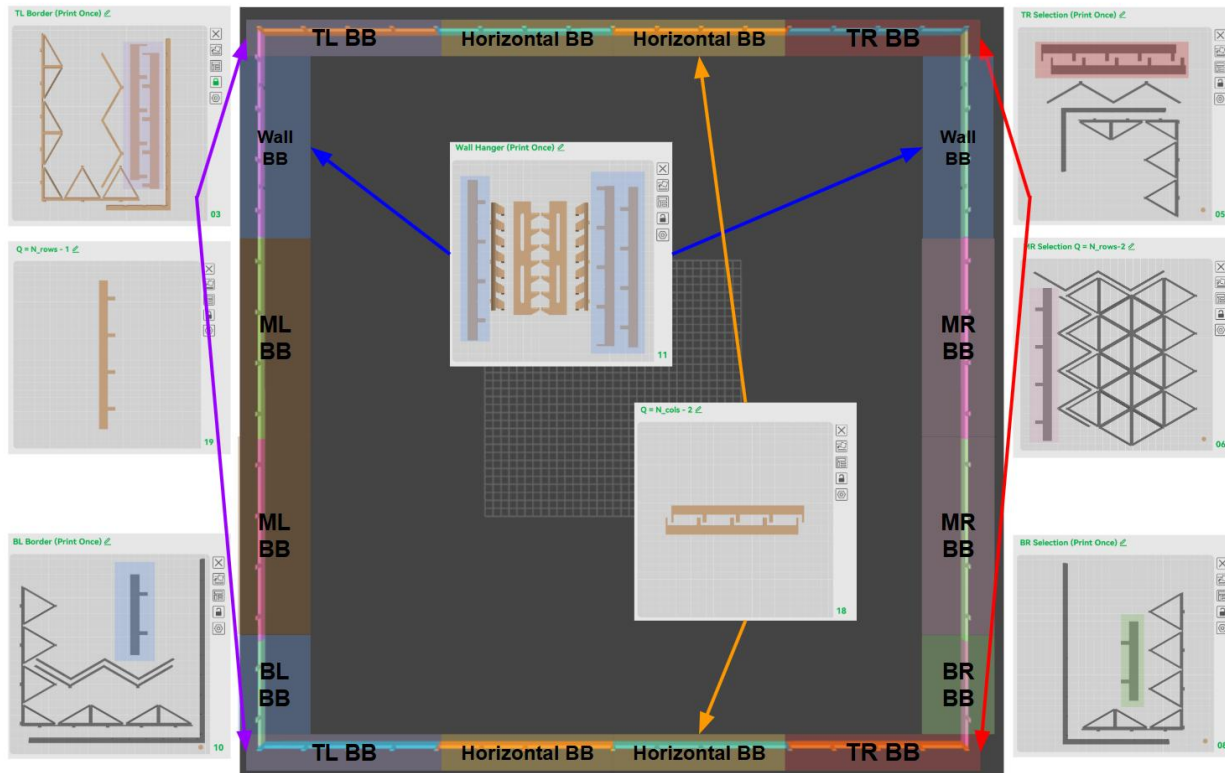
Subpanels



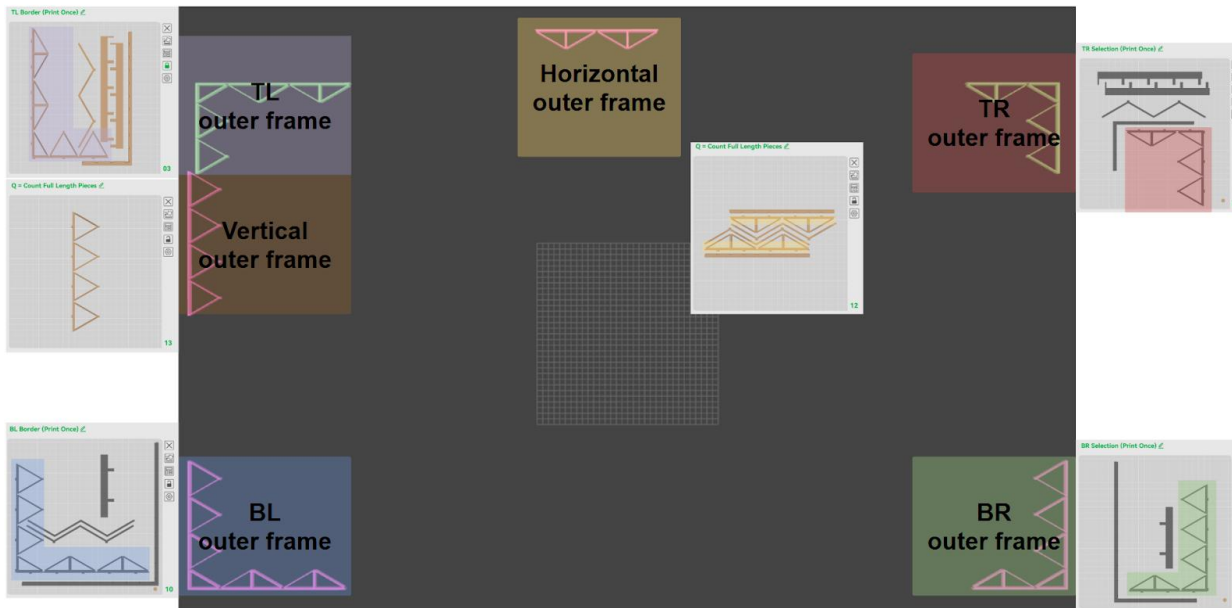
Border



Border background (BB)



Outer frame

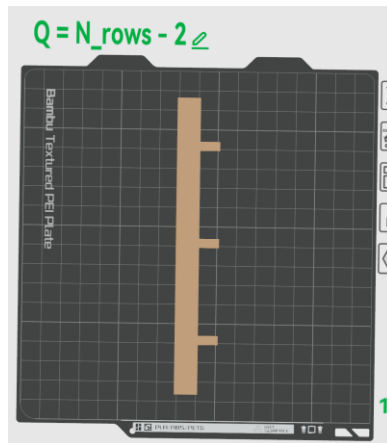


Changelog

General Changes

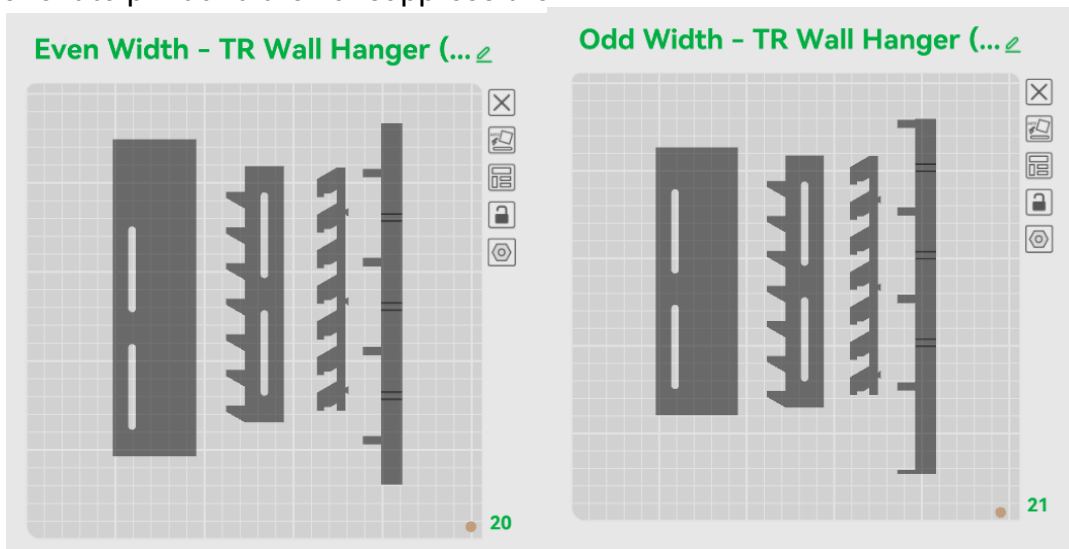
Sep 15: Updated formula

- Small change but updated the formula for the following component from $N_{\text{rows}} - 1$ to $N_{\text{rows}} - 2$ (for all print profiles)



Sep 14: Suppressed TR hanger components

- By default I suppressed the TR hanger components as you first have to determine which component to print and then unsuppress them



Sep 14: Added hanger marking templates

- See [here for instructions on how to use them](#)



Aug 3rd 12:30AM EST: Updated Insert Generator Code

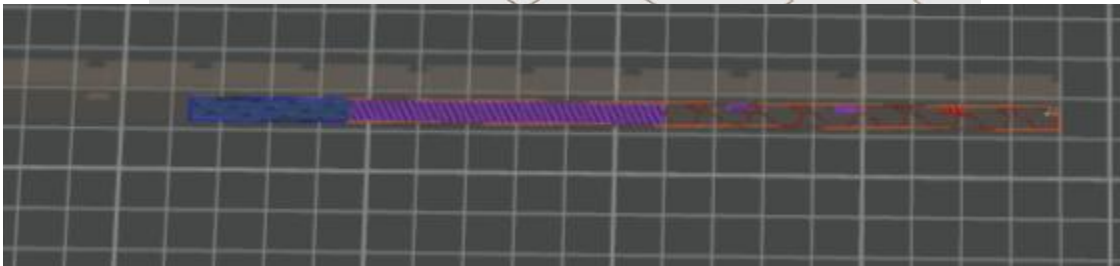
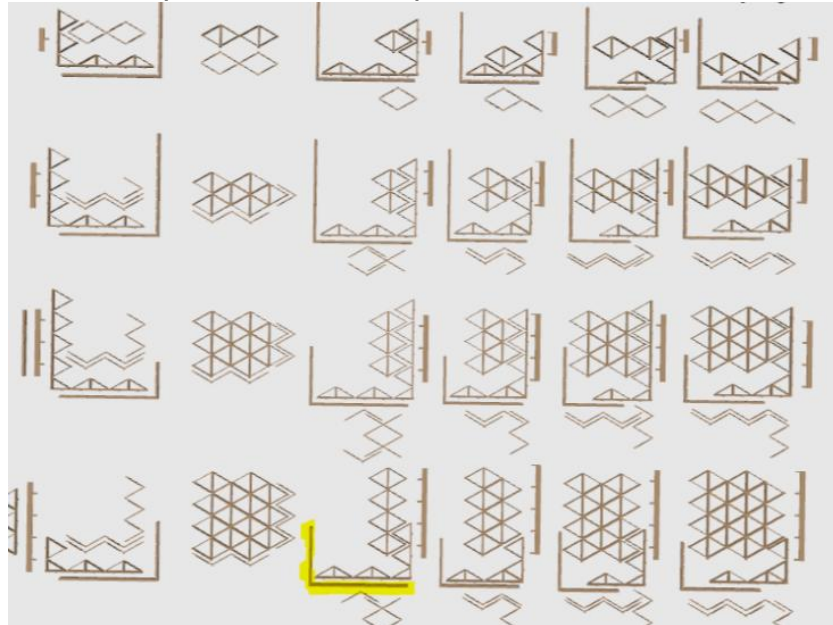
- Improved how inserts are laid out when generated
- "N Components" variable now impacts both the regular inserts and split inserts equivalently. I.e., if you require 25 split inserts, it generates 25 half-inserts rather than 50.

- Added additional usage instructions.

Corrected issues

BR Border not flat on print bed

- **Found:** Aug 16th ~7AM EST by Rinny0 (Tyty!)
- **Corrected:** corrected Sep 14. Flatted the object
- **Issue:** Seems to only affect the highlighted border below. Results in the border printing in a stepped fashion. I think this is just because I didn't rotate the object a full 180 degrees.
- **Solution:** Reorient border piece to bed so it prints flat.



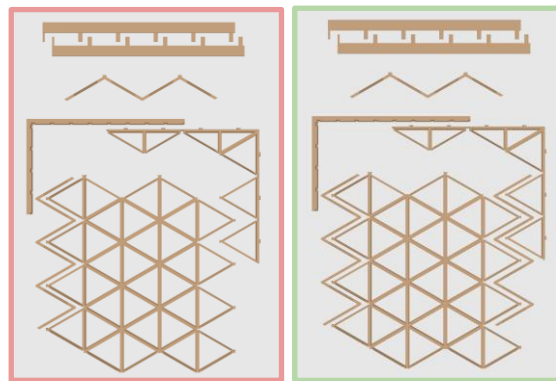
Wall Hanger Flexes Near the Top

- **Found:** Sep 2 ~ 11PM EST by choomba
- **Corrected:** fixed Sep 14th. Thickened the hanger to 3.6 mm feel free to slice it thinner if you think it's too thick
- **Issue:** Wall hanger component flexes near the top if screwed midway through the slot and can potentially cause the panel to fall if the weight is too high and not all the rungs are engaging.
- **Solution:** Will thicken up the wall hanger component so it flexes less. Make sure to engage all the rungs when hanging the panel and avoid hanging the panel only near the top of the wall component.



Missing TR Vertical Seam (only affects 4-Wide 4-Tall)

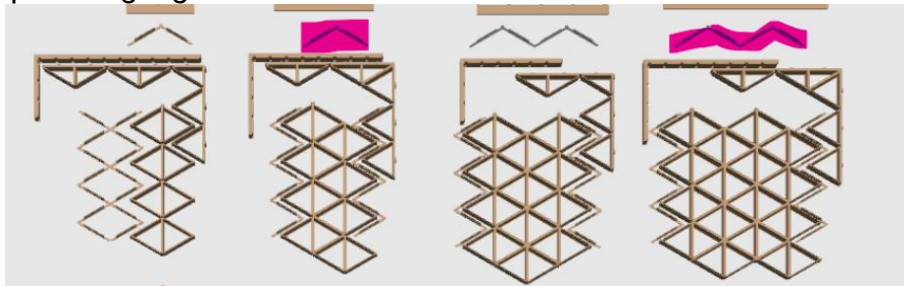
- **Found:** July 31st ~11:30PM EST by me :)
- **Corrected:** Fix implemented Aug 1st 12AM EST. Make sure to download the latest print profile.
- **Issue:** Vertical seam missing for 4-wide, 4-tall TR subpanel
- **Solution:** Add a second vertical seam for 4-wide, 4-tall TR subpanel



Before (Left) : missing second vertical seam After (Right): second vertical seam added

Incorrect TR Horizontal Seam (2-Wide & 4-Wide)

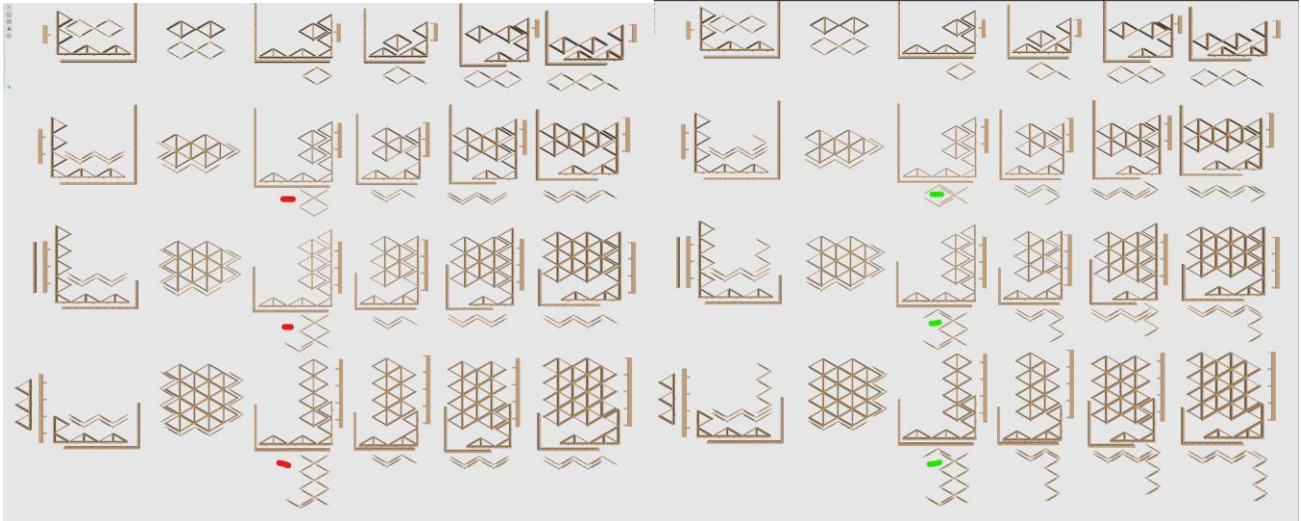
- **Found:** July 31st ~11:30PM EST by me :)
- **Corrected:** Fix implemented Aug 1st 12AM EST
- **Issue:** The top right seams highlighted below should be mirror images to their BR counterparts.
- **Solution:** Replace highlighted seams with mirrored versions of BR horizontal seams.



Horizontal seams that are not correctly mirrored.

Missing TR horizontal seam

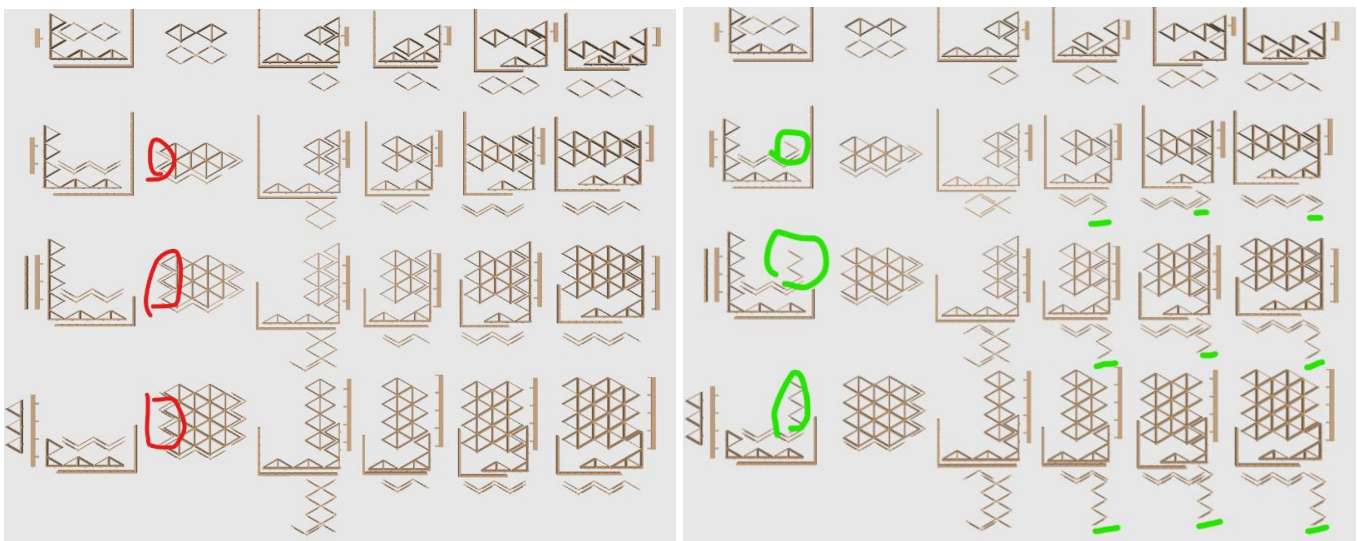
- **Found:** July 31st ~1PM EST by @iclx (Thank you!!)
- **Corrected:** Fix implemented July 31st ~7PM EST
- **Issue:** A horizontal seam will be missing with the current print settings for 1-wide BR subpanels
- **Solution:** Added horizontal seam needed to the respective BR subpanels.



Red (Left) = Before Missing Horizontal Seam Green (Right) = After Added Horizontal Seams

Extra BL vertical seam

- **Found:** July 31st ~3PM EST by @iclx (Thank you!!)
- **Corrected:** Fix implemented July 31st ~7PM EST
- **Issue:** An extra BL vertical seam would be printed.
- **Solution:** Removed vertical seam from BL subpanel and added it to BL border. Added an additional vertical seam to BR subpanel.



Red (Left) = Before

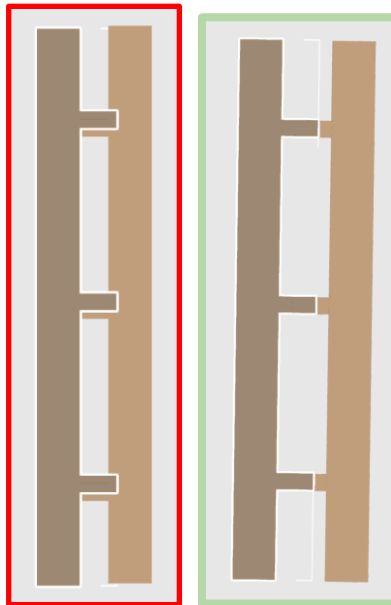
Green (Right) = After

Incorrect left/right background borders

- **Found:** July 30th ~11PM EST by @user_3965095218 (Thank you!)
- **Corrected:** Fix implemented July 31st ~7PM EST
- **Issue:** The vertical background borders should be mirrored with each other. After looking into it further, most vertical background borders are not mirrored with each other. Wait on printing

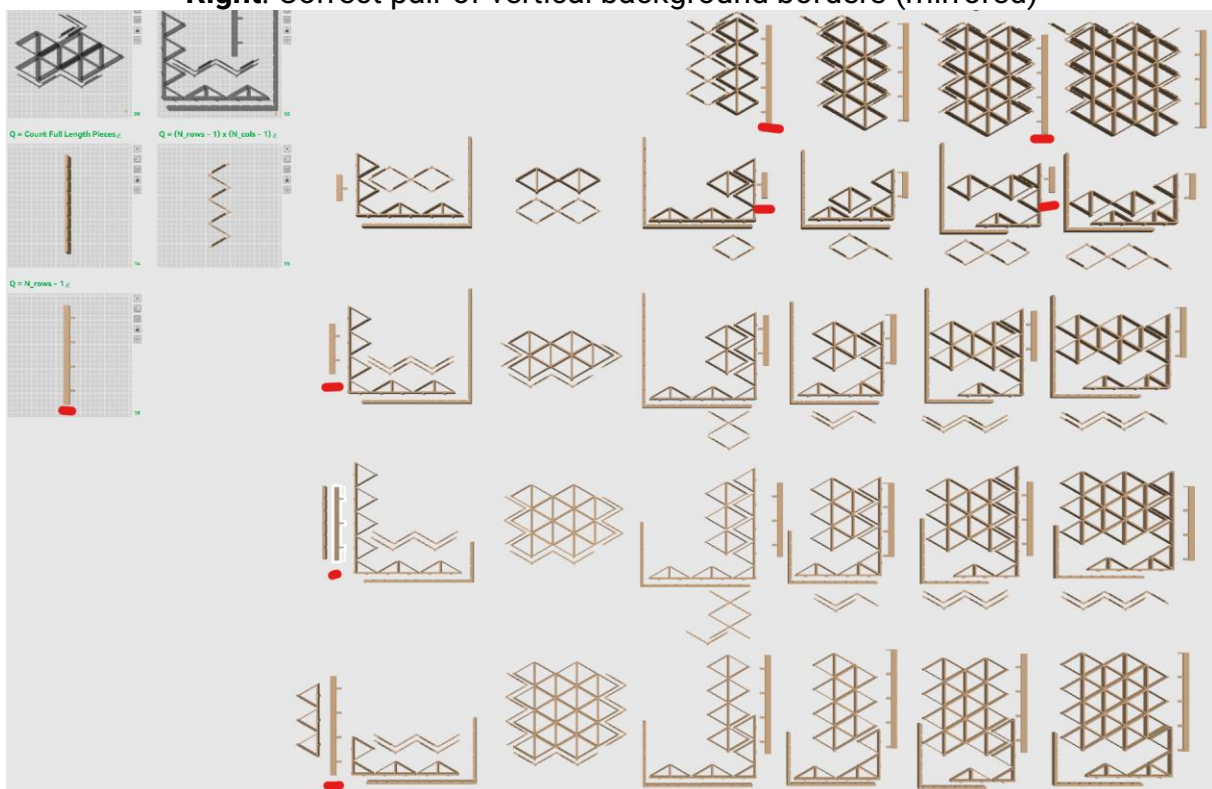
the left/right background borders for now until the fix is implemented. I also need to look into this more to figure out what happened. If you have already printed some left-right background borders, please wait until you finish printing the rest of the frame to see which remaining background borders you require. Alternatively, you can just not print the background borders (except for the hanger components) and still have a functional panel if you want to save filament. Sorry for the mistake!

- **Solution:** Added the correct background borders. See affected background border pieces below. **Download the newest print profile to get the correct vertical background border pieces.**



Left: Incorrect vertical background border pieces (not mirrored)

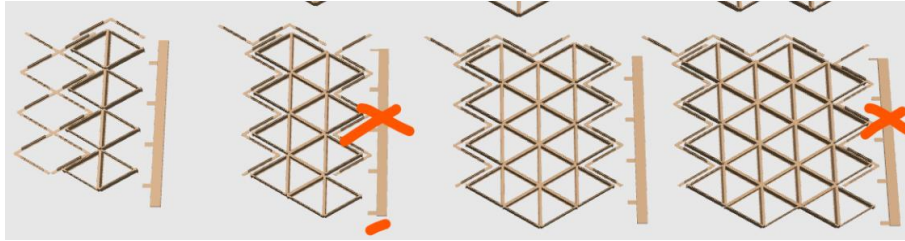
Right: Correct pair of vertical background borders (mirrored)



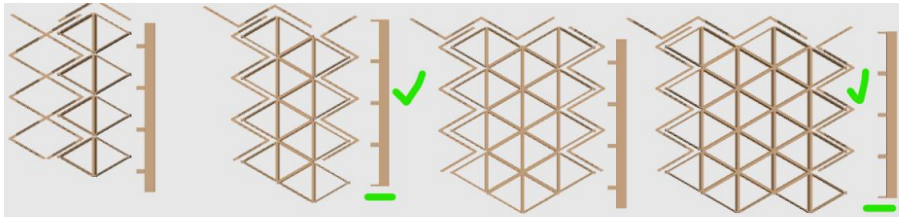
Affected background border pieces highlighted in red.

Incorrect border background

- **Found:** July 28th ~3:30 AM EST by @White273 (Thank you!)
- **Corrected:** July 28th 6:20 PM EST
- **Details:** Affects "Medium Printers (250x250)" print profile. The incorrect border background piece was paired with the odd-sized middle right subpanels. This issue should now be corrected with the newest print profile.



- do not print



- correct pieces

(Note the bottom of the pieces to see the difference)



- where they are in the print profile

Source Code Instructions

Source Code Files Included

Filename	Description	Instructions
modular_kumiko_panels.scad	The modular_kumiko_panels.scad file lets you fully customize your panel. It will let you create panels that fit exact dimensions, adjust the pitch size, change the frame depth, change the border thickness, and lets you generate all the components without having to go through the customizer on MakerWorld.	Click here for instructions
svg_insert_generator.scad	The svg_insert_generator.scad file will let you generate the background inserts and designs made using kumikodesigner.com in just a few steps. Faster than the manual generator alternative.	Click here for instructions
dual_insert_generator.scad	Lets you generate components individually (one pattern at a time). Additionally, it lets you generate the test tube holder components, and create inserts that can embed magnets.	Equivalent to MakerWorld Instructions Additional features that MakerWorld lacks: - Can generate test tube holders (Pattern) - Magnet Insert Generator

0. OpenScad Quickstart

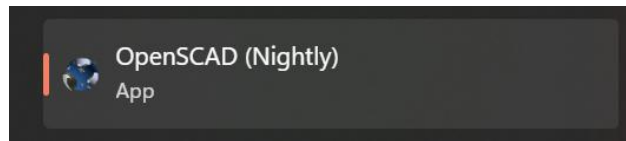
Message me on MakerWorld or Patreon if you have any questions, need help, or notice any bugs. I'm quite responsive but if there's a lot of individuals contacting me it'll take me longer to respond, so your patience is appreciated. **Before reaching out, please read the instructions thoroughly, as they may answer your question.**

BACKGROUND

The purpose of this page is to get OpenSCAD installed on your computer so you can start using the source code files available on Patreon.

STEPS

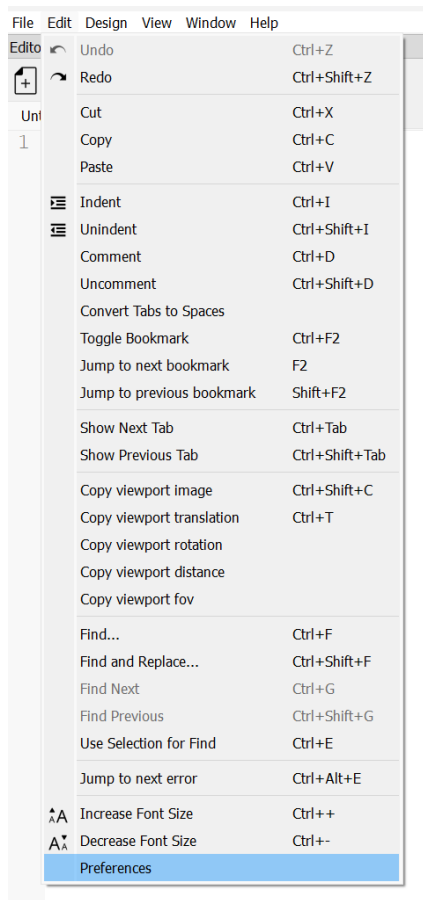
1. Download OpenSCAD via <https://openscad.org/downloads.html#snapshots>
 - I recommend downloading the nightly build since the stable release is 4 years old. Using the nightly build, you can enable some features that speed up the rendering process.
2. Install OpenSCAD
3. Install the BOSL2 package ([Instructions here](#) - Make sure to rename the folder to "BOSL2" as described in step 3 in the instructions)
4. Open OpenSCAD, click "New", and go to the "Edit" dropdown menu and select "Preferences"
5. Edit your preferences based on "RECOMMENDED PREFERENCES" at the bottom of this page.



What the nightly build icon looks like

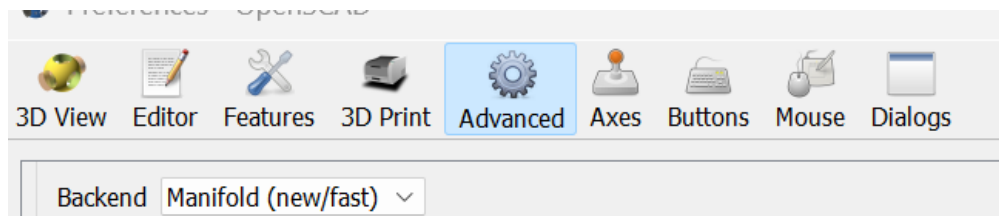


Click "New" for now so you can set your preferences before running any code.

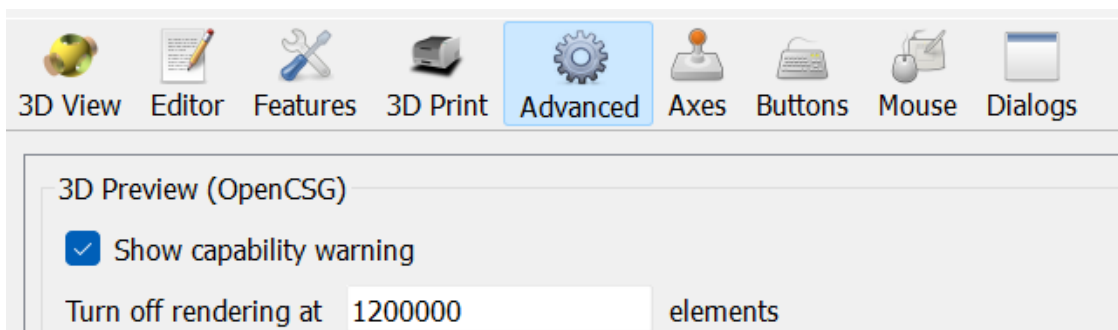


Open the Edit tab and select Preferences

RECOMMENDED PREFERENCES



Set Advanced > Backend to **“Manifold (new/fast)”**



Increase “Turn off rendering at ____ elements” to a large number.

Now you're ready to start using the source code. Please refer to the [Frame Generator](#) instructions next to start creating your custom frame.

1. Frame Generator

Message me on [MakerWorld](#) or [Patreon](#) if you have any questions, need help, or notice any bugs.
I'm quite responsive but if there's a lot of individuals contacting me it'll take me longer to respond, so your patience is appreciated. **Before reaching out, please read the instructions thoroughly, as they may answer your question.**

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BACKGROUND

The following instructions are for the file named "**modular_kumiko_panels.scad**".

The "**modular_kumiko_panels.scad**" file lets you fully customize your panel. It will let you create panels that fit exact dimensions, adjust the pitch size, change the frame depth, change the border thickness, and lets you generate all the components without having to go through the customizer on MakerWorld.

REQUIREMENTS

1. Download and install OpenScad. See OpenScad Quickstart ([OpenScad Quickstart](#))

STEPS

STEP 0: Complete the following before starting:

Step 0.1) Determine the size of the panel you'd like to generate. You can either do this in the generator or if you have a specific height and width you want to hit, determine the mm's needed.

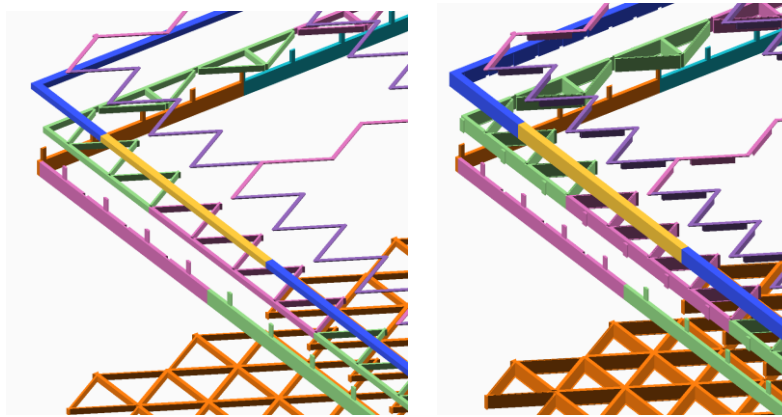
STEP 1: Understand what all the parameters do

Before you change any parameters for the frame it's important to understand what each of the parameters do. Below I've outlined all the available parameters.

GENERAL OPTIONS

Assembly_Option: whether to generate a reference or a printable version of the frame

- **"Reference Mode":** Generate a reference (do NOT print) version of the frame. This option is much faster than "Printable Mode" and is available so you can quickly prototype parameter changes.
- **"Printable Mode":** Generates a printable version of the frame. Once you've finalized the parameters, switch to this version to generate the printable components.



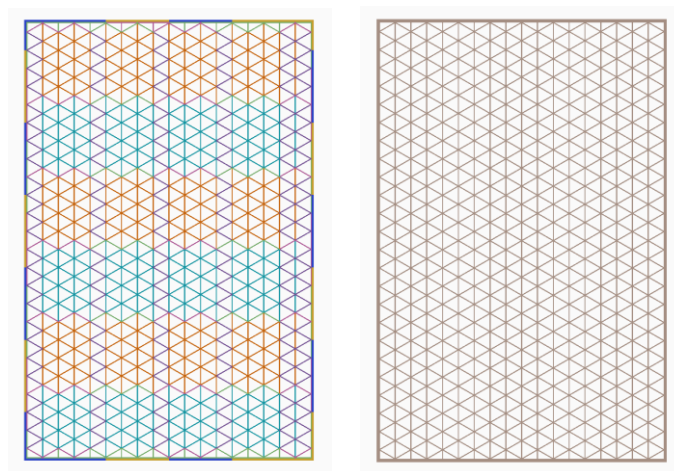
Assembly_Option

LEFT: "Reference Mode"

RIGHT: "Printable Mode"

Hide_Colors

- **If false:** Colors the frame components with alternating colors to easily visualize the differences between components. This is helpful as a reference when assembling the frame.
- **If true:** Uses only one colour for the frame. This is useful to visualize how the frame might look.



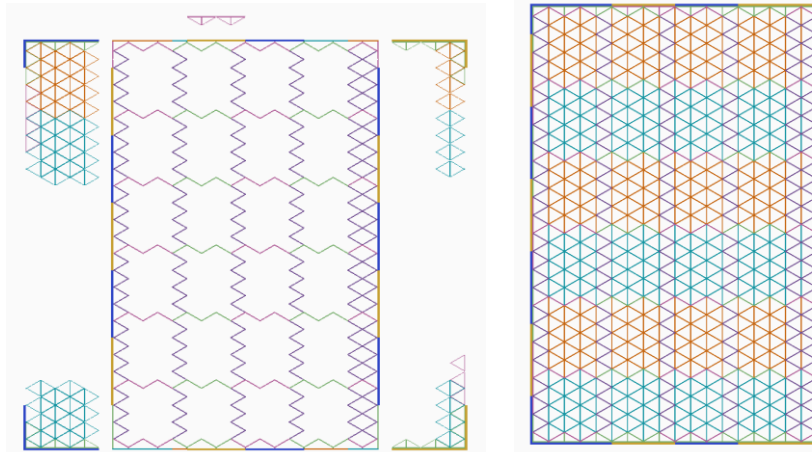
Hide_Colors

LEFT: false

RIGHT: true

All_Component_View

- **If false:** Generates a reduced # of components. This contains all necessary components but if you plan on printing this view, you'll need to calculate the correct # of each component variant. I don't provide details on how to print this view, but it's mainly here since it is faster to generate while retaining all required information for the frame and is useful as a reference.
- **If true:** Much slower to load but generates ALL objects for a component. I would recommend using this option when printing the frame as it's easier to organize in a slicer. Printing instructions are explained later in the instructions.



All_Component_View
LEFT: false RIGHT: true

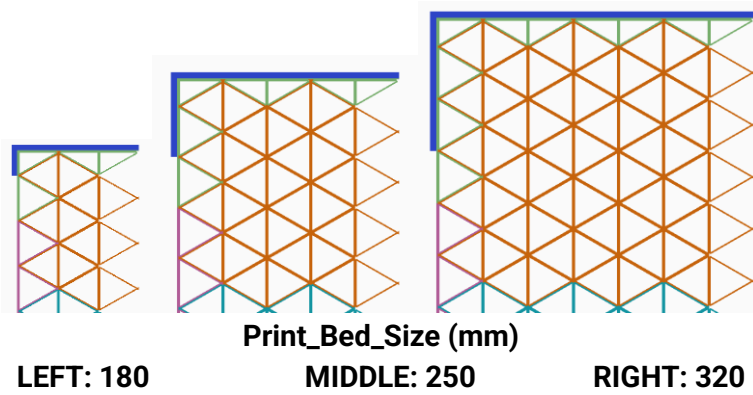
Hide_Text

- **If false:** Generates a tool tip that contains your selected parameters. It also warns you if you are in reference mode or printable mode. I recommend leaving this **false** so you know what your exported frame parameters were in case you forgot to screenshot/note it down.
- **If true:** removes the tool tip



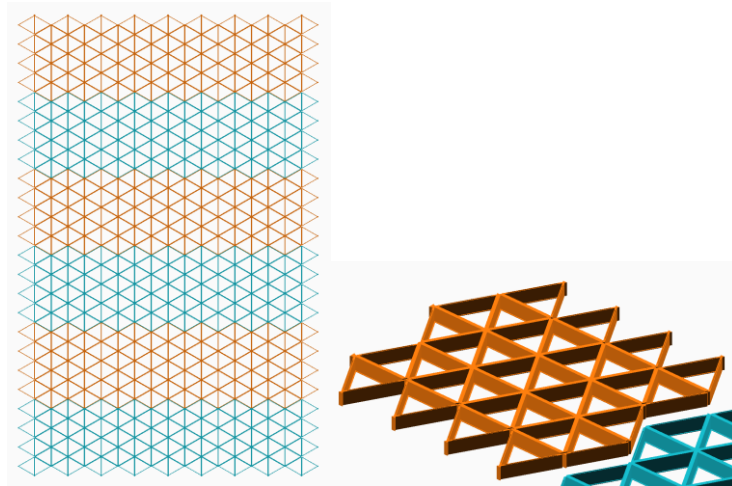
Hide_Text
LEFT: false RIGHT: true

Print_Bed_Size (mm): Constrains all objects to fit within this size. Set this to the smallest edge available on your print bed.

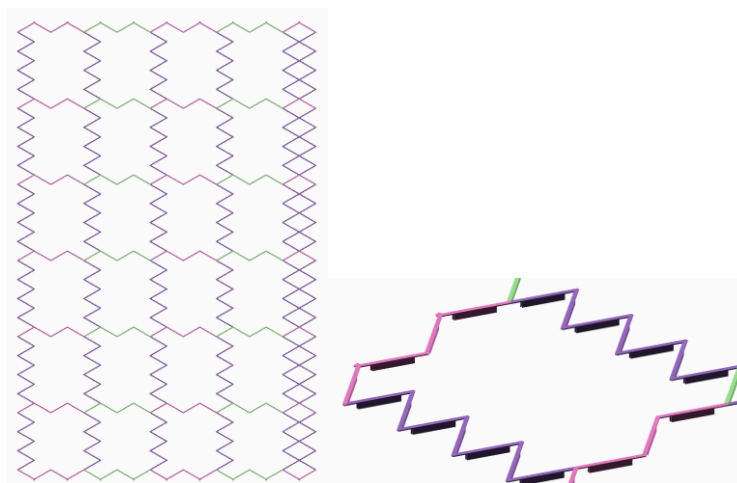


FRAME COMPONENTS

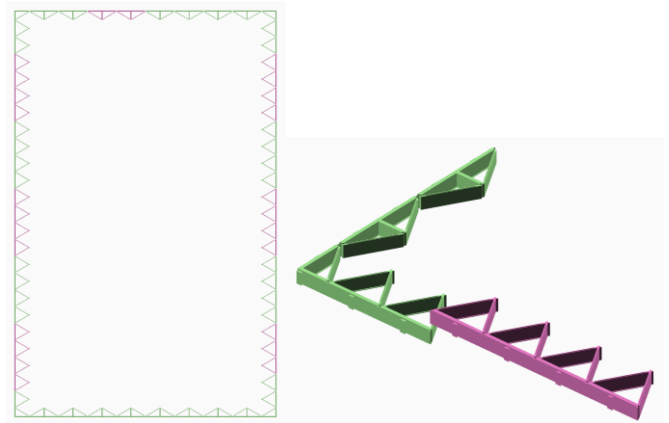
Turn on/off these components to view and/or export subsets of the components for a given frame.



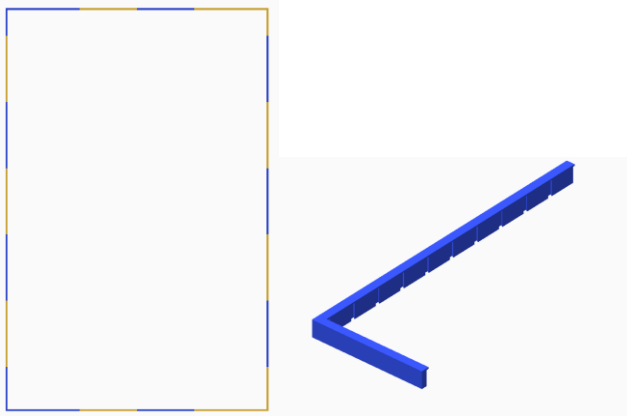
Subpanels



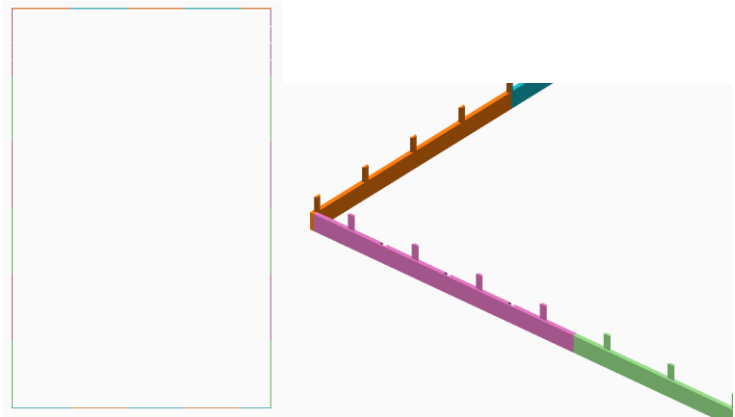
Seams



Outer_Frame



Border



Border_Background



Hanger

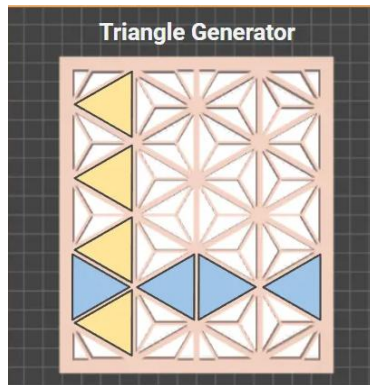
FRAME DIMENSIONS

Generator

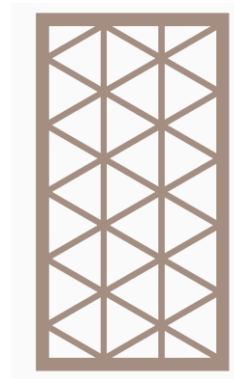
- Select to determine which method and parameter is used when generating your frame.
- **"Triangles"**: frame is generated based on the number of triangles horizontally and vertically inputted into the "Triangle_Generator" parameter.
- **"Dimensions"**: frame is generated based on the millimeter width and height inputted into the "Millimeter_Generator" parameter.

Triangle_Generator

- Formatted [N width triangles, N height triangles]



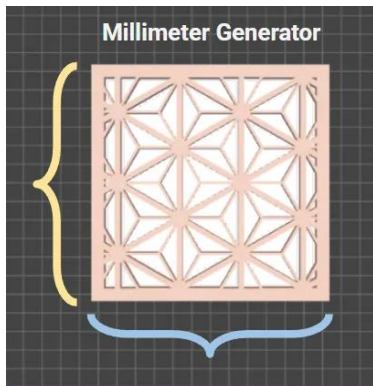
LEFT: How it's measured [4 - width, 4 - height]



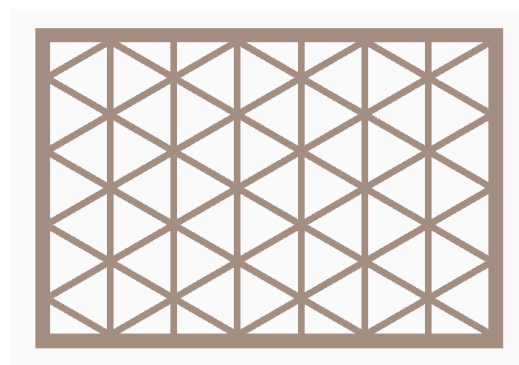
RIGHT: Triangle_Generator: [3, 5]

Millimeter_Generator

- Formatted [width mm, height mm]
- The frame is rounded down to fit your dimensions unless **"Make_Exact_Dimensions"** is true.



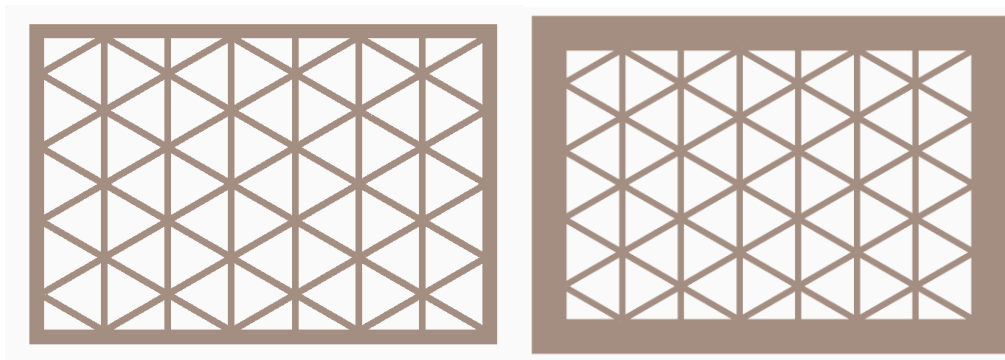
LEFT: How it's measured [width mm, height mm]



RIGHT: Millimeter_Generator: [350, 250]

Make_Exact_Dimensions

- Only works if “**Generator**” == “**Dimensions**”.
- **If false:** Fits the largest number of triangles vertically and horizontally to fit the given dimensions in “**Millimeter_Generator**” without going over.
- **If true:** Adjusts the frame border to exactly fit your dimensions inputted into “**Millimeter_Generator**”.



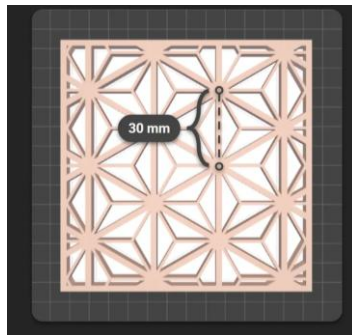
Triangle_Generator: [350, 250]

LEFT: Make_Exact_Dimensions: false – with a fixed border, the largest frame to fit is [318, 218]

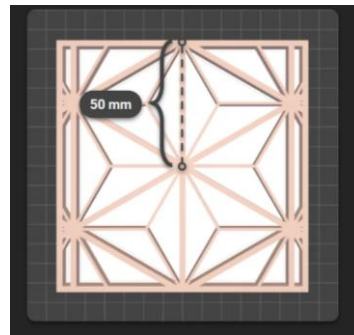
RIGHT: Make_Exact_Dimensions: true – the border is enlarged to fit exactly the dimensions [350, 250]

Grid_Pitch (mm)

- Distance between triangles in frame.



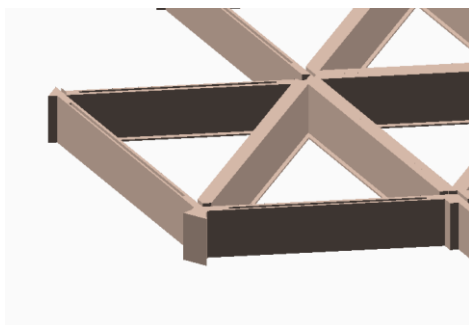
LEFT: Grid_Pitch = 30



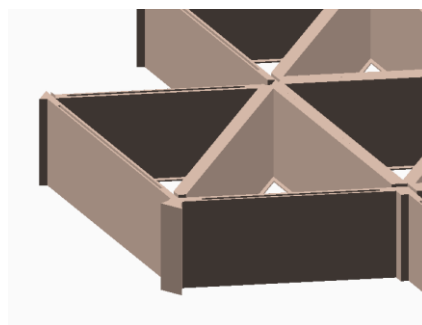
RIGHT: Grid_Pitch = 50

Frame_Depth (mm)

- Depth of the subpanels and borders.



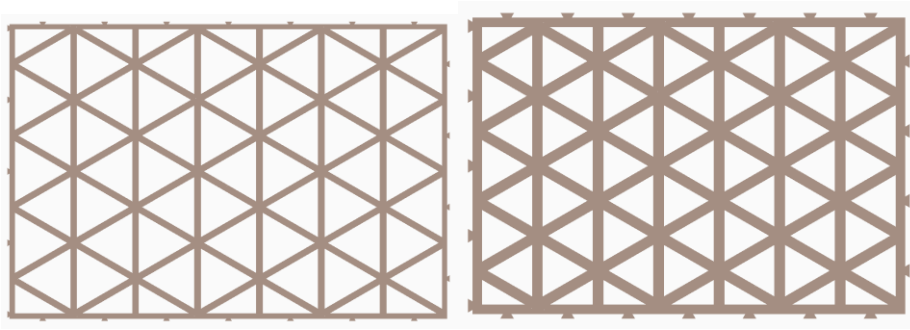
LEFT: Frame_Depth = 12



RIGHT: Frame_Depth = 24

Grid_Thickness (mm)

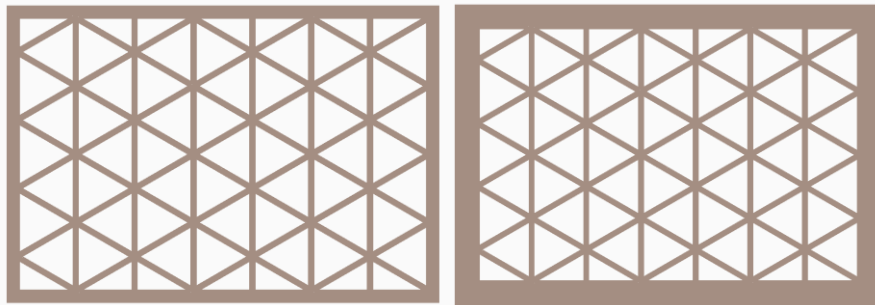
- Thickness of the frame



Border + Border_Background disabled for clarity
LEFT: Grid_Thickness = 3 **RIGHT: Grid_Thickness = 6**

Border_Thickness (mm)

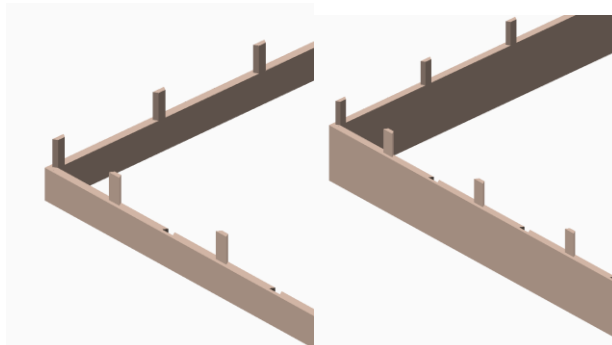
- Thickness of the border. This must be larger than $\text{Grid_Thickness} \times 2$, otherwise it will default to $\text{Grid_Thickness} \times 2$



LEFT: Border_Thickness = 0 (Defaults to $\text{Grid_Thickness} \times 2 = 3 \times 2 = 6$) **RIGHT: Border_Thickness = 16**

Background_Depth (mm)

- Depth of the “Border_Background” and “Hanger” components. If you plan on adding LEDs, make sure this is $\geq$ the LED width.



LEFT: Background_Depth = 12 **RIGHT: Background_Depth = 24**

STEP 2: Run the frame in "Reference Mode" (Only used for visualizing - NOT printing)

This step is meant to help you quickly adjust parameters in the frame to quickly prototype and visualize what the frame will look like and how the components come together. Do NOT print components in this step. This step is only meant to assist you until you finalize your parameters.

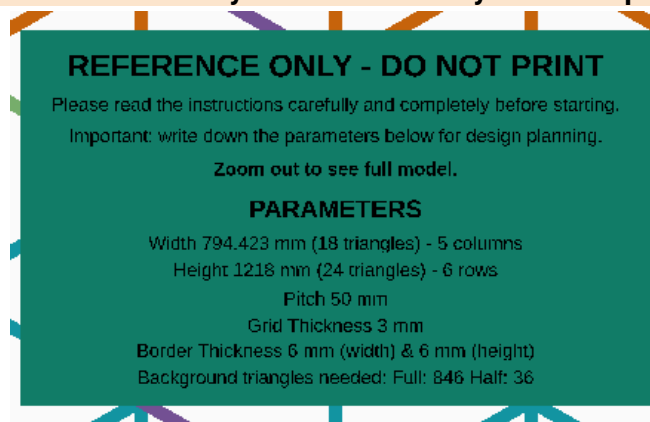
Step 2.1) Change "Assembly_Option" to "Reference Mode".

Optionally: also change "All_Component_View" to false. This is the fastest generation method as it minimizes duplicate components.

Step 2.2) Adjust the "Frame Dimensions" parameters until you're satisfied with the frame output.

Step 2.3) Change "Hide_Text" to false and verify that the frame dimensions match your requirements.

Step 2.4) Save a screenshot of the text box so you can reference your frame parameters in future steps.



Screenshot of the text box.

STEP 3: Run the frame in "Printable Mode", Render, and Export components

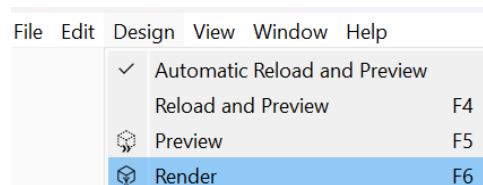
Once you've determined the frame parameters you're happy with, we can generate the printable version of the frame. Below is my preferred method of preparing print profiles.

This method involves exporting ALL objects for EACH component separately. This means that subpanels, seams, borders, etc will each get their own print profile. Doing this lets you arrange the objects faster in your print profile, and avoids having to do any calculations to determine the number of each component you need to print. The main drawback with this method is that there are many component variations which are difficult to spot, so by batch printing everything at random, you will have to sort the components after they're printed and find their corresponding location. For example, the top row and bottom row seams look identical but are mirror images of one another and can only be placed in their respective row. However, so long as you're patient with the assembly process, you'll be able to figure things out as you put things together. I typically print everything and then sort things afterwards.

Step 3.1) Set the parameters to match the following

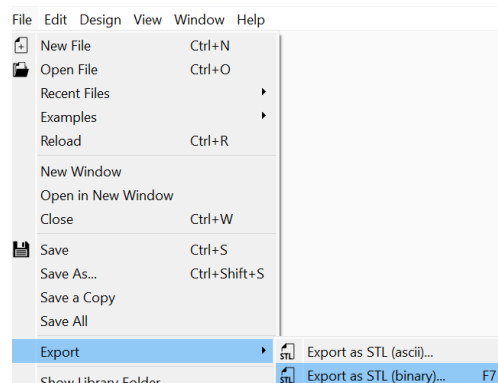
- **Assembly_Option = "Printable Mode";**
- **All_Component_View = true;**
- Set only 1 "Frame Components" parameter to true and leave the rest as false. Repeat the below steps for each component available so you have 6 separate stl files, one for each component (subpanel, seams, outer_frame, border, border_background, hanger).

Step 3.2) Render the object



Render the file

Step 3.3) Export as stl (binary). Make sure to rename the file to match the component you're exporting and add .stl to the end of the file name (for some reason when I rename an export, the .stl extension disappears).



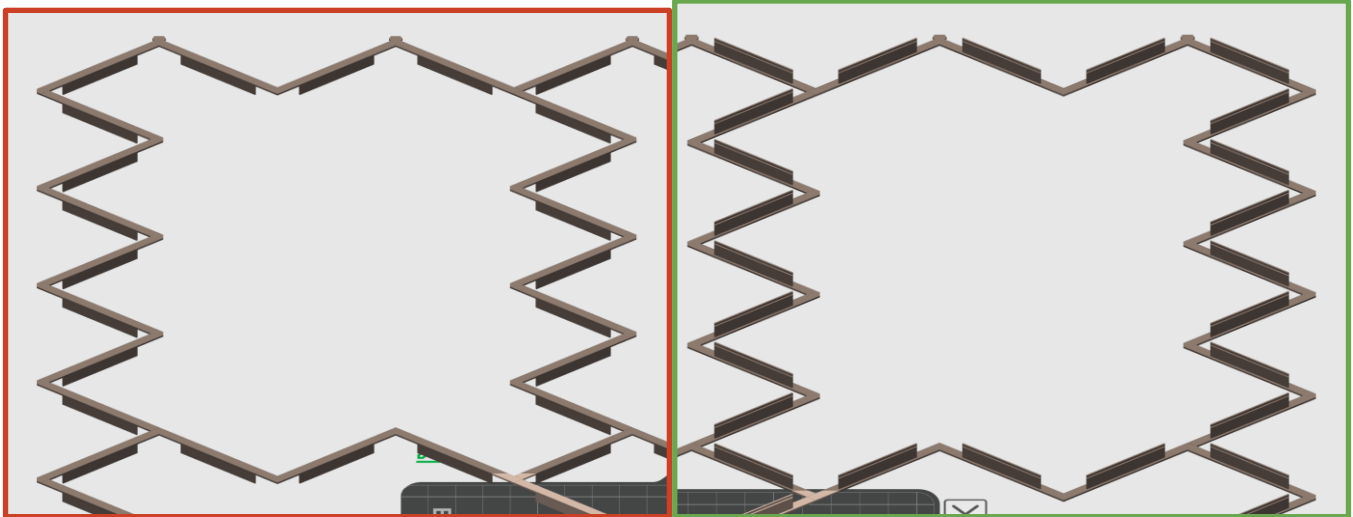
Export as STL (binary)

STEP 4: Prepare the print profiles

Finally we can prepare the print profiles for printing.

Step 4.1) Open stl in a slicer (the instructions below are using Bambu Studio)

Step 4.2) Flip over the following components over (Seams & Border components):



Seams

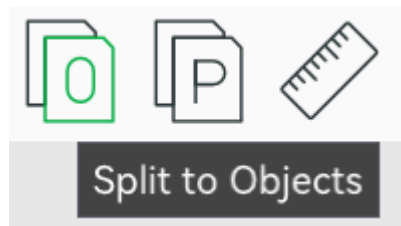
LEFT: Default orientation when loaded (**incorrect**) RIGHT: Orientation after flipped in slicer (**correct**)



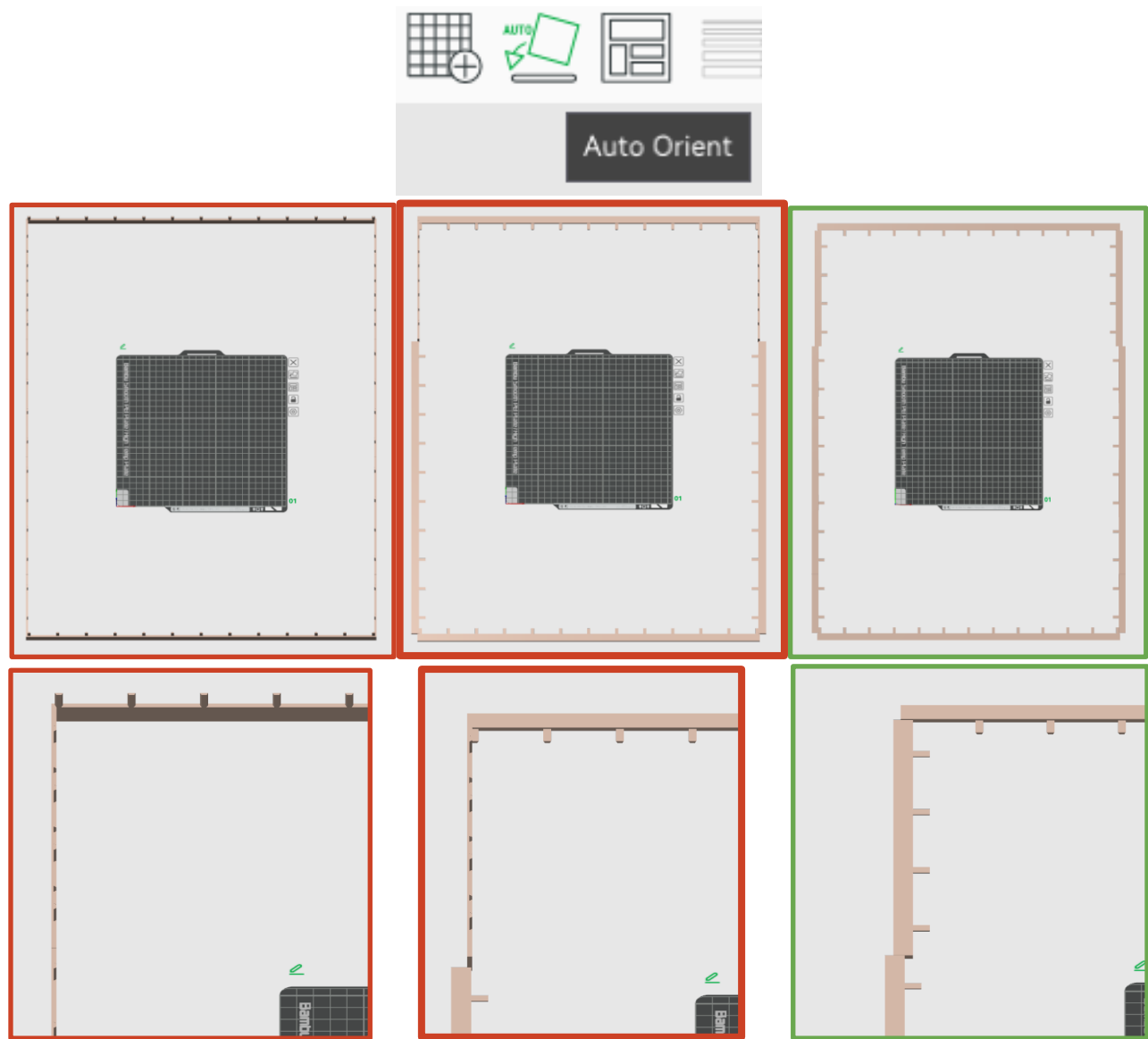
Border

LEFT: Default orientation when loaded (**incorrect**) RIGHT: Orientation after flipped in slicer (**correct**)

Step 4.3) Split component to objects



Step 4.4) For Border_Background components select "Auto Orient". This will reorient all the components EXCEPT for two components. It's extremely important that you reorient these two remaining components to match the others or else they will break and your frame will fall.

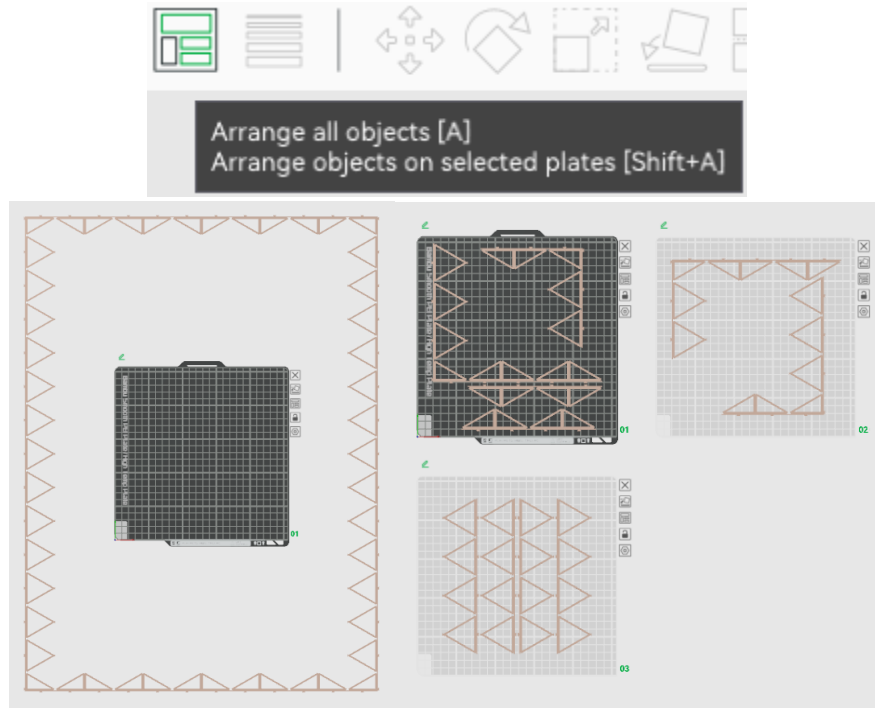


LEFT: Orientation when loaded (incorrect)

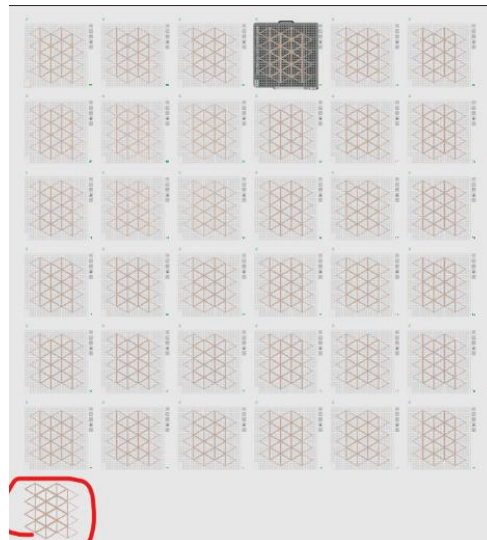
MIDDLE: Orientation after "Auto Orient" (incorrect)

RIGHT: Orientation after manually correcting corner components (see top left and top right of image to see difference) (correct)

Step 4.5) Click “Auto Arrange”. Note if you have a large number of components (mainly subpanels), you may run out of plates (Ex. Bambu Studio limits you to 36 plates), leaving leftover components in the corner. If this happens, you’ll have to swap out components after you’ve printed them until you’ve printed the leftovers.



LEFT: Before auto arrange RIGHT: After auto arrange



Example of leftover subpanels (See bottom left). If you have leftovers, you need to swap out components until you’ve printed all the leftovers.

Step 4.6) Set the print parameters for the corresponding component. See below for details:

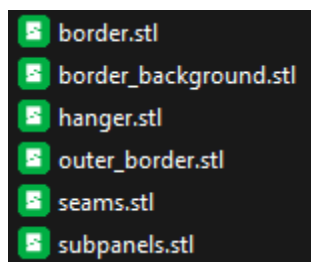
- **Note:** I start with the "0.20mm Standard" preset for a 0.4 nozzle and then make the following changes to the print profile and save the preset for future use.

Component	Walls	Infill Density	Infill Pattern	Wall Generator	Top shell Layers	Bottom Shell Layers
Subpanels	1	30%	Lightning (to save material)	Arachne *	>= 5	>= 3
Seams						
Outer_Frame		10%	Gyroid (for strength)			
Border						
Border_Background	>= 5	15%				
Hanger						

* It is essential the wall generator is set to "Arachne". The walls made using "Classic" are too thick.

4.7) Print all the plates.

4.8) Repeat for the remaining components.



All component files exported separately.

STEP 5: Design your panel in [kumikodesigner.com](https://www.kumikodesigner.com)

Before or while you start printing your frame components, start designing your panel on [kumikodesigner.com](https://www.kumikodesigner.com). It's important you follow these steps correctly or you risk having to redo your design with the correct settings. It's not a huge deal if you make a mistake, but you will lose some time correcting the mistakes.

Note: Please consider donating to **Douwe**, the creator of the Kumiko Designer tool below. I truly couldn't have created this project without the help of their tool. Click the "Donate" button on their website.

Support my work



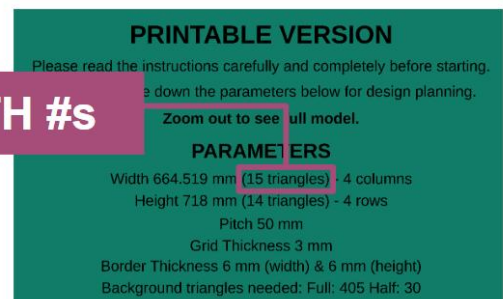
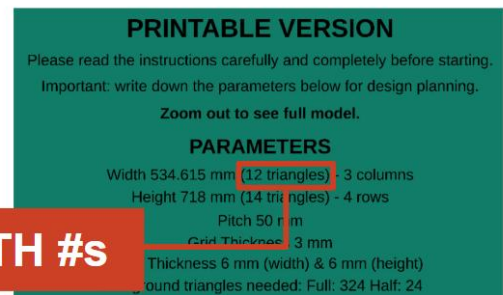
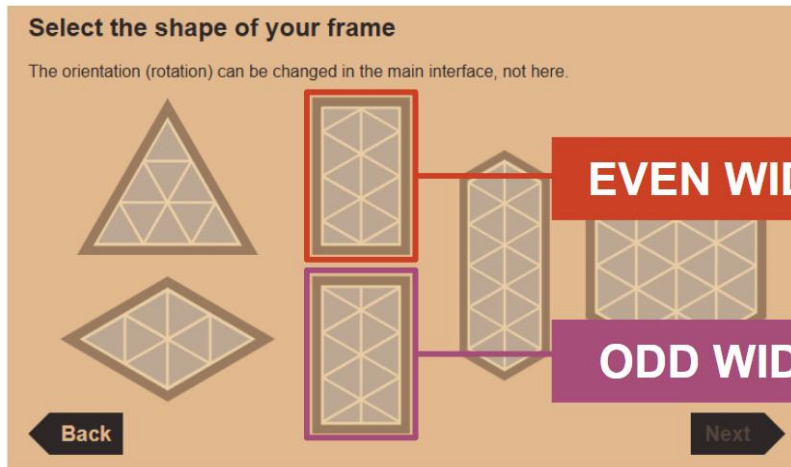
Creating and maintaining this website requires a significant time investment for me. If you enjoy using this application, please consider making a small donation in order for me to keep the website hosted.

Donate to Douwe at <https://www.kumikodesigner.com/>.

Step 5.1) Head over to Kumiko Designer website ([link here](https://www.kumikodesigner.com)). Click "Start designing", select "Framed panel".

Step 5.2) Refer to the screenshot of the text box made in Step 2.4, and refer to Width triangles.

- If N triangles width is even: select the top rectangle
- If N triangles width is odd: select the bottom rectangle



Step 5.3) Referring to the text box, input the correct "Size A" (equivalent to height) and "Size B" (equivalent to width), Pitch, and Mitsuke (equivalent to Grid Thickness) into kumiko designer. You can leave Frame width as default.

PRINTABLE VERSION

Please read the instructions carefully and completely before starting.
Important: write down the parameters below for design planning.

Zoom out to see full model.

PARAMETERS

Width 534.615 mm (12 triangles) 3 columns
Height 718 mm (14 triangles) 4 rows
Pitch 50 mm
Grid Thickness 3 mm
Border Thickness 6 mm (width) & 6 mm (height)
Background triangles needed: Full: 324 Half: 24

Frame dimensions

Height

Size A (number of triangles): 14
Size B (number of triangles): 12
Frame width (mm): 12

Frame dimensions

Width

Pitch (mm): 50
Mitsuke (mm): 3

Step 5.4) Validate the frame.

Step 5.4.1) Validate that the correct frame was selected in step 5.2. To do this, compare the top left corner of both frames and make sure they are identical.

LEFT: Frame from Step 5.2

MIDDLE: Matches (correct)

RIGHT: Incorrect

Step 5.4.2) Validate the correct parameters were inputted in step 5.3. To do this press "p" on your keyboard for the frame parameters you inputted. Compare what's on the screen to the parameters in your text box screenshot. Ignore Frame_Depth and compare everything else.

PRINTABLE VERSION

Please read the instructions carefully and completely before starting.
Important: write down the parameters below for design planning.

Zoom out to see full model.

PARAMETERS

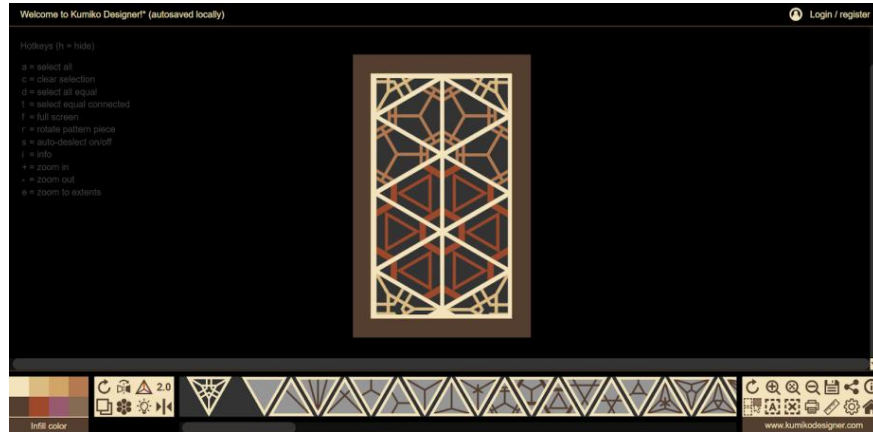
Width 534.615 mm (12 triangles) 3 columns
Height 718 mm (14 triangles) 4 rows
Pitch 50 mm
Grid Thickness 3 mm
Border Thickness 6 mm (width) & 6 mm (height)
Background triangles needed: Full: 324 Half: 24

```
// Uses triangles as dimensions (width, height)
Triangle_Generator = [12][14];
// Distance between triangles in frame.
Grid_Pitch = 50;
// Depth of frame
Frame_Depth = 12;
// Thickness of frame
Grid_Thickness = 3;
// Thickness of border around frame.
Border_Thickness = 12;
```

Validate the saved parameters (LEFT) with kumikodesigner parameters (RIGHT)

Step 5.4.3) If either 5.4.1 or 5.4.2 does not match, redo step 5 and re-verify.

Step 5.5) Design your frame. Using the tool you can customize the inserts and colors. Consider making an account to save your designs.



Step 5.6) Export the design SVG and head over to **svg_insert_generator.scad** to generate the inserts for your panel.



STEP 6: Follow SVG Insert Generator Instructions

Once you're done designing and have exported your design from kumikodesigner.com head over to the [SVG Insert Generator](#) instructions and using the **svg_insert_generator.scad** file, quickly generate all the inserts needed for your panel. Once done, head back here to follow the assembly instructions.

STEP 7: Assembly

Once all the components are printed, you can [follow the assembly instructions here](#).

2. SVG Insert Generator

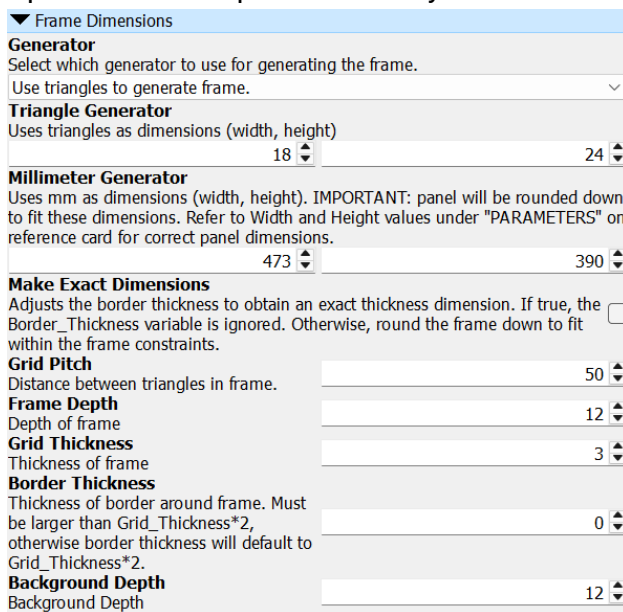
Message me on [MakerWorld](#) or [Patreon](#) if you have any questions, need help, or notice any bugs. I'm quite responsive but if there's a lot of individuals contacting me it'll take me longer to respond, so your patience is appreciated. **Before reaching out, please read the instructions thoroughly, as they may answer your question.**

BACKGROUND

The `svg_insert_generator.scad` file will let you generate the background inserts and designs made using kumikodesigner.com in just a few steps. Faster than the manual generator alternative.

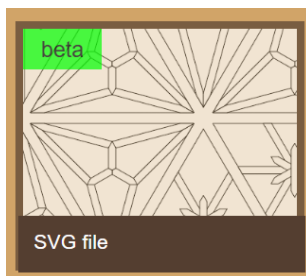
REQUIREMENTS

1. Your frame parameters (it's important this is inputted correctly to orient and trim the inserts)



The screenshot shows a web interface for generating frame dimensions. It has a dropdown menu for 'Generator' with 'Triangle Generator' selected. Below it are input fields for width (18) and height (24). The 'Millimeter Generator' section is also visible, with width (473) and height (390) inputs. There are checkboxes for 'Make Exact Dimensions' and 'Grid Pitch'. The 'Grid Pitch' section has a value of 50. The 'Frame Depth' section has a value of 12. The 'Grid Thickness' section has a value of 3. The 'Border Thickness' section has a value of 0. The 'Background Depth' section has a value of 12.

2. SVG export file from kumikodesigner.com



STEPS

STEP 0: Complete the following before starting:

Step 0.1) Finalize your frame via the frame generator.

- **IMPORTANT:** Make sure to note down the frame parameters used.

Step 0.2) Design your panel using kumikodesigner.com.

- **IMPORTANT:** Make sure you selected the correct frame type in kumikodesigner (depends on if your triangle width is odd or even). If you select the wrong frame, your design cannot be replicated 1:1. Refer to the frame generator instructions to understand how to tell if the frame you see in kumikodesigner is the correct one.

Step 0.3) Export the SVG from kumikodesigner.com

STEP 1: Prepare the SVG file

Step 1.1) Head over to **< redacted url - see svg_insert_generator.scad file for url >**

Step 1.2) Run the code

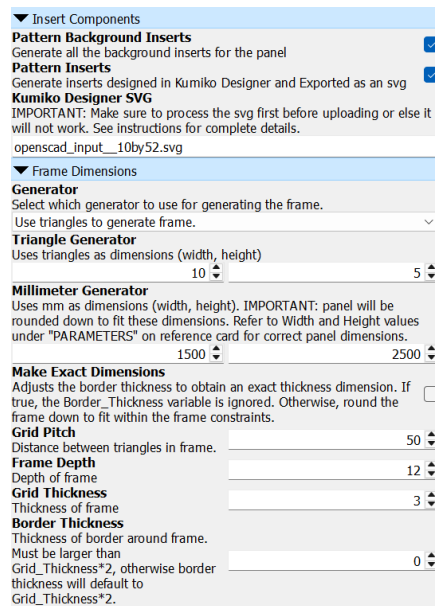
Step 1.3) Click "Choose Files" and select the SVG file downloaded from kumikodesigner.com

Note: make sure to only upload one SVG at a time since your browser will block additional downloads (at least mine does...).

Step 1.4) Download the modified SVG file and **add it to the same folder as the svg_insert_generator.scad**

Step 1.5) Come back here for the next step.

STEP 2: Edit the "Frame Dimensions" parameters below to exactly match the frame you generated.



The screenshot shows a software interface with a sidebar on the left containing a tree view with the following items: '▼ Insert Components', 'Pattern Background Inserts' (checked), 'Pattern Inserts' (checked), and 'Kumiko Designer SVG'. The main area displays the '▼ Frame Dimensions' section, which includes several sub-sections with adjustable parameters:

- Generator**: A dropdown menu set to 'Use triangles to generate frame.'
- Triangle Generator**: A note 'Uses triangles as dimensions (width, height)' followed by two spinners: '10' and '5'.
- Millimeter Generator**: A note 'Uses mm as dimensions (width, height). IMPORTANT: panel will be rounded down to fit these dimensions. Refer to Width and Height values under "PARAMETERS" on reference card for correct panel dimensions.' followed by two spinners: '1500' and '2500'.
- Make Exact Dimensions**: A checkbox that is currently unchecked, with a note 'Adjusts the border thickness to obtain an exact thickness dimension. If true, the Border_Thickness variable is ignored. Otherwise, round the frame down to fit within the frame constraints.'
- Grid Pitch**: A spinner set to '50'.
- Frame Depth**: A spinner set to '12'.
- Grid Thickness**: A spinner set to '3'.
- Border Thickness**: A spinner set to '0'.

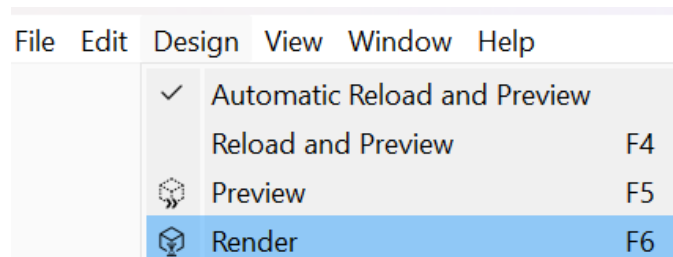
Update the values above to match your frame.

IMPORTANT: If you forgot to save the dimensions, you can click "p" on your keyboard when looking at your design on kumikodesigner.com and the parameters will show up. Ignore the "Frame_Depth" parameter since it'll always be set to 12 on kumikodesigner.com. The website creator was very kind to add this option to their site so please consider donating to support their work.

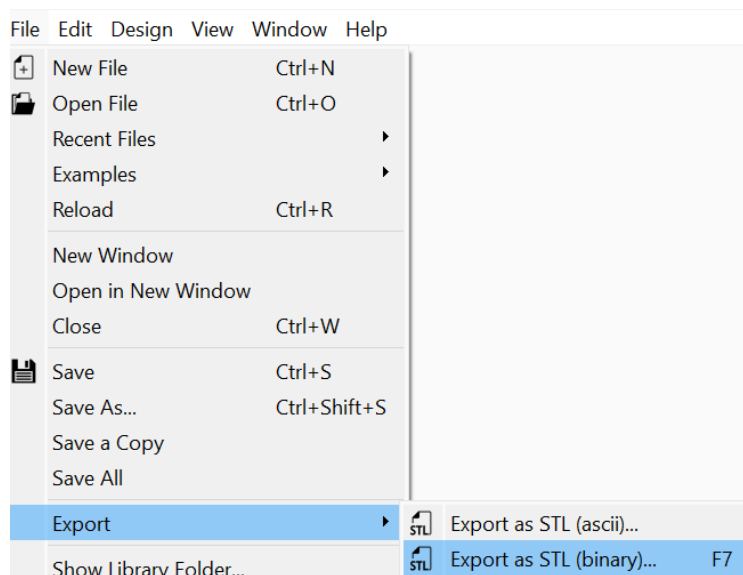
STEP 3: Update the "Insert Components" parameters. See below for details about the parameters:

- **Pattern_Background_Inserts** (true or false): Whether to print the background inserts that go behind each kumiko insert.
- **Pattern_Inserts** (true or false): Whether to print the kumiko inserts using the SVG file
- **Kumiko_Designer_SVG** (path): Path to the prepared SVG file from STEP 1.

STEP 4: Render and Export file as STL (binary).



Render the file



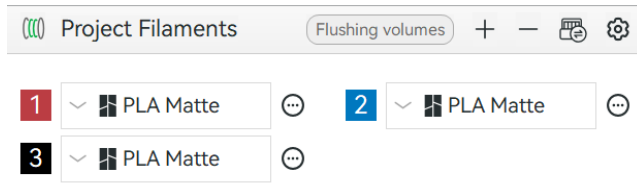
Export as STL (binary)

STEP 5: Open stl in your slicer, split the stl to parts, assign the color, arrange objects, set the print parameters, and print!

How to split to objects: <https://wiki.bambulab.com/en/software/bambu-studio/split-to-objects-parts>

How to assign colors: <https://wiki.bambulab.com/en/software/bambu-studio/multi-color-printing>

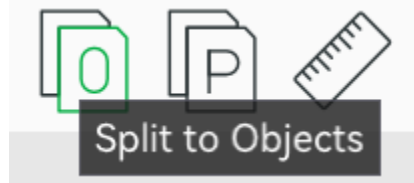
- I prefer the option of right clicking the object then clicking "Change Filament" and selecting the filament color. See below for details:



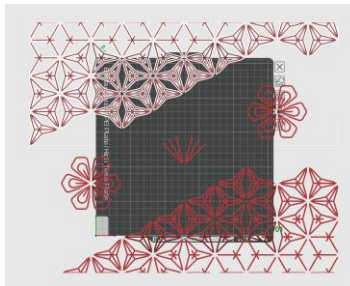
Add the relevant filaments



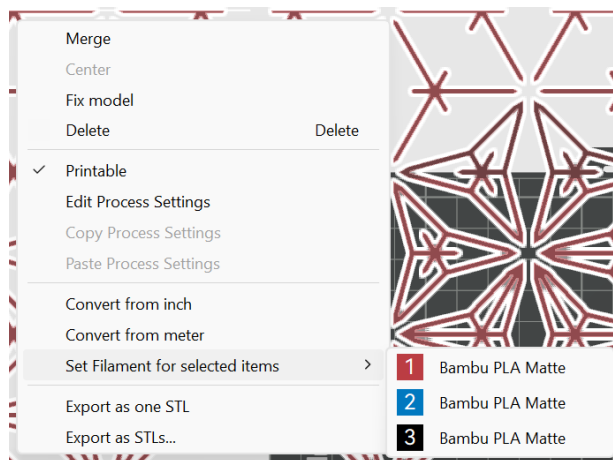
Adjust the filament color



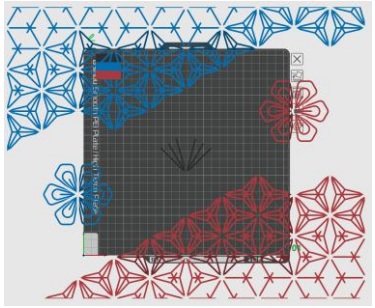
Select inserts and split to objects



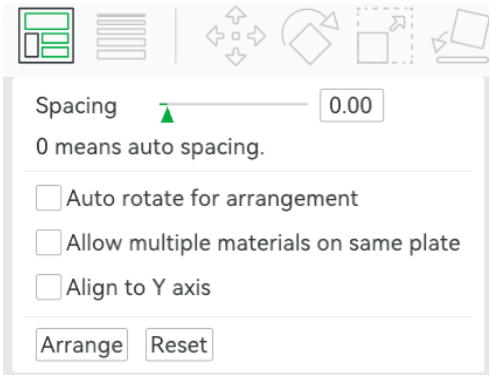
Highlight the components (Hold shift+ctrl and click and drag to select multiple inserts)



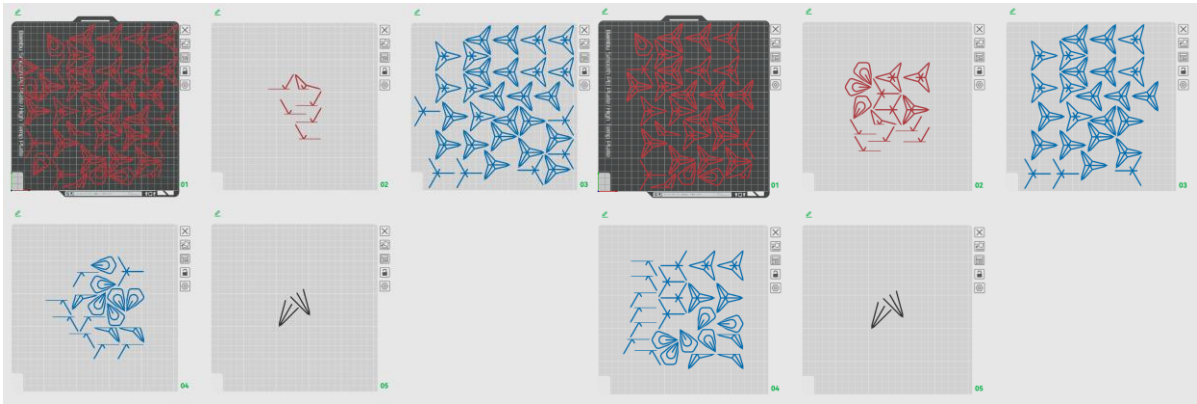
Right click on the inserts > Set Filament for selected items > Select the filament



Repeat until all colors are applied



Auto arrange inserts (**Make sure “Allow multiple materials on same plate” is deselected**)



Move any inserts close to the edges to a different plate and rearrange.
Left: before | Right: after

Recommended Print Parameters

- Note:** I start with the "0.20mm Standard" preset for a 0.4 nozzle and th

	Layer Height	Initial Layer Height	Walls	Infill Density	Infill Pattern	Wall Generator	Top shell Layers	Bottom Shell Layers	
serts	0.1	0.1	1	30%	Lightning (to save material)	Arachne	>= 5	>= 3	All ot de "0.20

	0.2	0.2							For a
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